Where the Error of the Factorised Cumulant Chain Lives, and What It Costs

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We locate where the error of a factorised cumulant chain for deep random ReLU MLPs is made, and what removing it would cost. The closures are accurate given true inputs; the error is input error made at each ReLU and compounded through depth; the κ_3 input and the per-neuron pattern (not the level) of the κ_4 diagonal are the decisive missing inputs; and a parity theorem shows why closures of old κ_3 built on even statistics lose it entirely. A Hermite ceiling ladder shows that second-order κ_4 carriers are capped near pattern correlation 0.5 while cubic births reach 0.8; the cubic-star feature built from it graded −4.4 %.

This write-up refers to submission #337457 (graded in Phase 2): adjusted score **2.7858e-9**, raw final-layer MSE **1.8592e-8**, util 0.150 (FLOPs as a fraction of the budget, §2.1). It is LordProcciuto's public e30_a chain (MIT; 3.1163e-9 as published, #335981) with kai_leal_gonzalez's (3,1) closure (MIT), plus our κ_4-pattern corrector (V55/V56, §5), our settings, and our refit and extension of e30_a's mean-correction table (§6.5). §2.3 separates their parts from ours. *For reference only:* our best result without e30_a's base chain is #337049 (e299safe, 3.2529e-9): our V29-based chain #334870 (v56As_fs, 3.6521e-9), on which §3–§5 are measured, plus mdhamidhosen's calibration and cost layer (§6.4) and e30_a's exact cost savings, fixed old basis and C_pre compaction with kai_leal_gonzalez's closure at mix 0.25 (§6.5).

21 October 2026. This is the main write-up. The full technical report is attached as the Appendix (full technical report), and "App. §x" refers to its sections. Each number and claim is labelled by its source:

- *graded: an official grade, with its submission id;*
- *measured: a run of our own scripts, with the result file named in App. §8.3;*
- *derived: algebra or arithmetic, with the derivation in the Appendix;*
- *estimate or untested: not yet measured;*
- *conjecture: an interpretation we have not tested, marked as such.*

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1. Summary

We started from 504aldo's public, MIT-licensed factorised third-cumulant chain ("V29"), which graded 5.418e-9 when submitted verbatim. Our changes take it to **3.6521e-9 (−32.6 %)**: raw MSE falls by 8.6 % (2.133e-8 → 1.950e-8) and util by 26 % (0.254 → 0.187) (derived). A final calibration and cost layer by mdhamidhosen on this chain (E282–E299, §6.4) takes it to **3.3946e-9 (#336377, −7.0 %)**; that layer is fitted and engineering, not new understanding. The submission this write-up refers to, **#337457 (2.7858e-9)**, moves our corrector onto LordProcciuto's cheaper public chain e30_a (MIT, 3.1163e-9 as published) with kai_leal_gonzalez's (3,1) closure; our corrector, our settings and our refit and extension of e30_a's correction table give −10.6 % there, −6.3 % of it from the fitted table (§2.3, §6.5). Of the contributions below, the V56 star (item 3) and the analysis behind it (items 1 and 3) act in #337457; the spherical identity and the parity theorem belong to our own chain.

Mechanistic contributions. These are what we consider our contribution to the understanding of white-box estimation:

1. **An error map with channel oracles** (§4; measured on $6{\cdot}10^7$-sample Monte-Carlo teachers). The closures are accurate when given true inputs; the error is input error made at each ReLU and accumulated through depth; the teacher's κ_3 and κ_4 inputs together would take the raw MSE to ×0.272 on MLP #0; the κ_4 gain lives in the **per-neuron pattern** of the κ_4 diagonal, not its level.
2. **A parity theorem** (§3.2; derived, verified numerically): old κ_3 is odd in the later weights, so every closure built on even statistics loses it.
3. **A ceiling ladder and the Hermite cubic-star feature V56** (§4.4, §5.2; identity derived and verified numerically to ≤ 8.1e-16; graded −4.4 %): second-order births cap the κ_4 pattern at 0.45–0.55, cubic births reach 0.80–0.85.
4. **An analysis of the κ_3-dressed C·T channel** (§4.5; oracle and component level, not graded): with the star it reaches the cubic-birth ceiling, and the chain's own legs reproduce it, but not yet at an affordable cost.
5. **Closure-input errors cancel** (§7.1 (2); measured at one step): the sum of their norms is 1.9–2.6× the norm of their sum (var against d3, cosine down to −0.83), so correcting one channel alone hurts.

6. **A spherical-input identity** (§3.1; derived, injection check measured, graded −1.1 % raw at zero cost).
7. **A negative-result atlas** (§7): 24 of our 60+ killed architecture and representation ideas, each with its gate and measured number, and three working conclusions that we label as empirical, not proven.

Engineering and fitted components. We state these plainly. Most of the gain from 5.418e-9 to 4.0784e-9 is **cost engineering**: Strassen placement, a transport join, dead-neuron compaction, skipping D21 (κ_3 slice, §2.2) entries of dead neurons, and an operation budget for the grader's hidden limits (§6). This is performance optimisation, not new understanding. The κ_4 gain from 4.0784e-9 to 3.6521e-9 (−10.5 %) passes through **V55, a linear map fitted offline** to our own Monte-Carlo teacher (§5.1); the V56 star enters as one more feature of that fitted map. Five constants are **tuned**, not derived: the feedback scale FB = 1.25 (§3.3), the adaptive-λ exponent β = 1.2, a ×1.25 on the V55 map's s3sq coefficient, the dense-age cut AGE_OLD = 5 and the old-basis rank R_OLD = 352. The −7.0 % of §6.4 is a Gauss-Newton calibration fitted on public MLPs plus mostly exact cost rewrites and two threshold approximations. The final −17.9 % (3.3946e-9 → 2.7858e-9, §6.5) comes from moving our corrector onto LordProcciuto's public chain, settings, and a refitted mean-correction table, which is a fitted component.

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2. Starting point: the reference method (not claimed)

2.1 Task and score

For a bias-free ReLU MLP of width n = 1024 and depth 16, with He-initialised weights and input $x \sim N(0, I)$, the estimator returns $E[h^{(l)}]$ for every layer. The score is

score = MSE_final × max(0.1, FLOPs / 2^{41}),

with the final-layer MSE measured against a 10^9-sample Monte-Carlo reference. FLOPs are metered by flopscope. We use **1 u = 2^{31} FLOPs**, about one dense 1024^3 product; the budget is then 1024 u, "util" is FLOPs/budget, and the 0.1 floor is 102.4 u. The grader caps each MLP at 120 s of wall time, 0.4 s of residual Python time outside metered operations, and 8 GB of address space. A capped MLP scores about 0.35, so a single cap in a hundred MLPs ruins a submission.

2.2 504aldo's V29 (MIT)

Our estimator modifies **504aldo's public V29** (`whest-p2-cumulant-k3`, forum post #18218), which implements the factorised K = 3 "simple" cumulant propagation of Wu et al. (arXiv:2605.05179, §4.3). Per layer it carries the mean, the covariance C, and the third cumulant κ_3 as a **sum of sources**, one per ReLU. Each source is stored as a propagator leg P_s, a birth leg A_s = P_s a_b, and thin n×16 feedback legs. The next ReLU needs two κ_3 slices, the diagonal d3 (d3_i = κ_iii) and the (2,1) slice D21 (D21_ic = κ_iic), read per source from contractions of Hadamard products of the legs. Sources older than four layers (the age cut AGE_OLD) live in a shared basis of rank R_OLD = 384 (the oldest in a nested rank-224 sub-basis). κ_4 is not carried: its off-diagonal part is regenerated each layer as G_off = λ_l C_off (the "λ law"), with a per-MLP adaptive λ. Products use pooled Strassen–Winograd.

The author prices V29 at 260 u and reports a tolerance law: extra final MSE ≈ 4.2e-6·ε^2, where ε is the relative RMS error of D21, so D21 must be right to about 2 %.

Graded baseline. V29, submitted verbatim, graded **5.418e-9** (raw 2.133e-8, util 0.254; **#331681**), matching the author's own submission.

Licence and scope of our claims. Every file we ship keeps 504aldo's MIT copyright notice and LICENSE, and carries a lineage header that names each change and its date. **Only the changes in §3–§6 are claimed as ours.** The sum-of-sources representation, the shared basis, the memoryless κ_4 law, the feedback legs and the pooled Strassen engine are the reference method.

2.3 The base of #337457: LordProcciuto's e30_a and kai_leal_gonzalez's closure (MIT; not claimed)

LordProcciuto's public **e30_a** (github.com/siming767-creator/whestbench-lordprocciuto; 3.1163e-9, #335981, cited) descends from V29. #337457 also uses **kai_leal_gonzalez's two-term closure of the (3,1) slice of κ_4** (github.com/al07-alt/whest-p2-k31-two-term; forum topic 18246).

Component of #337457	Origin
Sum-of-sources chain, shared basis, λ law, feedback legs, pooled Strassen	504aldo (V29), via e30_a
Removal of neurons with $\alpha < -2.85$, fixed-basis old level (rank 416/224) outside hot rows, Winograd leaf, exact fixed-cost savings, λ tilt, near-dead column cut τ_B (value 2.4 set by us)	LordProcciuto
Mean-correction table: 87 ridge features fitted on 70 public nets	LordProcciuto
Two-term (3,1) closure and its per-layer coefficients	kai_leal_gonzalez
κ_4-pattern corrector: V55 map and V56 cubic star (§5), coefficients refitted in mdhamidhosen's E-series (§6.4)	**ours**
Corrector scale 0.4, star/local split 0.7/0.2, closure mix 0.4, D21 ×0.98 at $L \geq 9$	**ours** (tuned)
Table refitted on our trajectory with 77 new features (164 in all; §6.5)	**ours** (fitted)

Not in #337457: V31 (§3.1; worse on e30_a, #337221), FB 1.25, our cost engineering (§6.1–§6.3), the E-series groups and rider. From e30_a's 3.1163e-9: corrector −2.1 %, settings −1.8 %, table −6.3 %, τ_B −0.8 % (−10.6 % in all; derived).

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3. Exact identities and derivations

3.1 Spherical-input identity (derived; V31)

Write the Gaussian input as $X = (R/\sqrt{n})\cdot Y$, with Y uniform on the sphere of radius $\sqrt{n}$ and $R \sim \chi_n$ independent of Y. A bias-free ReLU network is positively homogeneous of degree 1, so

$E[h(X)] = a_1 E[h(Y)]$, $a_1 = \sqrt{(2/n)}\,\Gamma((n+1)/2)/\Gamma(n/2) = 0.99975589$.

The sphere input has $C = I$ and $\kappa_3 = 0$, and its κ_4 is an exact harmonic core, $g = -6/(n+2)$ in the chain's representation. The estimator injects that core at layer 0, runs the existing κ_4 machinery unchanged, and rescales the output by a_1.

- **Measured:** the injection reproduces the predicted scale shift to 0.3 %.
- **Graded:** −1.1 % raw at zero cost (#331787).

The idea came from barnobarno666's public Phase-2 repository (idea only), pointed out to us by GPT-5.6 Sol in an outside consultation (App. §2.1). The derivation for this chain and its integration are ours.

3.2 Parity theorem: no mean-field part of the deep κ_3 (derived and verified; App. §2.2)

Take a source born at layer b. At any later layer l its legs are $P = W^{T(l)}\,D\,P'$ and $A = P\,a_b$, where P', a_b and the birth coefficients do not depend on $W^{(l)}$. Every quantity linear in the source's κ_3 (the slice products and d3) is therefore **odd** in $W^{(l)}$, while $C^{(l)}$, the leg Grams and the per-neuron variances are **even**. Because the weight ensemble is sign-symmetric,

$E[\,H_s^{(l)} \mid C^{(l)}, \mathrm{Gram}(A_s, P_s), \text{all state before } l\,] = 0$, exactly.

Consequences:

- Any closure that replaces old sources by a function of even statistics (mean-field closures, scale mixtures, polynomials in C) discards their entire contribution. This is consistent with the failures of such closures reported in 504aldo's public log (9.1e-7 and 2.3e-6) and with our learned residual correctors, which recover only 0.3–0.7 % of the residual (§7).
- Replacing a source by its conditional expectation given one fresh weight row has a relative-error floor of $\sqrt{(6/(r+8))}$ in the retained dimension r: 12 % at $r = 384$, far above the 2 % tolerance.
- κ_4 terms built from two κ_3 factors are even, and $C \otimes C$ is the only even invariant available. We read this as the reason the regenerated law $G_{off} = \lambda_l C_{off}$ works (conjecture beyond the derivation; not tested separately).

Status: derived, then verified (11 Oct; `experiments/probes/parity_check.py`). On the frozen V56 chain (width 128), flipping $W^{(l)}$ leaves every earlier layer and $C^{(l)}$, $\mathrm{var}^{(l)}$ bit-identical and flips μ, d3 and D21 exactly (difference 0.0; tested at $l = 5, 10$). The floor's algebra holds by Monte Carlo, and the bound $\|\tau\|^2 \leq (r+2)\|T\|^2/3$ is tight (it equals the exact operator norm) (eigen-computation, $r = 2$–8); a written proof is in App. §2.2. The floor is a statement about the idealised fresh-row

model. The mean $\mu^{(1)}$ is also odd, so closures that condition on μ are not covered. The argument originated in a consultation with Fable 5, an Anthropic Claude model run as a Claude Code subagent.

3.3 A half-carried feedback term (derived and measured; shipped as tuned FB 1.25)

Expanding the post-ReLU κ_3 to first order in C shows that the newborn source carries exactly **half** of two all-distinct K×C families (derived, App. §2.4). Write FB for the scale on these carried families: FB = 1 is V29, FB = 2 restores them in full. Regression on the slice residual gives coefficients of −0.55 to −0.9, against −1/2 predicted: consistent in sign and size, not exact.

Restoring the missing families in full makes the final mean worse: +3.4 % / +3.1 % on the teacher pair (MLP #0 and a synthetic net, §4), and ×1.024 on 100 MLPs at FB 2.0. A quarter restoration (FB 1.25) with the adaptive-λ exponent β = 1.2 scored ×0.9946 ± 0.0007 on the 100-MLP harness and graded 5.1284e-9 (#332110), at zero FLOPs. The shipped scale is therefore a **tuned constant**, not the derived one. Why the full restoration hurts is open; one explanation is ruled out (App. §6.5 (b)): the quadratic feedback cross term that also grows with FB is inert.

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4. The error map: where the chain's error lives

All results in this section are measured, using our Monte-Carlo teachers with $6{\cdot}10^7$ samples on public MLP #0 and a synthetic 1024×16 He network. They record the per-layer pre- and post-ReLU means, covariances, κ_3 slices and κ_4 diagonal. An **oracle** replaces one quantity inside the chain by the teacher's value and reruns the chain end to end. Two networks are a small sample; we state each result for the network it was measured on.

4.1 The closures are accurate; the error is accumulated input error

- **Mean closure.** Given the teacher's true marginals (μ, var, κ_{3iii}, κ_{4iiii}), the Edgeworth mean closure matches the teacher's ReLU mean to 5e-11–1.5e-10 at every layer.
- **Covariance closure.** Teacher-forced at one layer, the bivariate covariance closure falls to the teacher's noise.
- **Accumulation.** The chain's per-layer MSE grows smoothly, from 0.7e-9 at L1 to 2.29e-8 at L16.
- **Inherited share.** Forcing the true mean at every layer removes about 80 % of the final error.

We read these as: the error is **fresh input error** made at each ReLU and carried forward through depth.

4.2 Channel and joint oracles

The base is the plain chain at util 0.2476 (App. §4.8 gives the full table).

Teacher input injected (every layer)	MLP #0	Synthetic net
None (plain chain)	2.273e-8 (×1)	1.920e-8 (×1)
κ_3 slice + d3	×0.607	×0.716
κ_4 diagonal	×0.686	×0.885
κ_3 slice + d3 + κ_4 diagonal	**×0.272**	**×0.569**
κ_3 slice + d3 + κ_4 *pattern* at the chain's level	×0.294	×0.606

- **κ_3 is the larger single lever** on both networks.
- **The two inputs compound more than multiplicatively.** The product of the single-input ratios would be 0.416 and 0.634; jointly they give 0.272 and 0.569.
- **The remainder is consistent with the variance channel.** In the separate channel oracle of App. §4.2 (base 2.290e-8), the true variance, κ_3 diagonal and κ_4 diagonal together take MLP #0 to ×0.057, the truth's own noise floor. The two oracles inject different κ_3 quantities, so this is a consistency check, not a decomposition.

4.3 κ_3: where the error is real, and where the gain lives

We compared the chain's κ_3 with the teacher at every layer, and subtracted in quadrature a Gaussian estimate of the teacher's own sampling noise (derived; this assumes the two are independent):

Layer	D21 off-diagonal error	Teacher noise (est.)	Net D21 error	d3 norm ratio, chain/teacher
2	4.4 %	5.3 %	≈ 0	1.002
4	4.2 %	2.5 %	3.4 %	0.998
8	4.9 %	1.2 %	4.8 %	0.996
12	5.9 %	0.7 %	5.9 %	0.993
16	6.9 %	0.5 %	6.9 %	0.986

Under that noise estimate, the early "error" is mostly teacher noise, while the late slice error is real and grows with depth; the d3 level shrinks by 1.4 % at L16. Of the slice error, 91–96 % sits in the **all-distinct** post-ReLU entries $\kappa_3(h_i, h_j, h_k)$ (App. §4.5).

Localisation. Injecting the teacher's κ_3 only in a window of layers (MLP #0, base 2.2732e-8):

Window	Slice + d3	Slice only	d3 only
L1–4	×1.004	—	—
L5–8	×0.942	—	—
L9–12	×0.857	—	—
L13–16	×0.806	—	—
L1–8	—	×0.937	×1.012
L9–16	—	×0.834	×0.853

The gain lives in **L9–16**, where the slice and d3 matter about equally. Injecting the teacher's κ_3 into the feedback legs alone is worth ×1.001.

4.4 κ_4: the pattern, not the level

Injecting the teacher's κ_4 diagonal into the mean closure takes the raw MSE from 2.290e-8 to 1.609e-8. How that gain splits:

- **The teacher's pattern at the chain's own level** gives 1.551e-8, which is **108 %** of the gain.
- **The chain's pattern at the teacher's level** is *worse than no oracle* (+21.5 %).
- **In our oracles, κ_4 helps only through the mean closure's diagonal input.** True κ_4 everywhere except the mean closure gives ≈ ×1; the off-diagonal slices injected alone give ×1.03–1.22.
- **The chain's pattern is weak where it matters.** Over all neurons the chain's κ_4 diagonal correlates 0.78–0.83 with the truth; on the decisive neurons only 0.26–0.34. (*Pattern correlation* is the across-neuron correlation of predicted and teacher κ_4/v^2 on *decisive* neurons, $|\mu/\sigma| < 1.5$, which drive the mean closure.)

A ceiling ladder for κ_4 carriers. We measured the ceiling of whole *families* of birth laws by restarting a Gaussian h layers back and expanding each ReLU to Hermite order K (App. §5.4, $N = 4 \cdot 10^6$):

- **Quadratic births are capped in this ceiling model:** births that stop at second order reach pattern correlation 0.45–0.55. Every second-order carrier we and outside models proposed stalled in, near or below this band (−0.35 to 0.56, §7), consistent with the cap being the cause, though not a proof of it.
- **Cubic births suffice:** cubic births over six layers reach 0.80–0.85, and K = 4 adds ≤ 0.02.
- **Ages are complementary:** ages 1–6 are needed jointly. Ages 1–4 give 0.68–0.74, and ages 4–6 alone give only 0.37–0.41.

This ladder is why V56 uses a *cubic* feature.

4.5 The κ_3-dressed C·T channel (measured at oracle and component level; not graded)

The V56 star (§5.2) is the *Gaussian* cubic birth. A κ_3-dressed C·T readout, $ct_i = 24\, \Sigma_a\, g_a\, \bar{Y}_ai\, X_ai\, Q_ai$, reads the chain's κ_3 through the quadratic and cubic Hermite vertices. Here a runs over the neurons ("hubs") of a vertex layer, g_a is the ReLU's quadratic Hermite coefficient there, $\bar{Y}$ transports the hub to the target neuron i, X = C·(transport) is the covariance leg, and Q contracts the chain's κ_3 with two transports (App. §5.8; design from a 9 Oct Fable 5 consultation; every number below is ours):

- **Ceiling.** Sample-exact star + C·T reaches 0.82–0.85, matching the cubic-birth ceiling of §4.4. C·T alone sits at 0.42–0.44, where an earlier hub-pair carrier was killed; it is complementary to the star.
- **The chain's own legs reproduce it** at fidelity 0.993–0.996 on decisive neurons (component test).
- **End to end under an oracle map** (fitted to the teacher on the evaluation network, FLOPs not metered): ×0.773 with the sample-exact feature, ×0.819 with the chain's exact C·T on the available source subset.
- **Cost.** An exact age-collapse identity, Γ(a) = G(a) + Ystep(a) Γ(a+1), plus grouping by dense anchor, reduces it to two dense products per source and target (derived; matches the per-age route to 2.6–3.2e-7). At r = 64 a sketch keeps ×0.865 at about 40 u (estimate), roughly score-neutral. A rank-32 sketch keeps ×0.874 at an estimated 18–20 u. It is untested on V56 and is our main open lead.

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5. The κ_4-pattern programme (graded)

5.1 V55: a fitted, zero-FLOP κ_4-pattern map (fitted component)

Given §4.4, V55 corrects the pattern, not the level. At every layer ≥ 6 (≥ 4 from #334762 on), a **fitted linear map** turns seven per-neuron quantities the chain already holds into a predicted pattern of κ_4/v^2: s3sq = $(d3/v\text{^}(3/2))^2$; the row and column energies of the standardised D21; the correlation-row energy of C_pre; $|\mu/\sigma|$; the chain's own κ_4/v^2; and a constant. The pattern is rescaled to the chain's layer-mean level and fed **only** to the mean closure. The map is fitted offline, on decisive neurons, against our own teacher for MLP #0; the synthetic net is held out. It costs 0 u and +310 operations.

- **Pattern correlation:** 0.41–0.55 on MLP #0 against 0.24–0.41 for the chain, and 0.41–0.52 held out on the synthetic net.
- **100-MLP harness:** ×0.9455 [90 % CI 0.9410, 0.9498], with 97/100 MLPs improved.
- **Graded:** −5.4 % in both operation classes (A and B, §6.3). The harness projected 3.8764e-9 / 3.8562e-9; the grades were 3.8762e-9 (#334588) and 3.8582e-9 (#334591).

V55 is a regression on chain features, not a mechanism; its features saturate near pattern correlation 0.5 (App. §5.3).

5.2 V56: the Gaussian cubic-star feature

The cubic Hermite coefficient of the ReLU, c = $-\alpha\varphi(\alpha)/6$ in standardised units, together with the Wick identity $\kappa(u_a^3, u_b, u_c, u_d) = 6\rho_{ab}\,\rho_{ac}\,\rho_{ad}$, injects a **star**-shaped fourth cumulant downstream of every hub (derived; verified numerically to ≤ 8.1e-16, App. §2.5). V56 reads this star from the chain's **own** young dense source legs (ages ≤ 6), with the per-source coefficient $c_a = -\alpha_a\varphi(\alpha_a)/(6\sigma_a^2)$ recorded at birth (App. §5.8 notation):

$star_i = 24\,\Sigma_a\, c_a\, \bar{Y}_{ai}\, X_{ai}^3$.

The star enters as an eighth feature of the V55 map, at zero added FLOPs: util 0.1857, against 0.1855 for the V55 reference. Alone its ceiling is 0.47–0.52, but it is complementary to the V55 features: with teacher μ and C, leave-one-layer-out fits rise from 0.48–0.56 to 0.65–0.74. The star idea came from an outside-AI proposal, the "Hermite Tree Quartic Bank" (Round 18; unsigned), built on our own Hermite-family ceilings (App. §5.4–§5.5).

Graded. −4.4 % (#334805, B) and 3.6597e-9 (#334811, A). Refitting the map on the synthetic net gave a further −0.29 % / −0.21 %, reaching **3.6723e-9 (#334869) and 3.6521e-9 (#334870)**.

The κ_4 programme as a whole took the score from 4.0784e-9 to 3.6521e-9 (−10.5 %) at unchanged cost. Its coarse tuning is closed: scaling the star by 0.75 or 1.5 scored ×1.011, and limiting it to ages ≤ 4 scored ×1.0045; a 16 Oct sweep found only small further gains (K4C blend 1.1, ×0.995, graded #336382; App. §3.8).

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6. Engineering components: cost under the meter and the grader's limits

These sections are performance engineering. We report them because they carry most of the cost reduction and because the grader's hidden limits shaped every later design, not as contributions to understanding.

6.1 Graded trajectory

Step	Change (origin)	Score (id)	Raw MSE	Util
V29 verbatim (reference)	—	5.418e-9 (#331681)	2.133e-8	0.254
Strassen recursion floor 32 → 8; full depth on first call	Two more Strassen levels on the basis families	5.220e-9 (#331686); 5.195e-9 (#331732)	2.132e-8	0.2436
+ V31 spherical-input identity	§3.1, 0 FLOPs	5.137e-9 (#331787)	2.108e-8	0.2437
+ FB 1.25, β 1.2	§3.3 (tuned), 0 FLOPs	5.1284e-9 (#332110)	2.1046e-8	—
Transport join	Shared basis rebuilt at a join by transport, with no range finder	4.9519e-9 (#332257)	2.1788e-8	0.2273
+ AGE_OLD 5, R_FB 4	One more dense young age; feedback rank 16 → 4	4.8066e-9 (#332324)	2.1371e-8	0.2249
v48–v51 compaction	Dead neurons (ReLU slope Φ < 1e-2) leave the next transport; basis-side Strassen, aliased Grams; dead D21 columns and rows skipped (#332391, #332401, #332404, #332418; §6.2)	4.0902e-9 (#332420, A)	2.1636e-8	0.189
v52 operation budget; R_OLD 352 and exact trims	§6.3; old rank, QR-derived transport, feedback rank 2	B 4.0995e-9 (#332626); A 4.0784e-9 (#332621)	2.197e-8 (B)	—
V55 κ_4-pattern map; s3sq ×1.25, B / A	§5.1 (fitted; coefficient tuned)	3.8539e-9 (#334762) / 3.8331e-9 (#334763)	—	—
V56 cubic star, B / A	§5.2	3.6829e-9 (#334805) / 3.6597e-9 (#334811)	1.9706e-8 / 1.9539e-8	—
+ map fitted on the synthetic net, B / A		3.6723e-9 (#334869) / **3.6521e-9 (#334870)**	1.950e-8 (A)	0.187
E282–E299 (mdhamidhosen): 24-group calibration, rider, corrector refits, exact cost rewrites, thresholds	§6.4 (fitted; engineering)	3.5890e-9 (#335603) → 3.5085e-9 (#335729) → 3.3946e-9 (#336377)	1.8824e-8	0.180
+ e30_a's exact cost savings (LordProcciuto); kai_leal_gonzalez's (3,1) closure at mix 0.25	§6.5	3.3325e-9 (#336757)	1.8742e-8	0.178
+ e30_a's fixed old basis and C_pre compaction (LordProcciuto)	§6.5	3.2529e-9 (#337049)	1.9107e-8	0.170
Our corrector on LordProcciuto's e30_a (scale 0.4); kai_leal_gonzalez's (3,1) closure	§2.3, §6.5	3.0302e-9 (#337175)	1.9953e-8	0.152
+ D21 ×0.98, closure mix 0.4, corrector star/local split	§6.5 (tuned)	2.9956e-9 (#337258)	1.9723e-8	0.152
+ e30_a's 87-feature table refitted on our trajectory, + star modulators (N2)	§6.5 (fitted)	2.8475e-9 (#337301)	1.8747e-8	0.152
+ e30_a's near-dead column cut, τ_B set to 2.4	§6.5	2.8256e-9 (#337336)	1.8869e-8	0.150
+ table features: star^2, d3·star, corrector-local (N3, 122)	§6.5 (fitted)	2.7975e-9 (#337368)	1.8681e-8	0.150
+ D21/C row energies, carried delta and star (N4a, 164)	§6.5 (fitted)	**2.7858e-9 (#337457)**	**1.8592e-8**	**0.150**

Util is the grader's figure or adjusted ÷ raw. "—" means the value is not listed here. The B class keeps ≤ ~18k warm-call operations for a safety margin under the residual-time cap; the A class runs at about 20k.

6.2 Cost levers

The levers of the table: the transport join (−3.4 %); AGE_OLD 5 with feedback rank 4 (−2.9 %; on V56 the five nearby age/rank settings score ×1.026–1.104, App. §6.6 (c)); input-side dead-neuron compaction with skipped D21 columns and rows (−3.7 %, −3.7 %, −3.4 %; compacting the output side failed in every form, ×1.6 to ×10); Strassen floor 32 → 8 (−3.7 %); and basis-side Strassen with joiner Grams written as YY^T, billed at half (−4.6 %). Strassen level 6 exceeded 8 GB. Over the cost programme the trims removed 22 % of the metered FLOPs (util 0.2437 → 0.189) for +2.8 % raw.

6.3 The residual-time cap and the operation budget

Two variants that ran cleanly on our machines were graded catastrophically (#332402 at 0.0434, #332425 at 0.0204: about 12 % and 6 % of MLPs capped). The residual Python time tracks the number of metered operations (0.053–0.059 ms per op on Kaggle for every variant); the failing variants issued 21.7–22.3k operations. Our fix is an **operation budget**: Strassen only where it saves at least about 2 u per 1,000 operations. Class **B** (wider safety margin) keeps ≤ ∼18k operations, class **A** about 20k; neither has hit the residual-time cap since (#332566, #332567). A second hidden limit (an address-space cap) failed four 3-D forms of the V56 star (#334786–#334799) that passed in-process; `whest validate` catches neither limit, so our pre-submission check is an operation count plus a subprocess run.

6.4 Final calibration and cost layer (E282–E299, mdhamidhosen; fitted and engineering)

From 12 Oct, mdhamidhosen built on #334870 (numbers from his ledger, re-checked against the grader; App. §3.7):

- **Calibration (fitted).** A Gauss-Newton fit of 24 per-layer calibration groups (scales of the closure inputs and λ) on public mini MLPs 0–47 gave −1.66 % ± 0.20 % on unseen MLPs 48–99 (E282); a cubic final-layer ridge rider (900 full-split MLPs) and four end-to-end refits of the V56 κ_4 corrector followed (−0.77 / −0.48 / −0.33 / −0.37 % unseen; E291). Further steps gave ≤ 0.08 %.
- **Exact cost rewrites (engineering).** Grouped PK2 einsums (graded util −0.77 %), Strassen floor 4 and Strassen on the nested projections (−0.3 %, −0.48 % FLOPs in his ledger), and the Strassen-Winograd kernel (−0.77 % FLOPs) leave predictions unchanged up to rounding. With the D21 and dead-neuron thresholds raised (approximations; −0.73 % graded, E296 → E298) they give **3.3946e-9 (#336377, E299)**.

6.5 Final days: our corrector on a public chain, and a refitted table (17–21 Oct; measured and graded)

After the 17 Oct extension (grader on flopscope 0.12.3, which bills our chains identically on 100 Kaggle MLPs), we reviewed the new public write-ups and code and measured the candidates on the 100-MLP harness before grading; two (e299safe and V31 on e30_a) were checked on MLP #0 only (App. §3.9).

- **Exact cost savings from e30_a in our chain:** Gram-core skipping, R-factor rotation, folded constants: −1.4 % FLOPs, predictions unchanged up to rounding (#336711, #336719). kai_leal_gonzalez's two-term (3,1) closure (MIT, forum write-up) hurt at full strength (#336697, +0.7 %) but helped at mix 0.25 (#336757, −0.4 %). e30_a's fixed old basis and C_pre compaction gave our own-lineage best (#337049, 3.2529e-9).
- **Our corrector on LordProcciuto's public e30_a chain** (MIT; 158.8 u against 184.1 u for our e299xrk in the warm call on MLP #0, call-site ledger): at full strength it hurt (×1.085, MLP #0); at scale 0.3–0.4 it gave ×0.975–0.977 on 100 MLPs (94/100 at 0.3), graded ×0.979 against e30_a's published 3.1163e-9 (#336774). The gain comes from the V56 star (at scale 0.3: ×0.9768 with it, ×1.0024 without it); the local features duplicate e30_a's 87-feature mean-correction table.
- **Refitting e30_a's table on our trajectory (fitted).** The table was fitted on e30_a's own uncorrected trajectory. Refitted by ridge regression (grouped 5-fold CV) on teacher-forced dumps of public mini nets 0–69, with star modulators and the corrector's own κ_4 change added (N2, 101 features), it gave ×0.930 on held-out nets 70–99 (30/30) and graded −4.9 % (#337301). The original table transfers poorly to our trajectory, so the refit must follow larger changes of the chain. New non-local features paid further (N3: ×0.990 held-out, graded −1.0 %, #337368; N4a: ×0.9966 held-out, graded −0.42 %, #337457); fitting on 341 instead of 70 nets, an on-policy refit and the (3,1) orientation fix of forum topic 18246 did not (×0.9997, ×0.997–1.005, ×1.0005).
- **Measured and rejected:** V31 on e30_a (×0.985 on MLP #0, graded +3.9 %, #337221); e30_a's row compaction in our chain (×1.063, MLP #0); refitting the table after cheaper settings (rank 352: ×1.04 even with refit).
- **Operations:** the corrector adds about 0.8k warm operations to e30_a's 19.8k (20.6k on MLP #0, measured before the table extensions; A class); every e30_a-based submission graded with no capped MLP (a single cap would add about 3.5e-3 to the score).

- **Rules note:** the organisers allow building on public MIT code with attribution if the team adds substantial work of its own (forum topic 18240). In #337457 our own contribution is the corrector (V55/V56), the closure and corrector settings and the refitted, extended table; the base chain is LordProcciuto's e30_a and the (3,1) closure is kai_leal_gonzalez's (§2.3).

7. Negative-result atlas

"Gate" is the pre-registered pass criterion. Kill types: C, a ceiling or oracle (an upper bound); P, a teacher-forced component test; E, an end-to-end prototype; K, a knob on the 100-MLP harness. Ratios are final-MSE or score ratios against the chain. Details are in App. §6.

Idea	Gate	Measured	Kill
Compensating closure with no old tier (8 models proposed variants)	≤ chain MSE	7.43e-7, 32× the chain; every knob 6.8–9.9e-7	C
Mean-field / even-statistic closures of old κ_3	—	Zero by the parity theorem: old κ_3 is odd in later weights (§3.2)	derived, verified
Low-rank Hadamard products of source legs	low rank	A∘P needs 681–778 of 1024 directions for 99 % energy	C
Query surrogates: trace, diagonal Hermite, TensorSketch	D21 ≤ 2 %	98.2 %, 95.4 %, 790 % at L14	C/P
Source-free dressed hypertree (DHMP)	next D21 ≤ 2.5 %	36–40 %	P
Hermite-tree κ_3 replacing the chain	closure ≤ chain	10–20× worse	P
Every second-order κ_4 carrier (8 designs)	pattern corr ≥ 0.70	−0.35 to 0.56	C
Learned residual correctors (ridge, GBM; 102,400 neurons)	—	0.3–0.7 % of the residual recovered	E
Monte-Carlo control variates	—	Tangent-surrogate corr 0.15–0.29 at L16; variance reduction 1.0–2.2×	P
C·T channel via age-collapsed sketches (ACDS), r = 64	score gain at ≤ ∼30 u	Oracle E2E ×0.865 at ≈ 40 u (estimate): score about neutral. r = 32 (×0.874, est. 18–20 u) is open	E (parked)
Missing cubic-response C×κ_3 term (RLL)	ρ ≥ 0.1 with the residual at L ≥ 9	ρ ≈ 0; adding half the term worsens the slice residual by 15–42 %	P
γ-patch: FB 2.0 with the quadratic cross term held	ratio < 1	×1.0243, the same as unpatched FB 2.0 (×1.0244)	K
$\kappa_4 \to \kappa_3$ feed from the exact κ_4 ($N = 10^6$)	exact ≤ 0.80 pass, > 0.92 kill	×1.115	C/E
κ_3 level rescaling, d3 or D21 ×0.98–1.02	< 1 on both nets	×0.984 to ×1.074; the one gain does not transfer	K
Single shared pool for the whole κ_3	D21 ≤ 2 % at some r ≤ 512	3.4–6.3 % at r = 512; 0.2–1 % only at r = 768	C
Deferred exact join (youngest source exact)	D21 ≤ 2 %	2.5–8 % at r = 352–512	C
Age/rank re-split of V56 (5 settings)	score ratio < 1	×1.026 to ×1.104	K
Checkpointed birth archive (κ_3 skipped in a window, sources kept)	≈ ×1	×14.8 (L1–4) to ×125.6 (L1–12); no recovery after the window	E
Trace projection of old κ_3 (Wu et al.)	D21 ≤ 2 %	26–42 %	C
Cost-first κ_3 representation at 60 % of the cost	2 % channel accuracy	None found; slice-only states are not exact to first order (derived)	derived
Network-conditioned routing over 86 archived variants	≥ 3 % gain	Oracle ceiling 0.80 % before overhead	C
Learned pair-memory student PM364 (source-free, 364 parameters)	MSE to V56 ≤ 1e-9; D21, d3 ≤ 2 %	MSE −0.9 % after training; D21 58–70 %, d3 25–39 %	E
Exact old tier (no confinement)	score gain	−3.9 % MSE for +19 % cost	E

Idea	Gate	Measured	Kill
Strassen level 6	≤ 8 GB	8.13 GB	cap

7.1 Working conclusions (empirical; not proven)

1. Within the sum-of-sources family, cost and accuracy traded roughly one for one in every knob we swept. This is an empirical regularity over our sweeps, not a theorem. On the 100-MLP harness the shipped constants sit at or within about 0.5 % of the score optimum of the settings tested (App. §3.8): β 1.2 (×0.9993), λ ×0.9 (×1.0026), the FB × β grid (within +0.12 %), R_OLD 448 (×1.0113) and AGE_OLD 6 on V56 (×1.014). On the five-point age/rank grid on V56, the best point gains 0.5 % raw for 3.2 % more util, and saving 4.3 % util costs 10.3 % raw. The exceptions in our records are levers that are **exact or nearly exact** (transport join, compaction, Strassen placement, aliased Grams); these are where our cost gains came from. A 27 Sep audit of 74 earlier kills revived 8, all cost or knob levers killed on the wrong criterion; no architecture kill was revived.

2. Local corrections of one channel made the final mean worse in every case we tested: full FB restoration (+3.4 % / +3.1 %); the γ-patch (×1.0243); the RLL term (slice residual 15–42 % worse); scaling the chain's $\kappa_4 \to \kappa_3$ feed in either direction (−5 % costs 0.45 %; +10 %, toward the larger exact feed, costs 9.4 %); injecting the sampled exact feed (×1.115); raising d3 toward the teacher's level (×1.032 / ×1.015); single κ_4 slices (×1.03–1.22). Only **joint** input oracles give large gains (×0.272 against ×0.607 and ×0.686 alone). *Measured at one step (App. §7.5):* the chain's fresh closure-input errors cancel. The sum of their norms is 1.9–2.6× the norm of their sum, chiefly var against d3 (cosine down to −0.83). Correcting var alone (L2–L9) or d3 alone (L4–L10) makes the step worse by up to ×1.31 and ×1.13, while var and d3 together always help (×0.56–0.79). How this carries through depth to the final mean remains a conjecture, and for the feed, Monte-Carlo noise is a competing explanation. The practical rule we adopted: local tests are necessary gates, not sufficient ones; candidates are promoted only on the end-to-end mean.

3. Every source-free or pooled κ_3 state we built failed its accuracy gate. *Conjecture:* the common cause is that the all-distinct κ_3 entries, where 91–96 % of the slice error sits (§4.3), are information a cheap state does not hold. Evidence:

- **The slice-state argument (derived in an outside consultation; App. §6.6 (g)).** A perturbation H = ε Sym(e_a, e_b, e_c) with distinct a, b, c is invisible to the mean, the covariance and the slice, yet after one transport it produces a slice and a diagonal that are first order in ε. A recursion on the slice alone therefore cannot be exact to first order, and a linear state closed under every transport that returns every diagonal needs $C(n+2, 3) \approx 1.8 \cdot 10^8$ numbers. The argument does **not** exclude nonlinear, restricted-family or approximate encodings (the source chain itself is one).
- **Age memory (measured).** Keeping only sources of age ≤ h leaves D21 errors of 84 / 65 / 46 / 31 / 16 % at h = 2 / 4 / 6 / 8 / 10; six age modes still leave 8.5 %.
- **Pooling (measured).** Pooling only the old sources costs 1.2–1.3 % at r = 384, but pooling everything needs r = 768.
- **Skipping and learning (measured).** κ_3 could not be deferred to a checkpoint (×14.8 to ×125.6), and our per-neuron-feature student PM364 did not learn it: its D21 error at L2, the first transport, is 57.6 % and does not move under training (two seeds). This rules out that student design, not learned states in general.
- **Averaging (derived).** By the parity theorem (§3.2), closures built on even statistics lose old κ_3 entirely.

Where our cost goes. A V56 call-site ledger puts 193.8 u in one warm call: about 70 u in the D21 hub readout, 63 u in young-leg transport, 20 u in the old basis, and about 40 u elsewhere, which issues 11.3k of the 20.6k operations. The 133 u in the two per-source families grows with depth, and nothing we tested removes it without at least a comparable accuracy loss. For comparison, the reference algorithm of Wu et al. costs about 4,150 u at our size (derived from their cost table), about 21× V56.

Relation to the leaderboard. On the 10 Oct board the six highest entries have raw MSE 1.14–1.65e-8 at util 0.106–0.144. On 20–21 Oct our row shows 2.8e-9 at rank 34 (tied; the board rounds to two significant figures); the prize ranking comes from the organisers' private re-evaluation. One entry at rank 11 has V29-level raw MSE (2.10e-8) at util 0.102. At #337457's raw MSE (1.86e-8), util 0.10 would score about 1.9e-9 (derived). We do not know the leaders' mechanisms. Our measurements exclude only the specific κ_3 representations in the table above as cheaper drop-in replacements inside our chain.

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8. Reproducibility

Versions. whestbench 0.16.1, flopscope 0.12.1 (0.12.3 for the 17–21 Oct submissions, including #337457; identical billing on 100 MLPs, App. §3.9), dataset revision v2-phase2. Every graded number carries its AIcrowd submission id.

Path	Contents
`submissions/<variant>/estimator.py`	Every shipped estimator, with a lineage header naming each change: #337457 is `e30kc1n4/`, #334870 `v56As_fs/`, #334869 `v56s_fs/`, the reference `v29/`
`experiments/variants/e30k_x8.py` – `e30k_x14.py`, `kaggle/x8*`, `kaggle/x10` – `x14`, `external/`	The final-days line (§6.5): harness rounds, the self-contained Kaggle pipelines that dump, refit and evaluate the table, and copies of e30_a and the (3,1) closure (§2.3)
github.com/hamidhosen42/ARC-White-Box (private until release; available to the organisers on request)	mdhamidhosen's E-series (E299 = #336377), its ledgers and calibration tools
`experiments/variants/`, `experiments/probes/`, `experiments/student/`	Oracles (`r20_joint_oracle.py`, `r20_feed_oracle.py`, `v55_k4c.py`, `v56_star.py`), probes (`site_ledger.py`, `k3_error_profile.py`, `r17_hermite_family_ceiling.py`, `kaggle_pair_compare.py`), the PM364 student
`kaggle/`, `results/summaries/`, `consultations/`, `writeup/`	Kernel wrappers and teachers; result files; consultation notes and verdicts; this document, the Appendix and the PDF builder

App. §8.3 maps every Appendix section to its script and result file; known gaps in our records are in App. §8.4.

The 100-MLP harness. It runs the public mini split with its 10^9-sample truth on five Kaggle CPU runners. `kaggle_pair_compare.py` reports the paired geometric-mean ratio, wins, a bootstrap 90 % interval, the score and the residual-time profile. The base is bit-identical across runners, and paired standard errors are 0.01–0.15 %. Up to 16 Oct every predicted sign matched the grade; for single changes worth more than about 1 % the predicted size was within 0.05–0.7 percentage points; for gains under 0.5 % the size was unreliable. In the final days one non-significant joint prediction had the wrong sign (c1: ×1.0035 n.s., graded −1.1 %), and the refitted table N2 was over-predicted (×0.935, held-out ×0.930; graded ×0.951) (App. §3.9). The harness measures residual time but does not enforce the cap.

Building the PDFs. `python writeup/build_pdf.py` builds the Appendix; `python writeup/build_pdf.py writeup/WRITEUP_MAIN.md --out WRITEUP_MAIN --compact` builds this document. All scripts will be released under MIT after the competition closes.

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9. Disclosure of AI use

Public code we build on. #337457 runs on LordProcciuto's e30_a (MIT; its write-up discloses Claude use) and uses kai_leal_gonzalez's two-term (3,1) closure (MIT; its repository also discloses Claude use). e30_a's MIT notice (504aldo, LordProcciuto) is kept in the shipped files; kai_leal_gonzalez's closure is credited in the lineage header, and the released code adds his MIT notice. The corrector, the settings and the refitted, extended table are ours (§2.3, §6.5).

Code. Apart from the public code we build on (504aldo, LordProcciuto, kai_leal_gonzalez), essentially all code in our repository was written by AI coding assistants under the team's direction: every estimator variant, oracle, probe, harness runner and Kaggle pipeline (including the 17–21 Oct table dump-and-refit kernels) and the PM364 student. The assistants were Anthropic Claude models through Claude Code: mainly Claude Opus 5 and 5.5, with Fable 5, Opus and Sonnet subagents for specific derivations and measurements. The team chose what to build, ordered every run and reviewed the results. The new table features of §6.5 were proposed by Claude Code and selected by held-out measurement. The E-series layer of §6.4 (E282–E299) and its calibration fits were developed by mdhamidhosen in his own repository (§8); this statement covers our repository, and the E-series is covered by its author's own statement. Its graded numbers were re-checked against the grader.

Ideas. Many architecture proposals came from consultations with outside AI systems (GPT-5.6 (Sol, Terra), GPT Astra 6, Gemini 3.1 Pro, Grok, DeepSeek, Copilot, SciMaster/Bohrium, and several unsigned answers) over 20 numbered rounds (App. §6.4, §9). Of the shipped ideas:

- the spherical-input reparameterisation was pointed out in an outside consultation, which referred to a public repository;
- the cubic-star feature comes from an outside proposal built on our own ceilings;
- the parity argument and the FB-scale derivation came from Fable 5 subagent work with us, as did the κ_3-dressed C·T readout (§4.5) and the $\kappa_4 \to \kappa_3$ feed design (§7), 9–10 Oct;
- the slice-state argument (§7.1) came from an outside consultation.

Most were killed (§7); several outside documents reported measurements that no run supported (App. §6.4).

Write-up. Almost all of the text of this write-up and of the Appendix was drafted by Claude (Claude Code) from our result files and notes, and revised by the team. AI agents also reviewed the documents: a fact-check of every number against the result files (11 Oct) and a five-lane review of numbers, consistency, rules and attribution (21 Oct). The team checked each finding before accepting the edit and takes responsibility for every claim. (App. §9 lists the 19 Oct agent reviews.)

Validation. Apart from the cited figures of 504aldo, Wu et al. and the public leaderboard, and the numbers marked as estimates, every number here was produced by our own scripts (or derived from them) or by the official grader. Every outside claim was re-derived or re-measured before we used it. On 11 Oct 2026 an automated, AI-run fact-check compared every number then in this document with the result files; on 21 Oct a second review re-checked all 63 submission ids against the grader and about 165 measured numbers against their files. The errors found were corrected. No leaderboard feedback entered any fitted coefficient: the V55 map was fitted against our own teacher for MLP #0, the shipped V56 maps against our own teacher for the synthetic network, and the E-series calibration on public dataset MLPs with their published truth (mini 0–47, rider on full-split 100–999), validated on unseen mini MLPs 48–99; the refitted e30_a table (§6.5) on public mini nets 0–69, validated on unseen nets 70–99. Discrete choices (corrector scale, A or B class, which variant to keep) were made on the harness and, in a few cases, between graded submissions.

Understanding: what was verified numerically and what is only derived (several derivations were produced with AI assistance).

- *Verified numerically:* the cubic-star and Wick identities (to ≤ 8.1e-16) and their readout factors (to ≤ 9.0e-15); the spherical identity's scale shift (to 0.3 %); the age-collapse and anchor-grouping identities (to 2.6–3.2e-7); the closure accuracy given true inputs (§4.1); the implementation parity checks (PM364's flopscope and PyTorch versions agree to 2.8e-7); the parity theorem's sign flip on the real chain (exact) and the $\sqrt{(6/(r+8))}$ floor (Monte Carlo plus exact operator norm, with a written proof).
- *Derived, consistent with measurements but not exactly confirmed:* the half-carried feedback term (regression −0.55 to −0.9 against −1/2).
- *Derived only:* the slice-state argument and the C(n+2, 3) count, and the Wu et al. cost comparison.
- *Conjectures:* the depth propagation of the (one-step measured) error cancellation of §7.1 (2), the common cause in §7.1 (3), and the link from parity to the G_off = λ_l C_off law (§3.2).

References

1. Wilson Wu, Victor Lecomte, Michael Winer, George Robinson, Jacob Hilton, Paul Christiano. *Estimating the expected output of wide random MLPs more efficiently than sampling.* arXiv:2605.05179 (v2, 14 May 2026). The factorised K = 3 "simple" cumulant chain (§4.3), its cost table, and the reference implementations.
2. 504aldo, *whest-p2-cumulant-k3* (GitHub, MIT licence), and WhestBench Phase 2 forum post #18218. The V29 factorised third-cumulant chain used as our reference method.
3. WhestBench (whestbench 0.16.1) and flopscope (0.12.1), the ARC White-Box Estimation Challenge 2026 benchmark and FLOP meter (Alignment Research Center / AIcrowd). The organiser ruling of 11 Sep 2026 states that Strassen–Winograd is legal at any recursion depth.
4. Appendix (full technical report), Hack2Publish, 21 Oct 2026 (`writeup/ALGORITHM_TRACK_PAPER.md`).
5. LordProcciuto, *e30_a*, github.com/siming767-creator/whestbench-lordprocciuto (MIT), and Phase-2 write-up (graded #335981). The base chain of #337457.
6. kai_leal_gonzalez, two-term closure of the (3,1) slice of κ_4, github.com/al07-alt/whest-p2-k31-two-term (MIT), forum topic 18246.
7. barnobarno666, public Phase-2 repository (spherical-input reparameterisation).
8. Organiser ruling on building on other participants' public MIT code, forum topic 18240 (5 Oct 2026).

Where the Error of the Factorised Cumulant Chain Lives: Exact Identities, Cost Engineering, and a Cubic-Birth κ_4-Pattern Programme for Deep ReLU MLPs

WhestBench Phase 2, algorithmic-contribution write-up. Team Hack2Publish (mdhamidhosen, esfer_sami, foyshal_sk). **This write-up refers to submission #337457** (graded in Phase 2): adjusted score 2.7858e-9, raw final-layer MSE 1.8592e-8. It is LordProcciuto's public e30_a chain (MIT) with kai_leal_gonzalez's (3,1) closure (MIT), plus our κ_4-pattern corrector, our settings and our refit and extension of e30_a's mean-correction table; §1.4 separates their parts from ours. *For reference only:* our best result without e30_a's base chain is #337049 (3.2529e-9), built on #334870 (v56As_fs, 3.6521e-9), the chain this paper analyses. *Version of 21 October 2026. Every number is either a graded leaderboard result (submission id given) or a measurement whose result file is named in §8. Items still open are marked [OPEN]. Unmeasured quantities are marked (estimate) or (untested). Claims are labelled derived (algebra), measured (run by us) or graded (official grader).*

Abstract

We start from the public, MIT-licensed factorised third-cumulant chain of team 504aldo ("V29", raw 2.13e-8 at 0.254 of the budget) and pursue two questions: where its error is made, and what an estimator must contain to remove that error at an affordable cost.

Contributions.

1. **An exact identity, shipped in our own chain (not in #337457).** A spherical-input identity turns the Gaussian input into a uniform-sphere input plus one exact scalar, $a_1 = 0.99975589$. It is free and graded −1.1 % raw (#331787).
2. **A parity theorem** (derived, with a written proof of its floor, and verified numerically on the real chain). It shows that closures built only on even statistics (mean-field closures, scale mixtures, polynomials in C) carry none of the old κ_3; closures that condition on the mean μ are not covered.
3. **Cubic curvature-birth identities.** These come with readout factors, verified to ≤ 1e-14.
4. **Cost engineering under a metered budget.** Our tools are dead-neuron compaction, basis-side Strassen, aliased Grams, and an operation budget derived from a residual-time grader failure. Together they remove 22 % of the metered FLOPs for +2.8 % raw (util 0.2437 → 0.189, #331787 → #332420). With the zero-cost accuracy fixes they take the score from 5.418e-9 to 4.078e-9.
5. **An error map built from $6{\cdot}10^7$-sample Monte-Carlo teachers.** The closures are accurate given true inputs (mean closure to 5e-11–1.5e-10; covariance closure down to the teacher's noise). The error is fresh input error that accumulates through depth. The single most valuable missing input is the **per-neuron pattern** of the κ_4 diagonal: the teacher's pattern at the chain's own level would take the raw MSE from 2.290e-8 to 1.551e-8, which is the leaders' raw range.
6. **A κ_4-pattern programme with graded gains.**
   - A zero-FLOP linear corrector (V55) graded −5.4 % (#334591).
   - A cubic Hermite-star feature built from the chain's own young legs (V56) graded a further −4.4 % (#334805), reaching 3.6521e-9 (#334870).

7. **A ceiling ladder consistent with every earlier κ_4 carrier stalling near correlation 0.45** (consistent with, not a proof of, the cause).
    - Carriers whose births stop at second order in the Hermite expansion of the ReLU are capped at 0.45–0.55.
    - Cubic births over a 6-layer horizon reach 0.80–0.85.
    - An age-resolved ceiling (9 Oct) shows that ages 1–6 are jointly needed. Ages 1–4 give 0.68–0.74, ages 2–6 give 0.69–0.71, and ages 4–6 give only 0.37–0.41.
8. **A table of the more than 60 architecture and representation ideas we killed, with their ceiling numbers.** It includes outside-AI proposals and states which of them came with no measurement.
9. **Derived constraints on any better method.** The next design target is stated quantitatively.
10. **The κ_3 input and the κ_3-dressed C·T channel (9–10 Oct, oracle and component level; not graded).**
    - A joint oracle shows that the teacher's κ_3 input is the larger single lever: ×0.607 against ×0.686 for the κ_4 diagonal on MLP #0, and ×0.716 against ×0.885 on the synthetic net. Together the two give ×0.272.
    - A C·T readout built from the chain's own κ_3 legs, $ct_i = 24\,\Sigma_a\, g_a\, \bar{Y}_{ai}\, X_{ai}\, Q_{ai}$, lifts the star's ceiling from 0.47–0.52 to 0.82–0.85. Used as a feature in a teacher-fitted (oracle) affine map, it takes the plain chain to ×0.773, which is about 75 % of the perfect-κ_4 gain.
    - An exact age-collapse identity, $\Gamma(a) = G(a) + Ystep(a)\,\Gamma(a+1)$, together with grouping into dense A/P anchors plus exact low-rank anchors, cuts the source contraction to two dense products per source and target, plus one shared backward recursion. Low-rank sketches keep only ×0.843–0.885 (round 2; ×0.874–0.913 for the round-3 variants). At an estimated 40 u this is about score-neutral, so the channel is real but not yet affordable. A rank-32 eigen-sketch (10 Oct) keeps ×0.874 at half the rank-64 thin-product count, an estimated 3–4 % score gain on the plain chain under the same oracle map (untested on V56).
11. **Where the κ_3 error is, and three mechanisms killed (10 Oct; oracle and component level; not graded).**
    - Measured against an estimate of the teacher's own sampling noise, the chain's early-layer κ_3 "error" is mostly teacher noise. The real (2,1)-slice error grows from about 3 % at L4 to about 7 % at L16, and the d3 level shrinks by 1.4 % at L16. The teacher's κ_3 helps only at L9–16 (§4.10).
    - A missing cubic-response $C\times\kappa_3$ term, a feedback-scale confound and the $\kappa_4\to\kappa_3$ feed channel were each tested and killed. The last kill holds only for the tested sample size, $N = 10^6$ (§6.5).
    - Local corrections of one channel repeatedly make the final mean worse; only joint input oracles give large gains (§7.5).
12. **Where the shipped cost goes, and why the cost-side architectures fail (10–11 Oct; measured, partly on 100 MLPs).**
    - A call-site ledger of V56 (§3.6) puts 193.8 u and 20.6k operations in one warm call. About 70 u goes to the D21 hub readout, 63 u to young-leg transport and 20 u to the old basis; the rest is about 40 u.
    - One shared basis cannot hold the whole κ_3: at rank 512 the D21 error is 3.4–6.3 % at L11–15, and only rank 768 reaches 0.2–1 % (§6.6 (a)). Keeping only the youngest source exact (a deferred exact join) leaves 2.5–8 % (§6.6 (b)).
    - On 100 MLPs every age/rank setting near V56's scores worse (×1.026 to ×1.104), so the one-for-one law of §3.5 holds on V56 (§6.6 (c)).
    - Zeroing κ_3 in the closures for a window of layers costs ×15 to ×125, and the damage does not recover after the window (§6.6 (d)). A ±2 % change in the κ_3 level is not a gain either (§6.6 (e)).
    - The factorised K = 3 algorithm of Wu et al., priced from their cost table with naive products, costs about 4,150 u at our size, about 21× V56 (derived, §3.6); the released ARC reference code measured 1523 u (§6.3). Its trace projection of old κ_3 leaves 26–42 % D21 error (§6.6 (f)).

13. **A learned pair-memory student, PM364 (11 Oct; killed).** We distilled V56 into a source-free state with two n×n pair memories, an exact port of V56's closure and 364 learned parameters. The student fits under the 0.1 floor (about 93 u, estimate). It trained on 448 V56-labelled networks, and its held-out error moved by only −0.9 %. Its D21 error stays at 58–70 %, against a 2 % gate (§6.7). Per-row heads cannot rebuild the all-distinct post-ReLU κ_3, which agrees with a derived first-order obstruction for slice-only states (§6.6 (g)).
14. **The submission this paper refers to (#337457, 2.7858e-9; §1.4, §3.9; graded).** Our V55/V56 corrector, our settings and our refit and extension of e30_a's mean-correction table, on LordProcciuto's public e30_a chain (MIT) with kai_leal_gonzalez's two-term (3,1) closure (MIT): −10.6 % on e30_a's published 3.1163e-9, of which −6.3 % comes from the fitted table. Before that, mdhamidhosen's calibration and cost layer took #334870 to 3.3946e-9 (#336377, §3.7). Our best result without e30_a's base chain is 3.2529e-9 (#337049).

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1. Setting and baseline

1.1 Task and scoring

For a bias-free ReLU MLP of width n = 1024 and depth 16, with He-initialised weights and input x ∼ N(0, I), the estimator returns E[h^(l)] for every layer. The score is

score = MSE_final × max(0.1, FLOPs / 2^41).

The final-layer MSE is measured against a 10^9-sample Monte-Carlo reference. FLOPs are metered analytically by flopscope.

We use the unit **1 u = 2^31 FLOPs**, which is about one dense 1024^3 product, so the budget is B = 1024 u. "util" denotes FLOPs/B, and the 0.1 floor equals 102.4 u.

The grader enforces three hard limits per MLP:

- 120 s wall time;
- 0.4 s of **residual** Python time outside metered operations;
- an 8 GB address space.

A capped MLP receives zero predictions with multiplier 1, which scores about mean(target^2) ≈ 0.35. A single capped MLP in a hundred therefore ruins a submission.

1.2 The public chain (V29, 504aldo, MIT)

Our estimator is a modification of 504aldo's public "V29". We use it under the MIT licence; its copyright header and LICENSE are retained in every file we ship.

V29 implements the factorised K = 3 "simple" cumulant propagation of Wu et al. (arXiv:2605.05179, §4.3). Per layer it carries:

- the mean;
- the covariance C;
- the third cumulant κ_3, as a **sum of sources**.

Each ReLU injects one source. A source is stored as:

- a propagator leg P_s (n×n, the identity at birth, updated $P \leftarrow W^T \operatorname{diag}(w_1)\, P$);
- a birth leg A_s = P_s a_b;
- thin n×16 feedback legs.

The next ReLU needs the slices D3_i = κ_iii and D21_ic = κ_iic. These are obtained per source from contractions of Hadamard products of the legs.

Further features of V29:

- **Shared basis for old sources.** Sources older than four layers are confined to a shared basis of rank 384, and the oldest to a nested rank-224 sub-basis.
- **κ_4 handling.** κ_4 is not carried. It is regenerated each layer by a memoryless law, G_off = λ_l C_off.
- **D21 feedback.** D21 enters through feedback legs.
- **Pooled Strassen–Winograd.** Products use pooled, batched Strassen–Winograd at level 5.

The author's ledger prices V29 at 260 u, split as:

Component	Cost
Young sources	116 u
Old sources	107 u
Thin and elementwise	25 u
Covariance	7 u
Closure	6 u

The author also reports a tolerance law: extra final MSE ≈ 4.2e-6·ε^2, where ε is the relative RMS error of D21. D21 must therefore be right to about 2 %.

Graded baseline. V29 verbatim graded 5.418e-9 (raw 2.133e-8, util 0.254; #331681), matching the author's own submission.

1.3 Measurement instruments

All of the following are described in §8.

- **Monte-Carlo teachers** with $6 \cdot 10^7$ samples on public mini MLP #0 (logan-fitzgerald) and on a synthetic 1024×16 He network. They record, per layer, the pre- and post-ReLU means, covariances, κ_3 slices and κ_4 diagonal and slices.
- **Oracles** that replace one internal quantity of the chain by the teacher's value and rerun the chain end to end.
- **A deterministic 100-MLP paired harness** on Kaggle, using the public mini split with its 10^9-sample truth.

The project κ_4 metric. Throughout §5, the metric is the correlation of the standardised κ_4 diagonal, κ_4/v^2, with the teacher's, computed on *decisive* neurons ($|\mu/\sigma| < 1.5$) at layers 12, 15 and 16. Raw (unstandardised) correlations inflate every candidate because they contain v^2, which the chain already gets right. One outside proposal scored 0.80–0.87 on raw correlation and 0.43 on this metric.

1.4 The base of submission #337457 (LordProcciuto's e30_a and kai_leal_gonzalez's closure; MIT; not claimed)

The submission this write-up refers to, #337457, is built on **LordProcciuto's public e30_a** (github.com/siming767-creator/whestbench-lordprocciuto, MIT, Copyright 504aldo and LordProcciuto; graded 3.1163e-9 as #335981, cited). e30_a descends from V29 (§1.2). Its author's changes are: physical removal of neurons with $\alpha < -2.85$; a fixed-basis old level (rank 416/224) used only in non-hot rows; Strassen with a Winograd leaf; exact fixed-cost savings; a λ tilt; a near-dead column cut (τ_B); and an **87-feature ridge mean-correction table** fitted on 70 public nets. #337457 also uses **kai_leal_gonzalez's two-term closure of the (3,1) slice of κ_4** (github.com/al07-alt/whest-p2-k31-two-term, MIT; forum topic 18246) with the author's per-layer coefficients.

Component of #337457	Origin
Sum-of-sources K = 3 chain, shared basis, memoryless κ_4 law, feedback legs, pooled Strassen	504aldo (V29), via e30_a
Dead-neuron removal, fixed-basis old level, hot rows, Winograd leaf, exact fixed-cost savings, λ tilt, τ_B column cut (value 2.4 chosen by us)	LordProcciuto (e30_a)
Mean-correction table: the 87 original features and the ridge-table design	LordProcciuto (e30_a)
Two-term (3,1) closure and its coefficients	kai_leal_gonzalez
κ_4-pattern corrector: V55 map and V56 cubic star (§5), coefficients refitted end to end in mdhamidhosen's E-series (§3.7)	**ours**
Corrector scale 0.4, star/local split 0.7/0.2, closure mix 0.4, D21 ×0.98 at L ≥ 9	**ours** (tuned constants)
Table refitted on our trajectory, plus 77 new features (star modulators, the corrector's κ_4 change, star^2, d3·star, corrector-local, D21/C row energies, carried delta and star; 164 in all)	**ours** (fitted)

Not in #337457: V31 (§2.1; graded worse on e30_a, #337221), FB 1.25 (§2.4; neutral on e30_a), our cost engineering (§3.1–§3.6; e30_a has its own) and the E-series calibration groups and rider (§3.7). Our changes take e30_a from 3.1163e-9 to 2.7858e-9 (−10.6 %, derived): corrector −2.1 %, closure and corrector settings −1.8 %, the fitted table refit and extension −6.3 %, and e30_a's own τ_B knob −0.8 %. Of our mechanistic contributions, the V56 star (§5.5) and the analysis that motivated it (§5.1–§5.4) act in #337457: at scale 0.3 on 100 MLPs the corrector gave ×0.9768 with the star and ×1.0024 without it. The rest of §2–§7 was measured on our own V29-based chain #334870, where the corrector was developed.

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2. Exact identities and derivations

2.1 Spherical-input identity (derived; shipped as V31)

Write $X = (R/\sqrt{n})\cdot Y$, where Y is uniform on the unit sphere scaled to radius $\sqrt{n}$ and $R \sim \chi_n$ is independent of Y.

A bias-free ReLU network h is positively homogeneous of degree 1, so

$E[h(X)] = E[R/\sqrt{n}] \cdot E[h(Y)] = a_1\, E[h(Y)]$, with $a_1 = \sqrt{(2/n)}\,\Gamma((n+1)/2) \,/\, \Gamma(n/2) = 0.99975589$ for $n = 1024$.

The sphere input has $C = I$ and $\kappa_3 = 0$. Its fourth cumulant is $\kappa_4 = c\cdot \mathrm{Sym}_3(I\otimes I)$, with $c = -2/(n+2)$. In the chain's harmonic-core representation this is a previous-layer core $g = -6/(n+2)$.

The estimator therefore does three things:

- it injects that core at layer 0;
- it runs the existing κ_4 machinery unchanged;
- it rescales the output by a_1.

Checks.

- **Measured:** the injection reproduces the identity's scale shift to 0.3 % (+2.450e-4 against +2.442e-4 predicted). The control without a_1 gives 8.02e-8.
- **Graded:** raw 2.132e-8 → 2.108e-8 (−1.1 %), score 5.137e-9 (#331787), at zero cost.

The reparameterisation idea comes from a public Phase-2 repository (barnobarno666), surfaced in an outside-AI consultation (§9). On that repository's K3 chain it was worth −7.9 %. On V29, most of that gain is already absorbed by the fitted κ_4 regeneration.

2.2 Parity theorem: no mean-field part of the deep κ_3 (derived and verified)

Take a source born at layer b. At any later layer l its legs are $P = W^{T(l)}\, D\, P'$ and $A = P\, a_b$, where P', a_b and the birth coefficients do not depend on $W^{(l)}$.

- Every quantity that is linear in the source's κ_3 (the products H, T2, T3 and T4, and the diagonal D3) is therefore an **odd** function of $W^{(l)}$.
- $C^{(l)}$, the leg Grams and the per-neuron variances are **even** functions of $W^{(l)}$.

Because the weight ensemble is sign-symmetric,

$E[\, H_s^{(l)} \mid C^{(l)}, \mathrm{Gram}(A_s, P_s), \text{all state before } l \,] = 0$, exactly.

Consequences.

1. Any closure that replaces old sources by a function of even statistics discards the entire contribution. This covers mean-field closures, scale mixtures and polynomials in C. The failures of such closures reported in the public log (9.1e-7 and 2.3e-6) are consistent with this (we did not rerun those closures).
2. Replacing a source by its conditional expectation given one fresh weight row has a relative-error floor of $\sqrt{(6/(r+8))}$ in the retained support dimension r. That is 12 % at r = 384 and 16 % at r = 224, six to eight times the 2 % tolerance.
3. κ_4 terms built from two κ_3 factors are even, and $C \otimes C$ is the only even invariant available. We conjecture that this is why the empirical law $G_{off} = \lambda_l C_{off}$ works (not proven).

Proof of the floor (written out, 11 Oct). Absorb the gates and the fresh-row covariance square root into a symmetric 3-tensor T, so that two distinct fresh weight rows become independent $g, h \sim N(0, I_r)$ and the source contributes $y = T(g, g, h)$. Let $\tau_c = \Sigma_a T_{aac}$.

1. $E[y \mid h] = \Sigma_{abc} T_{abc} E[g_a g_b] h_c = \tau \cdot h$, so $\mathrm{Var}\, E[y \mid h] = \|\tau\|^2$.
2. $y - E[y \mid h] = \Sigma_{abc} T_{abc} (g_a g_b - \delta_{ab}) h_c$. Using $E[(g_a g_b - \delta_{ab})(g_{a'} g_{b'} - \delta_{a'b'})] = \delta_{aa'} \delta_{bb'} + \delta_{ab'} \delta_{ba'}$ and the symmetry of T, the residual variance is $2\|T\|^2$.
3. The map $T \mapsto \tau$ is the adjoint of $u \mapsto \mathrm{sym}(I \otimes u)$: $\langle T, \mathrm{sym}(I \otimes u) \rangle = \tau \cdot u$. Write $T = T_0 + \mathrm{sym}(I \otimes u)$ with T_0 orthogonal to that range; then $\tau(T_0) = 0$, $\tau(\mathrm{sym}(I \otimes u)) = (r+2)u/3$ and $\|\mathrm{sym}(I \otimes u)\|^2 = (r+2)\|u\|^2/3$. Hence $\|\tau\|^2 = (r+2)/3 \cdot \|\mathrm{sym}(I \otimes u)\|^2 \le (r+2)\|T\|^2/3$, with equality at $T = \mathrm{sym}(I \otimes v)$.
4. The relative error^2 of the replacement is $2\|T\|^2 / (2\|T\|^2 + \|\tau\|^2) \ge 2 / (2 + (r+2)/3) = 6/(r+8)$. ■

The floor is therefore exact for this idealised fresh-row model. The modelling step (gates and covariance absorbed into a Gaussian T) is an approximation of the real chain.

Numerical verification (11 Oct; `experiments/probes/parity_check.py`, `results/summaries/parity_check.json`).

- **Sign flip on the real chain.** On the frozen V56 chain (width 128, depth 16), flipping $W^{(l)} \to -W^{(l)}$ at l = 5 and l = 10 leaves every earlier layer bit-identical. At layer l, C and var are unchanged and μ, d3 and the full D21 flip sign, with difference exactly 0.0. The (W, −W) average of d3 and D21 is therefore exactly zero.
- **Floor.** Monte Carlo ($4 \cdot 10^5$ draws, r = 8–32, random and extremal T) reproduces $E[y^2] = \|\tau\|^2 + 2\|T\|^2$ and the residual $2\|T\|^2$ to within sampling error (≤ 0.7 %). The extremal T attains relative errors 0.5000 (r = 16) and 0.3873 (r = 32), equal to $\sqrt{(6/(r+8))}$.
- **Bound.** The largest eigenvalue of the trace map on the symmetric-tensor space is (r+2)/3 to six digits for r = 2, 3, 4, 6 and 8, so the bound is the exact operator norm.

Scope. The mean $\mu^{(l)}$ is also odd in $W^{(l)}$. The theorem therefore covers closures built on even statistics (C, Grams, variances, earlier state), not closures that condition on $\mu^{(l)}$. Sources born at layer l itself, after the ReLU, are not old sources and are outside the statement.

The argument originated in a consultation with Fable 5 (an Anthropic Claude model, run as a Claude Code subagent). We checked it, wrote out the proof above and verified it numerically. Two other models derived the floor independently with the same constant.

The learned-residual experiment in §4.7 is the data-side counterpart of this theorem.

2.3 Exact structure of the births (measured to rounding)

- **Birth leg.** The birth leg satisfies $A_s = P_s \cdot \mathrm{diag}(w_1, b) \cdot C_{off}^{(b)}$ to 1e-5 at every slot and layer: the birth operator is the row-scaled off-diagonal post-ReLU covariance. It has a zero diagonal and a flat spectrum, needing 310–809 singular directions for 99 % of its energy, so it is not compressible.
- **Born tensor.** The born tensor splits into two families:
  - AAP, with |AAP|/|total| = 0.36–0.37;
  - MPP, with 0.74–0.75.

  Their mutual cosine is +0.56–0.57.

2.4 Edgeworth port deficit (derived and measured; shipped as FB 1.25)

Expanding the post-ReLU κ_3 to first order in C shows that the newborn $\mathrm{Sym}(X_1 \otimes I \otimes Y_1)$ carries exactly **half** of both K×C all-distinct families.

Measurement. Regression coefficients on the slice residual are −0.55 to −0.69 and −0.6 to −0.9, against −1/2 predicted.

Restoring the missing families fully does not help, but a partial restoration does:

Restoration	Effect on the teacher pair
Full (S = 2)	+3.4 % / +3.1 % (worse)
Quarter (S = 1.25)	−0.8 % / −0.35 %

The fix is zero-FLOP. It was shipped together with an adaptive-λ exponent of 1.2. On the 100-MLP harness this scored ×0.9946 ± 0.0007, and it graded 5.1284e-9 (#332110). Why the full fix hurts the final mean even though it improves the slice at layers 4–10 is an open paradox **[OPEN]**. A 10 Oct test removed one explanation: the quadratic feedback cross term that also grows with the scale is inert, and the linear restoration itself is what hurts (§6.5).

2.5 Cubic curvature-birth cumulant identities (derived; numerically verified)

Expand the ReLU around a neuron's pre-activation mean μ and standard deviation σ, with $\alpha = \mu/\sigma$ and standardised input u:

$\mathrm{relu}(\mu + \sigma u) = \Sigma_k\, a_k\, He_k(u)$, $a_k = \sigma^k\, E[\mathrm{relu}^{(k)}(\mu+\sigma u)] / k!$.

Since $\mathrm{relu}'' = \delta$:

- $a_2 = \sigma\varphi(\alpha)/2$;
- $a_3 = -\sigma\alpha\varphi(\alpha)/6$.

The second coefficient a_2 is the "curvature" coefficient. The third coefficient a_3 is the cubic coefficient used by the Round-18 star and by V56, where it appears as $c = -\alpha\varphi/6$ in standardised units.

For jointly Gaussian standardised u, Wick's theorem gives the cubic-birth identity

$\kappa(He_3(u_a), u_b, u_c, u_d) = \kappa(u_a^3, u_b, u_c, u_d) = 6\, \rho_{ab}\, \rho_{ac}\, \rho_{ad}$.

A cubic Hermite term at hub a thus injects a **star**-shaped fourth cumulant into all downstream neurons. Quadratic terms pair up into **path** (He_2–He_2) and C·T shapes.

Numerical verification (`r17_curvature_identities.json`):

- the cubic identity and its He_3 form hold to 8.1e-16 and 6.3e-16;
- the quadratic Q1 and Q2 identities hold to 5.1e-16 and 2.4e-16;
- the combinatorial readout factors hold to 9.0e-15 for the star (24), 3.5e-15 for C·T (24) and 4.1e-15 for the path (48).

These identities are the algebraic basis of §5.4–§5.5.

2.6 Counting results on the sum-of-sources family (derived or enumerated)

- **Young-tier floor.** An exact enumeration of contraction orders gives 193 square products for the young tier in every order: 39 P + 54 A + 100 D21 = 98.4 u at the measured 0.505 u per product. Two outside AI models (Round 10) reproduced the count with counting scripts.
- **Duality.** Storing only birth matrices and layer maps and reading slices by reverse-mode differentiation (the "lazy cubic circuit") costs about 4l products at depth l, about 245 u in total, against 249 u for the source chain. The two are dual contraction orders of one tensor network.
- **Irreducibility.** Sym^3 is an irreducible GL(n) representation. A linear state closed under every dense $M^{\otimes 3}$ therefore needs $C(n+2, 3) \approx 1.8\cdot 10^8$ numbers.
    - *Caveat:* with fixed, known weights, only what future readouts can observe matters, which is at most n^2 per remaining layer.
    - The statement is a count over known contraction orders, not a lower bound on arithmetic cost.

- **Algebraic floor per source.** Over the contraction orders we enumerated, exactly evaluating one old source's D21 costs at least $\min(2n^2r, 4nr^3)$ per source-layer (not a lower bound on arithmetic cost in general).

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3. Cost engineering under a metered, op-capped budget

3.1 Where the FLOPs go

A per-call-site ledger of metered FLOPs (`m1_matmul_ledger.py`) on the V29-era chain gives 249 u per network. Of this, 220 u (88 %) is matrix products.

Family	Cost
Young-source transport	≈ 54 u
Young-source D21 contractions	≈ 50 u
Joins (10 per network)	41–46 u
Old sources in the basis	≈ 41 u
Thin feedback	≈ 15 u
Strassen additions	≈ 12 u
Einsum	9 u
Covariance and closure	≈ 8 u

The young tier sits at its enumerated floor (§2.6), so cost had to come from elsewhere: joins, dead neurons, the basis side, and the meter's pricing rules.

3.2 Graded cost steps

Each step below is exact or near-exact. "Util" is the grader's figure, or adjusted ÷ raw where that is the only source.

Step (date)	Mechanism	Graded score (id)	Raw	Util
V29 verbatim (20 Sep)	—	5.418e-9 (#331681)	2.133e-8	0.254
STRASSEN_MIN 32→8	Two more Strassen levels on the rank-384/224 families. Prediction drift 4.9e-13, peak 5.97 GB	5.220e-9 (#331686)	2.132e-8	0.245
+ STRASSEN_FIRST 5	First call per worker at full depth	5.195e-9 (#331732)	2.132e-8	0.2436
+ V31 sphere (§2.1)	Accuracy, 0 FLOPs	5.137e-9 (#331787)	2.108e-8	0.2437
+ FB 1.25, β 1.2 (§2.4)	Accuracy, 0 FLOPs	5.1284e-9 (#332110)	2.1046e-8	—
Transport join (25 Sep)	Rebuild the shared basis at a join by transporting the previous basis (no range finder)	4.9519e-9 (#332257)	2.1788e-8	0.2273
+ AGE_OLD 5, R_FB 4	One more dense young age; feedback rank 16→4	4.8066e-9 (#332324)	2.1371e-8	0.2249
v48 compaction (26 Sep)	$w_1 := 0$ for $\Phi < 1e\text{-}2$; live-row contraction, widths padded to multiples of 32	4.6294e-9 (#332391)	2.1462e-8	—

Step (date)	Mechanism	Graded score (id)	Raw	Util
v49 basis-side Strassen + aliased Grams	Projections, rotations, Qc and the D21 lift through Strassen; joiner Grams as YY^T (billed at half)	4.4157e-9 (#332401)	2.1463e-8	—
v50 D21 live columns	D21 columns of dead neurons never computed	4.2512e-9 (#332404)	2.1537e-8	—
v51 D21 rows, $\Phi < 3e\text{-}3$ / 7e-3	D21 rows of very dead neurons never computed	4.1081e-9 (#332418) / 4.0902e-9 (#332420)	2.1545e-8 / 2.1636e-8	0.191 / 0.189
v52 op budget, safe B / A (27 Sep)	Strassen only where it saves ≥ ∼2 u per 1,000 ops (§3.3)	4.1696e-9 (#332566) / 4.1228e-9 (#332567)	—	—
+ R_OLD 352, Tq from the QR R factor, R_FB 2	Exact or near-exact trims	B: 4.0995e-9 (#332626); A (R_OLD 352 only): 4.0784e-9 (#332621)	2.197e-8 (B)	—

Net effect. Over the whole cost programme, the trims removed 22 % of the metered FLOPs (util 0.2437 → 0.189, #331787 → #332420) for +2.8 % raw (2.105e-8 → 2.164e-8). On 26 Sep alone, the score fell from 4.8066e-9 to 4.1081e-9 (−14.5 %).

What could not be compacted. Only the *input* side of dead neurons can be compacted. Each of the following loses, measured on the teacher pair:

Attempt	Effect
Removing dead neurons as hubs of the new source (their own κ_3 must propagate)	×6.40 / ×6.74
Zeroing their entries of the transported covariance	×1.59 / ×1.46
Zeroing rows with $\lvert\alpha\rvert > 3$	+6.5 % / +19.6 %
Treating always-on neurons as hubs	×10.0 / ×8.0

3.3 The residual-time cap and the operation budget (graded failures)

Two variants that ran cleanly on our machines graded catastrophically:

Submission	Grade	Share of MLPs capped
#332402 (v49 + R_OLD 320)	0.0434	≈ 12 %
#332425 (v51, row threshold 1e-2)	0.0204	≈ 6 %

Memory was not the cause: the local peak was 5.51 GiB. We measured that the residual Python time tracks the number of metered operations, with no other dependence on the variant:

- about 0.07 ms per op on our laptop;
- 0.053–0.059 ms per op on Kaggle, for every variant;
- about 0.02 ms per op inferred for the grader.

The failing variants issued 21.7k–22.3k operations; v48, at ≈ 16.4k, had passed.

The fix was an op budget.

- Strassen is used only where it saves at least about 2 u per 1,000 ops. Each top-level Strassen call costs 110–240 ops.
- The covariance slot rides inside the compacted family.

- The op ledger names the expensive groups:

Op group	Ops	Cost saved
Nested projections	1.3k	0.7 u
Qc	1.25k	1.7 u
D21 lift	1.2k	1.6 u
C slot	1.27k	0 u

Both classes graded with no capped MLP (#332566 and #332567). This produced our two submission classes:

- **B (wider safety margin):** ≤ ∼18k warm-call ops.
- **A:** ∼19.9k–20k ops.

3.4 The 8 GB limit (graded failures)

The first V56 implementation batched the cubic star as a 3-D broadcast. It failed the grader's smoke test four times (#334786, #334787, #334798, #334799).

Kaggle reproduced the failure only under `whest run --runner subprocess`, which gave "Unable to allocate 450. MiB for an array with shape (118013952,)". In-process runs passed, with maxrss 5371 MB against 5346 MB for the V55 base.

A bisection submission with the star disabled graded the same raw MSE as V55 (2.0648e-8, bit for bit, as #334762; score 3.8546e-9, #334803). The per-slot 2-D-plus-mat-vec form then passed (#334805). Strassen level 6 was ruled out earlier for the same reason, with a peak of 8.13 GB.

Lesson. `whest validate` catches neither the residual cap nor the address-space limit. Subprocess runs on the mini split are the minimum pre-submission check.

3.5 The one-for-one law, and where it breaks

Inside the chain, every *accuracy knob* we swept on the 100-MLP harness trades about 1 % cost for 1 % accuracy, and the shipped constants sit at, or within about 0.5 % of, the score optimum (§3.8):

Knob	Score ratio
β 1.2	×0.9993
λ ×0.9	×1.0026
Two tier-1 passes	×1.073
FB × β grid	within +0.12 %
R_OLD 448	×1.0113
AGE_OLD 6 on V56	raw ×0.978 but util +3.7 %, score ×1.014

The law does **not** hold for **cost levers that are exact or nearly exact**: compaction, Strassen placement, aliased Grams and the transport join. Our 27 Sep audit of 74 earlier kills found that all 8 kills that had to be revived were cost or knob levers. They had wrongly been killed on MSE alone, on mixed sub-populations, in one direction only, or on a different base.

No architecture kill was revived. Every one had missed its gate by a factor of 5 to 50.

A 100-MLP grid over the young-age gate and the old-basis rank on V56 (11 Oct, §6.6 (c)) confirms the law on the shipped estimator. Every one of the five settings scores worse than the shipped AGE_OLD 5, R_OLD 352, by ×1.026 to ×1.104.

3.6 Call-site ledger of the shipped V56 (measured 10 Oct)

`experiments/probes/site_ledger.py` wraps every public flopscope call. It charges each outermost call its metered FLOPs and one operation, and attributes both to the estimator line that requested it (the Strassen helpers are looked through). The ledger records the second, warm `predict()` call of the A-class V56 (`submissions/v56As_fs/estimator.py`) on one network. Result: `site_ledger_v56As_fs.json`.

The warm call costs **193.8 u** in **20,586 operations**. We group the call sites as follows (the grouping is ours):

Group	Call sites	Cost	Ops
D21 hub readout	Strassen leaf products of the hub (`prod`, 59.6 u) and the live-row D21 readout `_d21_live` (10.2 u)	≈ 70 u	4,707
Young-leg transport	The compacted transport family (52.9 u), which also carries the newborn's A leg and the covariance slot, and its uncompacted form in the first layers (10.1 u)	≈ 63 u	1,629
Old basis	Dense legs of the old sources formed from their basis factors: tier 1 (7.4 u), tier 2 through the sub-basis (9.6 u), and the projection at a join (3.2 u)	≈ 20 u	2,925
Everything else	Symmetric product of C_pre (6.2 u), thin feedback and D slices, rotations, QR, Grams, closure and κ_4 terms	≈ 40 u	11,325
Total		**193.8 u**	**20,586**

Reading the ledger:

- **Two families dominate.** The D21 readout and the young transport are 133 u, about 69 % of the bill. These are the per-source contractions whose number grows with depth.
- **The old tier is now cheap.** The old basis costs about 20 u, against 107 u in the author's V29 ledger (§1.2) and about 41 u in our V29-era ledger (§3.1). Compaction, Strassen placement and the nested basis removed most of it.
- **Most operations are not in the expensive families.** The 40 u remainder issues 11.3k of the 20.6k operations. Removing FLOPs from the hot families therefore does not by itself relieve the residual-time budget (§3.3).
- **What a 60 % budget would need (derived).** At 0.6 × 194 u = 116.4 u, removing the whole 70 u readout still leaves 124 u, and removing the whole 63 u transport leaves 131 u. Any design at that budget must cut both hot families by more than half, or cut the remainder too.

This ledger is one network. The harness mean util of the same estimator is 0.1861, about 190.6 u (§6.5 (b)).

Comparison with the reference algorithm's cost table (derived; the released reference code measured 1523 u, §6.3). Wu et al. price their factorised K = 3 algorithm at about 4,150 u for n = 1024 and L = 16 (their Table 1, naive matrix products, as read in `consultations/wu_paper_deep_read_2026-10-10.md`). V56 is therefore about 21× cheaper than the reference algorithm it descends from. The paper's factor count grows as $6n\ell$, and it has no source merging, truncation or compression. The savings from V29's shared basis and from our trims (§3.2) are what make the method fit the budget at all.

3.7 Final calibration and cost layer: E282–E299 (mdhamidhosen, 12–16 Oct; fitted and engineering)

From 12 Oct mdhamidhosen continued from #334870 in his E-series. Each step was screened on public mini MLPs (fit 0–47, validation on unseen 48–99) and then graded. Scores below were re-read from the grader on 16 Oct (round 1429); util is adjusted ÷ raw.

Submission	Estimator	Change	Adjusted	Raw	Util
#334870	v56As_fs	start (this paper)	3.6521e-9	1.9497e-8	0.1873
#335603	E282	Gauss-Newton fit of 24 calibration groups (closure-input and λ scales per layer, GN9; unseen −1.66 % ± 0.20)	3.5890e-9	1.9154e-8	0.1874
#335674	E284	+ GN10 + cubic final-layer rider (ridge, trained on full-split MLPs 100–999)	failed: plain-numpy test hooks ("Prohibited file")	—	—
#335695	E285	E284 without those hooks	3.5707e-9	1.9058e-8	0.1874
#335720	E288	+ GN11 + end-to-end refit of the V56 κ_4 corrector (GN13; unseen −0.77 %)	3.5421e-9	1.8903e-8	0.1874
#335724	E290	+ corrector refits GN14, GN15 (−0.48 %, −0.33 %)	3.5156e-9	1.8766e-8	0.1873
#335729	E291	+ GN16 (−0.37 %)	3.5085e-9	1.8722e-8	0.1874
#335805	E294	+ grouped PK2 einsums (exact; util −0.77 %)	3.4815e-9	1.8722e-8	0.1860
#335816	E295	+ Strassen floor 4 (exact up to rounding)	3.4663e-9	1.8723e-8	0.1851
#335834	E296	+ Strassen on the nested projections (exact; +1.55k ops)	3.4452e-9	1.8722e-8	0.1840
#335843	E297	+ D21 dead-column threshold 0.03 (approximation)	3.4319e-9	1.8915e-8	0.1814
#336176	E298	+ dead-neuron threshold 0.0125, D21 dead-row threshold 0.005	3.4201e-9	1.8819e-8	0.1817
#336377	E299	+ Strassen-Winograd kernel (exact; −0.77 % FLOPs)	**3.3946e-9**	**1.8824e-8**	**0.1803**

From #334870 to #336377 the score falls 7.0 %: raw −3.5 %, util −3.7 % (derived). Every lever of this layer converged on unseen MLPs (his ledger): further steps of the 24 groups (GN12, GN18), of the corrector (GN17, GN20), a rider refit and α-dependent scales (GN19) each gave ≤ 0.08 %, and the rank, age and QPASS screens on E297 found no further gain. The calibration is fitted on public dataset MLPs and their published truth; no leaderboard feedback entered it. It adds no understanding beyond §4–§5: it is the fitted end point of the chain.

3.8 The 16 Oct knob sweep and E299-B (measured and graded)

Knob sweep on #334870. Twenty single-knob variants of v56As_fs were run on the 100-MLP harness against the shipped base (`kaggle_tune_compare.txt`; score ratios, base 3.7586e-9 on the harness):

Variant	Score ratio	Wins
K4C blend 1 → 1.1	**0.9953** (0.9947–0.9959)	89/100
D21 column threshold 0.02	0.9950 (raw ×1.0037, util −0.9 %)	22/100

Variant	Score ratio	Wins
star coefficient ×1.15	0.9979	64/100
λ block 8–11 ×0.95 / λ ×0.93 / β 1.3 / K4C from layer 3	0.9987 / 0.9990 / 0.9993 / 0.9994	58–70/100
ages 5, star ×1.25, λ block 12–15 ×0.95	1.0000 / 0.9996 / 0.9996	—
FB 1.125 / 1.375, β 1.1, λ ×0.97, λ blocks ×1.05	1.0004–1.0022	26–49/100
K4C blend 0.9, dead threshold 0.02 / 0.005, D21 row threshold 0.012	1.0017–1.0129	6–53/100

Blend 1.1 graded ×0.9953 on the A class (#336382, 3.6348e-9) and ×0.9948 on the B class (#336383, 3.6533e-9). Of the combinations, only one runner finished before the sweep was stopped: blend 1.1 with λ block 8–11 ×0.95 gave 0.9939 and with star ×1.15 gave 0.9947. By then E299 was 7 % ahead, and its corrector refits (§3.7) adjust the same levels, so the sweep was closed without a test on E299.

E299-B: E299 in the safe operation class. E299 issues 21,953 warm-call operations (measured, MLP #0), close to the 21.7–22.3k at which v49–v51 variants were capped (§3.3). Turning off three Strassen paths (nested projections, Qc, D21 lift) and raising the Strassen floor to 8 are exact rewrites: on MLP #0 the final MSE moves by ×1.0005 (rounding), the warm operations fall to 18,095 and the warm FLOPs rise by 1.9 % (`e299b_warm_check.json`). Peak working set on Windows, two calls on MLP #0 (`e299b_mem_check.json`): E299 4.87 GB, E299-B 5.22 GB, E295 5.19 GB, all below the 8 GB limit (mdhamidhosen's macOS figures for E295–E299 were 6.2–8.3 GB, so the platform matters). E299-B graded 3.4719e-9 (#336440): raw 1.8825e-8 against E299's 1.8824e-8 (×1.00005, rounding), util 0.1844 (+2.3 %), no failed MLP. An intermediate variant with only the nested projections off (20,523 operations, +0.5 % FLOPs) is prepared as `submissions/e299b1/`.

3.9 Final days, 17–21 Oct: a public base chain, our corrector, and a refitted table (measured and graded)

On 17 Oct the organisers extended Phase 2 to 21 Oct and moved the grader to flopscope 0.12.3. On 100 Kaggle MLPs E299 and E299-B bill exactly as under 0.12.1 (util 0.1788 and 0.1830, identical under both versions; `data/kaggle_out/fs123_*`). We then read the new forum write-ups and public code and ran ten parallel AI-agent reviews (Claude subagents through Claude Code; outside-AI documents, forum, e30_a's write-up, our kill table, corrector refit, cost, mdhamidhosen's ledger, error map, reference paper; brief in `consultations/AGENT_BRIEF_2026-10-19.md`). All of them judged a top-10 jump (≈ −37 %) infeasible in the remaining time; every candidate below except e299safe and e30kang (MLP #0 only) was measured on the 100-MLP harness (paired, same nets) before grading.

Step	Change	Harness (paired)	Graded
e299x / e299xr	e30_a's exact cost savings in E299 (Gram core skipped, QR-factor rotation, folded constants, …)	−0.97 % / −1.4 % FLOPs, predictions unchanged up to rounding	3.3619e-9 (#336711), 3.3461e-9 (#336719)
e299k31	kai_leal_gonzalez's two-term (3,1) closure (MIT), full strength	×1.0075	3.4192e-9 (#336697, worse)
e299xrk	the same at mix 0.25	×0.9960 (76/100; 73/100 on the E299 base)	3.3325e-9 (#336757)
e299safe	+ e30_a's fixed old basis and C_pre compaction	×0.972 (MLP #0; `kaggle/safe/out/safe_results.json`)	3.2529e-9 (#337049)
e30ak4c	our V55/V56 corrector on e30_a (LordProcciuto, MIT; published 3.1163e-9), scale 0.3	×0.9768 (94/100)	3.0519e-9 (#336774)

Step	Change	Harness (paired)	Graded
e30ak4c40	scale 0.4	×0.9752 (87/100)	3.0506e-9 (#337041)
e30ak4c40k31	+ (3,1) closure at mix 0.25	×0.9945 (87/100)	3.0302e-9 (#337175)
e30kang	+ V31 spherical identity	×0.985 (MLP #0 only)	3.1473e-9 (#337221, worse)
c1	+ D21 ×0.98 at L ≥ 9, closure mix 0.4, corrector local 0.2 / star 0.7	singly ×0.9949 / ×0.9974 / ×0.9967; jointly ×1.0035 (n.s.)	2.9956e-9 (#337258)
c1 + N2	e30_a's mean-correction table refitted on this trajectory (§ below)	×0.9346 (99/100); held-out nets 70–99 ×0.9304 (30/30)	2.8475e-9 (#337301)
+ τ_B 2.4	e30_a's near-dead column cut	×0.9930 (21/30 held-out)	2.8256e-9 (#337336)
N3	table refit with 21 more modulators (h_k·st², h_k·D3·st/var^1.5, h_k·corrector-local), 122 features	×0.9901 held-out (24/30)	2.7975e-9 (#337368)
N4a	+ 42 modulators (standardised D21 row/column energies, C_pre correlation-row energy, |α|·st, previous-layer delta and star carried through W), 164 features	×0.9966 held-out (21/30; CI 0.994–0.999)	**2.7858e-9 (#337457)**

Why the corrector transfers only partly. Call-site ledgers on MLP #0 put e30_a at 158.8 u against our 184.1 u (`data/kaggle_out/ledger/`). On e30_a the corrector at full strength costs ×1.085 (MLP #0); the gain comes from the V56 star (at scale 0.3 on 100 MLPs: ×0.9768 with the star, ×1.0024 without it), because e30_a's 87-feature ridge mean-correction table already contains the local κ_4-pattern features and was fitted on e30_a's uncorrected trajectory.

Table refit (fitted). We dumped teacher-forced per-layer features on public mini nets 0–69 under our trajectory and refitted the table per layer by ridge regression with grouped 5-fold CV over nets, adding star modulators (N1, 94 features) and the corrector's own κ_4 change (N2, 101 features). One-step CV error against the original table: 0.96 (all layers). End to end on 100 MLPs: N0 ×0.9496, N1 ×0.9369, N2 ×0.9346; the gain on held-out nets 70–99 is as large as on nets 0–69, so the fit does not overfit the public set. The original table transfers poorly to our trajectory (refitting the same 87 features, N0, alone gives ×0.9496), so the table should be refitted after larger changes of the chain (for the small τ_B 2.4 change a refit added nothing, ×0.9999 held-out). Pipelines: `kaggle/x10/`, `kaggle/x11/` (self-contained Kaggle kernels).

Further table experiments (20–21 Oct). Refitting N2 on 341 nets (mini 0–69 + 271 full-split nets) gave ×0.9997 on held-out mini nets 70–99 (13/30) and ×0.9993 on 32 held-out full-split nets (19/32): the remaining error is a feature limit, not a data limit. An on-policy refit (free-running dumps) gave ×0.997–1.005; the (3,1) orientation fix (T21) ×1.0005. Pipelines: `kaggle/x12/` – `x14/`.

Measured and rejected (17–21 Oct): separate star/local corrector scales other than 0.7/0.2 (×1.004–1.019); table reading the corrected κ_4 (×1.021); FB 1.25/1.4 on e30_a (×0.999/×1.002); D21 ×0.96/0.97 (×1.055/×1.022); λ tilt, hot rows 2.5, dead τ 2.6/3.1 (×1.000–1.16); e30_a's strided Strassen views (`X_STV`, all 100 MLPs failed); e30_a's row compaction in E299 (×1.063, MLP #0); "cheaper setting, then refit" (rank 352: ×1.04 with or without refit; τ_B 2.4: refit adds nothing). Results: `data/kaggle_out/{exp5_*,e30k31_*,e30tune_*,x8_*,x8c_*,x10_eval,x11_*}`.

Rules. The organisers allow building on another participant's public MIT code with attribution, provided the team adds substantial work of its own (forum topic 18240). #337457's base chain is LordProcciuto's and its (3,1) closure is kai_leal_gonzalez's; ours are the corrector (§5), the closure and corrector settings and the refitted, extended table (§1.4). Our best result without e30_a's base chain is #337049 (for reference; it is not the submission this write-up refers to).

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4. The error map: where the error lives

All results in this section are measured on MLP #0 against the $6{\cdot}10^7$-sample teacher, and partly replicated on the synthetic network. The plain chain's final MSE in these runs is 2.290e-8.

4.1 The closures are accurate; the error is accumulated input error

- **Mean closure.** The Edgeworth mean closure, evaluated on the teacher's true marginals (μ, var, κ_{3iii}, κ_{4iiii}), matches the teacher's ReLU mean to 5e-11–1.5e-10 at every layer (9.3e-11 at L16).
- **Bivariate closure.** Teacher-forced at one layer, the bivariate covariance closure falls to the teacher's noise:

	L7	L14
RMS error per entry	7.04e-5 → 2.61e-5 (noise 2.5e-5)	8.84e-5 → 2.76e-5
Sensitivity-weighted next-layer variance error removed	87 %	91 %

- **Accumulation.** The chain's per-layer MSE grows smoothly, from 0.7e-9 at L1 to 2.29e-8 at L16, adding about 1–2.5e-9 per layer.
- **Gain.** The inherited mean error passes each layer with a gain of $|W\ \text{row}|^2 E[\Phi^2] = 0.65$ at layer 2, rising to 0.8–0.96 at depth.
- **Inherited share.** Forcing the true mean at every layer takes the final MSE from 2.24e-8 to 4.2e-9, so about 80 % of the final error is inherited mean error.

4.2 Channel oracles

The true value of one channel is injected into the mean closure at every layer:

Exact input	Final MSE	Ratio
Variance	1.220e-8	0.533
κ_4 diagonal	1.609e-8 [1]	0.702
κ_3 diagonal	1.965e-8	0.858
Variance + κ_3	0.817e-8	0.357
Variance + κ_3 + κ_4	0.130e-8	0.057 (the truth's noise floor)
True κ_4 everywhere *except* the mean closure	2.311e-8	≈ 1
κ_4 diagonal + (2,2) + (3,1) slices	1.738e-8	0.759
(3,1) only / (2,2) only / both	2.353 / 2.796 / 2.680e-8	1.03 / 1.22 / 1.17

[1] Two runs give 1.600e-8 and 1.609e-8 (`k4oracle2_target_base.json`: 1.6086e-8). An independent replication by a subagent gave ratios 0.532 / 0.865 / 0.702 / 0.057.

Single-channel ratios are upper bounds on each channel's own share, because each includes error inherited from the other channels.

Two facts follow from the table:

- κ_4 acts **only** through the mean closure, and only through its **diagonal**.
- The off-diagonal κ_4 slices are harmful when injected alone.

A first-order attribution of the L16 error gives:

Source	Share
Pre-activation mean	1.84e-8 (80 %)
Variance	2.0e-9
κ_{3iii}	1.0e-9
κ_{4iiii}	1.0e-9

At depth, the *fresh* error per layer splits about 50 % variance, 25 % κ_3 and 25 % κ_4. The fitted coefficients are −0.99, −0.98 and −1.04, against −1 in theory, with R^2 0.93–0.97 from L9 on.

4.3 The chain's per-channel error against the teacher

Teacher noise is given in parentheses where it was measured.

Layer	Mean (σ units)	Variance	d3	κ_4 diagonal (slope, corr over all neurons)	C off-diagonal	D21 off-diagonal
7	1.9e-4	0.04 % (0.02)	3.2 % (1.1)	15 % (0.95, 0.78)	0.39 %	4.6 %
11	3.1e-4	0.09 %	3.8 %	22 % (0.89, 0.78)	0.59 %	5.7 %
15	4.6e-4	0.14 %	4.6 %	32 % (0.78, 0.83)	0.73 %	6.6 %
16	5.1e-4	0.14 %	4.7 %	34 % (0.75, 0.82)	—	—

Over all neurons the chain's κ_4 diagonal correlates 0.78–0.83 with the truth. On the **decisive** neurons that drive the mean closure it correlates only 0.26–0.34, at a level ratio of about 1.0. This gap between the two populations is why §5 uses the decisive-neuron metric.

4.4 Variance error is inherited covariance error

About 95 % of the variance error is $\mathrm{dvar}_i = w_i^T\, \delta C\, w_i$, a random projection of the off-diagonal post-ReLU covariance error. That covariance error has two parts:

- an inherited part, with slope 1.01 and R^2 rising from 0.52 to 0.79 with depth;
- a fresh part of about 4.2e-5 RMS per entry.

The audited missing covariance terms explain little:

- the C^3, C^4 and $C\times\kappa_{31}$ patches have RMS 2–3e-6, against an error of 6.5–9e-5;
- the chain's own (1,1) terms explain only 3–8 %.

There is no missing covariance term. The variance channel's 47 % comes from accumulated κ_3 and κ_4 input error.

4.5 κ_3: the all-distinct entries, and memory

The chain's $\kappa_3(i,i,j)$ slice has slope 1.00 and R^2 0.996–0.998, but a random error that grows from 4.1 % at L3 to 6.7 % at L15. Of that error:

- 91–96 % sits in the **all-distinct** post-ReLU entries $\kappa_3(h_i,h_j,h_k)$, which are themselves 9–16 % wrong;
- it is not truncation: with no confinement the slice error stays at 6.6 % and the final MSE moves only −3.9 %, for +19 % cost;
- about 55 % of it is unexplained even at second order in K, since the twelve K×K terms have RMS 3e-6 to 2e-5 against an error of 2.2e-4.

Refinement (10 Oct, §4.10). These percentages are measured against the $6{\cdot}10^7$-sample teacher and include its sampling noise. A Gaussian estimate puts that noise at 2.5 % (L4) to 5.3 % (L2) on the slice, so most of the 4.1 % at L3 is teacher noise. Net of noise, the chain's slice error is about 3 % at L4 and grows to about 7 % at L16.

Perfect κ_3 oracle. A perfect κ_3 input (slice + d3) is worth ×0.6075 on MLP #0 (1.382e-8 vs 2.275e-8) and ×0.715 on the synthetic net. The synthetic figure is partial: 8 of 14 layers. Applied in layer windows on MLP #0 / synthetic:

Window	Ratio
Layers ≥ 8	0.637 / 0.754
Layers ≥ 12	0.704 / 0.863
Layers ≤ 11	0.884 / 0.837

The windows compose multiplicatively: 0.884 × 0.704 ≈ 0.62.

Memory. The κ_3 memory is long and coherent but not low-rank in age. Keeping only sources of age ≤ h leaves this relative D21 error at L15:

h	2	4	6	8	10
Error	84 %	65 %	46 %	31 %	16 %

- The age contributions are coherent: mean pairwise cosine 0.29–0.39.
- They are not low-rank: projecting on six age-modes still leaves 8.5 %.
- Removing the old tier entirely costs 32× (7.43e-7). Every age gate follows MSE ≈ 7.4e-7 × 0.66^(g−4).

4.6 κ_4: level transported, pattern regenerated

- **One-step transport.** Pairwise transport of the true post-ReLU slices gets the next layer's κ_4 diagonal **level** right (slope 1.00) but not its **pattern** (corr 0.24–0.59).
- **Exact slices are not enough.** Even with the exact (4), (3,1) and (2,2) slices, the next diagonal is 14 / 22 / 33 / 43 / 46 % wrong at L4 / L8 / L12 / L15 / L16.
- **Where the content lives.** The next layer's diagonal is about 90 % the (2,2) pair term. The next d3 comes 90–99 % from off-diagonal κ_3, so the diagonal-to-diagonal channel carries only 18 % (L4) down to 1.7 % (L16) in norm.

4.7 What does not carry the missing information

Gaussian and copula closures. The one-step closure floors (2e8 samples per layer) give these RMS errors of the next-layer mean:

	L4	L8	L12	L15
Gaussian closure	5.2e-4	5.1e-4	4.6e-4	4.5e-4
Gaussian copula with exact marginals	1.14e-3	9.2e-4	8.1e-4	7.5e-4

The Gaussian closure errs at about 3× V29's final error. The copula with exact marginals is worse than the plain Gaussian closure at every layer.

Learned residual correctors. On 100 MLPs (102,400 output neurons), learned correctors recover almost nothing:

Model	Residual MSE recovered
Ridge, 19 features	0.3 %
Ridge, 189-feature quadratic expansion	0.7 %
Gradient boosting	0.5 %

- Per-network offsets explain 0.44 % of the residual variance.
- Every single feature correlates ≤ 0.04 with the residual.
- This is the parity theorem (§2.2) seen from the data side.

Monte-Carlo routes.

- A 10^4-sample forward pass costs 156 u and delivers the variance only to 1.4 %, against a 0.1 % need.
- Control-variate surrogates decorrelate with depth:
  - the tangent network falls to 0.15–0.29 at L16;
  - the second-order Taylor surrogate falls from 0.967 at L2 to 0.00 at L16.
- The resulting variance reductions are only 1.0–2.2×.

4.8 The κ_3 and κ_4 inputs jointly (joint oracle, measured 9 Oct)

The oracle of §4.2 and §4.5 is rerun with both inputs at once (`experiments/variants/r20_joint_oracle.py`). The teacher's off-diagonal κ_3 (2,1) slice replaces the chain's D21 everywhere after it is computed, in the closure terms and in the feedback legs. The teacher's d3 = κ_{3iii} replaces the chain's D3, and the teacher's κ_4 diagonal enters the mean closure.

The base is the plain V40-family chain (FB 1.25, β 1.2, spherical identity off) at util 0.2476. Its final MSE is 2.273e-8 on MLP #0 and 1.920e-8 on the synthetic net, slightly different from the 2.290e-8 base used in §4.1–§4.7. The synthetic teacher covers layers 2, 4, …, 14 and 15 only; at its other layers the chain keeps its own values, so the synthetic column is partial.

Teacher input injected (every layer unless noted)	MLP #0 final MSE	Ratio	Synthetic final MSE	Ratio
None (plain chain)	2.273e-8	1	1.920e-8	1
κ_3 slice + d3	1.380e-8	0.607	1.376e-8	0.716
κ_4 diagonal	1.561e-8	0.686	1.700e-8	0.885
d3 only + κ_4 diagonal	1.204e-8	0.530	1.394e-8	0.726
κ_3 slice (chain's own d3) + κ_4 diagonal	0.995e-8	0.438	1.429e-8	0.744
κ_3 slice (chain's own d3) + κ_4 pattern at the chain's level	0.977e-8	0.430	1.441e-8	0.751
κ_3 slice + d3 + κ_4 diagonal	**0.619e-8**	**0.272**	**1.092e-8**	**0.569**

Teacher input injected (every layer unless noted)	MLP #0 final MSE	Ratio	Synthetic final MSE	Ratio
κ_3 slice + d3 + κ_4 pattern at the chain's level	0.669e-8	0.294	1.163e-8	0.606
κ_3 slice + d3 at every layer, κ_4 diagonal from L8	0.653e-8	0.287	1.053e-8	0.548
Both inputs from L8 only	0.746e-8	0.328	1.102e-8	0.574

Reading the table:

- **κ_3 is the larger single lever on both networks:** ×0.607 against ×0.686 on MLP #0, and ×0.716 against ×0.885 on the synthetic net.
- **The two inputs compound more than multiplicatively.** The product of the single-input ratios would be 0.416 on MLP #0 and 0.634 on the synthetic net; jointly they give 0.272 and 0.569. Each channel's error is partly inherited from the other (§4.2).
- **With the true κ_4, the true d3 matters.** On MLP #0, adding the teacher's d3 to the teacher's slice takes 0.438 to 0.272.
- **The κ_4 pattern is again enough.** The teacher's pattern at the chain's level gives 0.294 against 0.272 for the full diagonal (compare §5.1). Injecting κ_4 only from L8 loses little (0.287).
- **What remains is mostly the variance channel.** After κ_3 and κ_4, the remaining ×0.27 is consistent with the channel oracle of §4.2 (base 2.290e-8), where the true variance, κ_3 diagonal and κ_4 diagonal together take MLP #0 to the noise floor (×0.057), and with §4.4, where the variance error is inherited covariance error. (The two oracles differ: §4.2 injects the κ_3 diagonal, not the slice + d3.)

4.9 Which Hermite order each channel needs, one step at a time (measured 9 Oct)

`r20_onestep_channels.py` runs $2{\cdot}10^6$ exact samples of z_l through MLP #0. At one layer l, the ReLU is replaced by its Hermite expansion of order K around the teacher's (μ, σ) (§2.5), with common random numbers, and z_{l+1} = act(z_l) $W^{(l)}$. Every next-layer statistic is compared with the **exact** ReLU step on the same samples. A chain whose births stop at order K makes at least this error in each channel per step.

Values are given at source layer L2 → L15 (layers 2, 4, 8, 12 and 15 were measured; the trend is monotone unless a range is given):

Order K	C off-diagonal	Variance	D21 off-diagonal	κ_3 diagonal	κ_4 pattern corr	κ_4 level ratio
1 (linear)	13.4 → 6.4 %	19.0 → 4.9 %	76.8 → 19.7 %	75.0 → 16.1 %	0.63 → 0.84	0.38 → 0.87
2	2.17 → 0.42 %	3.26 → 0.99 %	20.1 → 9.9 %	21.2 → 10.0 %	0.85–0.90	1.95 → 1.43
3	1.49 → 0.32 %	2.14 → 0.63 %	8.85 → 2.05 %	9.26 → 2.05 %	0.95 → 0.99	1.52 → 1.10
4	0.88 → 0.20 %	1.29 → 0.36 %	4.61 → 1.34 %	4.30 → 1.38 %	0.98 → 0.999	1.16 → 1.06

Errors are relative Frobenius errors against the exact step; the κ_4 columns use the project metric. For scale, this run's own distance to the $6{\cdot}10^7$-sample teacher is 1.5 % → 0.53 % (C off-diagonal) and 0.10 % (variance). The common random numbers make the K-against-exact differences much more precise than that.

Findings:

1. **κ_3 needs at least the cubic response.** With quadratic births (K = 2), the one-step D21 and d3 errors are 10–21 %, against a D21 tolerance of about 2 % (§1.2). K = 3 reaches 2.1–2.7 % only at L12–L15 and is 7–9 % at L2–L4. K = 4 is 1.3–4.6 %.
2. **The covariance needs at least the quadratic response.** The linear step alone leaves 6–13 % of the off-diagonal covariance.

3. **One κ_4 step is easy; compounding is not.** K = 2 gets the one-step κ_4 pattern to 0.85–0.90 but overshoots the level by 1.4–1.9×. Compounded over six layers, the same truncation falls to 0.45–0.55 (§5.4, set A), while K = 3 holds.

4.10 Where the κ_3 error is real, and where its gain lives (measured 10 Oct)

Error profile against teacher noise. `chain_k3_error_profile.json` compares the plain chain's κ_3 (the captures used in §5.8) with the $6{\cdot}10^7$-sample teacher at every layer. `teacher_k3_noise_estimate.json` gives a Gaussian approximation of the teacher's own sampling noise on the same quantities (an estimate). The "net" columns subtract that noise in quadrature, $\sqrt{(\text{error}^2 - \text{noise}^2)}$. This step is derived and assumes the noise is independent of the chain's error.

Layer	D21 off-diagonal error	Teacher noise (est.)	Net D21 error	d3 error	d3 noise (est.)	Net d3 error	d3 norm ratio, chain/teacher
2	4.4 %	5.3 %	≈ 0	3.3 %	5.4 %	≈ 0	1.002
4	4.2 %	2.5 %	3.4 %	2.9 %	2.2 %	1.9 %	0.998
8	4.9 %	1.2 %	4.8 %	3.1 %	0.9 %	3.0 %	0.996
12	5.9 %	0.7 %	5.9 %	3.8 %	0.6 %	3.8 %	0.993
16	6.9 %	0.5 %	6.9 %	4.6 %	0.4 %	4.6 %	0.986

Errors are relative Frobenius errors against the teacher.

Reading the table:

- **The early "error" is mostly teacher noise.** At L2 the estimated noise exceeds the measured difference on both the slice and d3.
- **The real slice error is carried and grows with depth**, from about 3 % at L4 to about 7 % at L16. At L7, L11 and L15 the profile gives 4.6 / 5.7 / 6.6 %, the same values as §4.3.
- **d3 has a level bias, not only scatter.** Its decisive-neuron correlation stays at 0.9965–0.9994, but its norm ratio falls steadily from 1.002 (L2) to 0.986 (L16), a 1.4 % shrink.

Where the κ_3 gain lives (`results/summaries/oct10/k3loc_{1,2,3}.json`). The teacher's κ_3 is injected only in a layer window, with the oracle of §4.8, on MLP #0. "Slice" is the off-diagonal (2,1) slice, "d3" the diagonal; "feedback legs" injects the slice into the feedback legs only. The base is the plain chain of §4.8 (2.2732e-8).

Teacher κ_3 injected	Window	Final MSE	Ratio
Slice + d3 (§4.8)	L1–16	1.380e-8	0.607
Slice + d3	L1–4	2.282e-8	1.004
Slice + d3	L5–8	2.141e-8	0.942
Slice + d3	L9–12	1.949e-8	0.857
Slice + d3	L13–16	1.832e-8	0.806
Slice only	L1–8	2.129e-8	0.937
Slice only	L9–16	1.897e-8	0.834
d3 only	L1–8	2.300e-8	1.012
d3 only	L9–16	1.938e-8	0.853
Feedback legs only	L1–16	2.276e-8	1.001

Reading the table:

- **The gain is in L9–16.** The product of the four slice + d3 windows is 0.653 (derived), against 0.607 when every layer is injected at once. L9–16 alone would give about 0.857 × 0.806 ≈ 0.69 (derived; not run as one window).
- **Early injection does not help.** Slice + d3 at L1–4 gives ×1.004, and d3 alone at L1–8 gives ×1.012. This fits the profile above: at L2–L4 the teacher's κ_3 is about as noisy as the chain's is wrong.
- **At depth, the slice and d3 matter about equally**: ×0.834 and ×0.853 at L9–16.
- **The feedback legs are not the lever**: ×1.001 (compare ×0.9999 in §6.3).

A κ_3 fix must therefore remove a carried late-layer error: 5–7 % in the slice and a 1.4 % shrink of the d3 level at L16. The three candidate mechanisms tested on 10 Oct are reported in §6.5.

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5. The κ_4-diagonal pattern programme

5.1 The target is the pattern, not the level

Oracle (`k4_oracle_run2.py`): the teacher's κ_4 diagonal is injected into the mean closure of the full chain. "Share" is measured against the full gain, none (2.290e-8) → diagonal (1.609e-8).

Injected target	Final MSE	Share of the gain
Teacher **pattern** at the chain's level	**1.551e-8**	**108 %**
Teacher pattern, layers ≥ 8 only	1.707e-8	86 %
Teacher pattern, layers ≥ 12 only	1.813e-8	70 %
Teacher pattern, layers ≤ 11 only	2.064e-8	33 %
50 % blend of teacher and chain patterns	1.789e-8	74 %
Chain pattern at the teacher's **level**	2.783e-8	worse than none (+21.5 %)

The whole κ_4 gap is the per-neuron *pattern* of the deep diagonal. The level the chain already produces is adequate.

Joint oracle: cheap κ_3 plus true κ_4. The same oracle on cheaper κ_3 chains shows that the gain sits on the full κ_3 chain only:

κ_3 configuration	Util	Own κ_4	True κ_4 diagonal	Ratio
Full chain	0.2476	2.290e-8	1.609e-8	×0.702
AGE_OLD 2	0.2303	5.240e-8	4.590e-8	×0.876
AGE_OLD 2, R_OLD 256/128	0.1784	1.288e-7	1.244e-7	×0.966
AGE_OLD 1, R_OLD 192/96	0.1381	6.511e-7	6.665e-7	×1.024

In these three configurations on MLP #0, cutting κ_3 cost and recovering with κ_4 did not pay, and we did not pursue it; the κ_4 improvements we shipped ride on the full κ_3 chain.

5.2 The deep pattern is carried memory

One-step carry of the true state (`opus12_k4_carry.json`, project metric):

Carrier	L12	L15	L16
Chain	0.339	0.264	0.324
Exact linear carry of the true κ_4	0.771	0.785	0.797
Linear carry + quadratic ReLU curvature	0.820	0.855	0.863

Gaussian physics alone is not enough. A Gaussian input with the true mean and covariance gives only 1–4 % of the deep κ_4 level from L8 on.

Memory horizon. Non-Gaussian history older than h layers is erased by a Gaussian restart, and the true network is run forward from there (`q18_k4_horizon.json`, N = 1e6). Pattern correlation at L12 / L16:

h	1	2	3	4	6
L12 / L16	0.23 / 0.20	0.40 / 0.41	0.53 / 0.59	0.67 / 0.65	0.78 / 0.77

Correcting for the teacher's noise raises h = 6 to 0.85 and h = 2 to 0.52. The later N = 4e6 runs (§5.4, set B) give somewhat higher values at h = 3–4, which is consistent with lower noise attenuation.

A young carrier spanning 2–3 layers is therefore dead from the start. An ideal one, tested out of sample, moves the G4-weighted error only from 0.210 to 0.203.

5.3 V55: a zero-FLOP linear pattern corrector (graded)

Construction. At every layer ≥ LMIN (LMIN = 6; LMIN = 4 from #334762 on), a fixed linear map turns per-neuron quantities the chain already holds into a predicted pattern of κ_4/v^2. The inputs are:

- s3sq = $(d3/v\text{^}(3/2))^2$;
- the row and column energies of the standardised D21;
- the correlation-row energy of C_pre;
- $|\mu/\sigma|$;
- the chain's own κ_4/v^2;
- a constant.

The pattern is rescaled to the chain's layer-mean level and fed to the **mean closure only**; the pair closure keeps the chain's κ_4.

The map is fitted once, offline, against our MLP #0 teacher on decisive neurons ($|\mu/\sigma| < 1.5$); the synthetic net is held out. The last layer uses a reduced map. The cost is 0 u and +310 metered ops (17,961 in total for the B class).

Pattern correlation.

- MLP #0: 0.41–0.55, against 0.24–0.41 for the chain (layers 6–15).
- Held out on the synthetic network: 0.41–0.52, against 0.25–0.35.
- The map transfers between networks: fitted on the synthetic net, it gives 0.43–0.56 on MLP #0.

Teacher pair.

Class	MLP #0 (fit network)	Synthetic net (held out)
B	2.1962e-8 → 2.0381e-8 (−7.2 %)	1.8856e-8 → 1.8343e-8 (−2.7 %)

Class	MLP #0 (fit network)	Synthetic net (held out)
A	−7.0 %	−2.7 %

Harness and grader. On 100 MLPs: ×0.9455 [90 % CI 0.9410, 0.9498], with 97/100 MLPs improved.

Class	Projected grade	Actual grade
B	3.8764e-9	3.8762e-9 (#334588)
A	3.8562e-9	3.8582e-9 (#334591)

Each is −5.4 % against its base. The s3sq coefficient ×1.25 then graded 3.8539e-9 (#334762) and 3.8331e-9 (#334763).

Sweeps place the map at its optimum. Ratios are on the 100-MLP harness.

Change	Ratio
s3sq ×1.5 / ×1.75 / ×2.0	×1.0068 / ×1.0207 / ×1.0396
c2 ×1.5 / ×0	×1.0563 / ×1.0131
Chain coefficient ×0.9–×1.2	within ×1.0011
Last-layer corrector off	×1.0116
Lever-weighted "lean" fit	worse: the decisive-neuron map scores ×0.9957 against it

Saturation. Squares, products and logs of the features add only +0.02–0.03 in sample. Previous-layer features carried by $(W \circ W)^T$ add nothing held out.

The cause is the feature set. With the same layer's teacher quantities, a least-squares combination reaches 0.65–0.70 out of layer, but almost all of that comes from the k22 row sum, a teacher-only feature that correlates 0.62–0.70 on its own. Without k22, out-of-layer fits fall to 0.39–0.54. **Per-layer chain features saturate near 0.5.**

5.4 Why carriers stalled: the Hermite-family ceilings

Between 27 Sep and 8 Oct we tested many κ_4 *carriers*: states transported from layer to layer with births at each ReLU. Every one stalled. Pattern correlation at L12 / L15 / L16:

Carrier (birth content)	Correlation	Source
Skew-lift Σ_s Sym4($T_s \otimes r_s$)	0.03 (best LSQR fit)	R13
Dense pair core born at the first ReLU	−0.35 to −0.22	R13
Signed bilinear Sym($H \otimes C$), H carried by $W^T H W$	0.351 / 0.321 / 0.371	R16
Diagonal-only version of the same	0.336 / 0.294 / 0.351	R16
True (4) + (3,1) + (2,2) slices carried one step	0.30–0.45	R16
Pair-innovation births, all ages	0.433 / 0.362 / 0.411	R16b
Hub-pair C·T curvature births, h = 6	0.442 / 0.432 / 0.431	R17
Gaussian hypertree bank (GHQ-6)	0.559 / 0.471 / 0.499	R17

The pair-innovation carrier also made the closure error ×1.12–1.43 worse than the chain.

To find out why, we measured the ceiling of entire *families* of birth laws (`r17_hermite_family_ceiling.py`, Kaggle GPU, N = $4 \cdot 10^6$). The construction:

1. Start from a Gaussian restart N(μ, C) of the teacher h layers back.
2. At each ReLU, replace the ReLU by its Hermite expansion of order K around the teacher's (μ, σ).
3. Measure the resulting κ_4 pattern.

Set A and set B: which Hermite orders are needed.

Mode	L12	L15	L16
Sanity: true ReLU, no restart	0.967	0.973	0.980
h = 6, true ReLU (restart ceiling)	0.859	0.820	0.818
h = 6, K = 2 (quadratic births)	0.548	0.449	0.530
h = 6, orders 0/1/3 only	0.511	0.498	0.469
h = 6, K = 3 (cubic births)	**0.848**	**0.799**	**0.817**
h = 6, K = 4	0.865	0.816	0.822
h = 4, K = 3 / true ReLU	0.749 / 0.753	0.687 / 0.702	0.711 / 0.710
h = 3, K = 3 / true ReLU	0.652 / 0.663	0.588 / 0.600	0.643 / 0.647
h = 2, K = 3 / true ReLU	0.517 / 0.525	0.436 / 0.434	0.485 / 0.494
True state 1 back, K = 2	0.850	0.875	0.881
True state 3 back, K = 2	0.687	0.689	0.738
True state 6 back, K = 2	0.527	0.477	0.578
True state 6 back, K = 3	0.915	0.927	0.949

Findings:

1. **Quadratic births are capped near 0.45–0.55.** Every second-order carrier landed in, near or below this band (−0.35 to 0.56).
2. **K = 3 is necessary and sufficient.** K = 3 tracks the true ReLU at every horizon, and K = 4 adds ≤ 0.02.
3. **Both pieces are needed.** Dropping the quadratic term loses as much as dropping the cubic one.
4. **Compounding K = 2 diverges.** Applied over all 16 layers, it blows the level up by ×1e12. Even from the true state, quadratic births lose the pattern step by step: 0.85–0.88 after one step, 0.69–0.74 after three and 0.48–0.58 after six (one-step channel errors in §4.9).
5. **Horizon.** The 0.70 gate needs a horizon of about 5–6 layers.

Set C: which ages need cubic births (9 Oct, `r17_hermite_family_ceiling_C.json`). The restart is at h = 6. The full cubic response is applied only at the listed ages, and the response is linear elsewhere.

Ages with cubic births	L12	L15	L16	Level ratio L12 / L15 / L16
1–6 (all)	0.848	0.799	0.817	0.81 / 0.53 / 0.48
2–6	0.709	0.708	0.691	0.61 / 0.44 / 0.35
1–4	0.736	0.682	0.713	0.28 / 0.19 / 0.19
1–3	0.647	0.582	0.637	0.13 / 0.09 / 0.09
1–2	0.525	0.458	0.456	0.05 / 0.03 / 0.04
1	0.343	0.239	0.231	0.02 / 0.00 / 0.02

Ages with cubic births	L12	L15	L16	Level ratio L12 / L15 / L16
4–6	0.367	0.397	0.409	0.30 / 0.19 / 0.20
6	0.026	0.116	0.187	0.05 / 0.04 / 0.03

Reading the table:

- **Ages are complementary.** No single age carries the pattern, and no late window does either: ages 4–6 give 0.37–0.41.
- **Dropping age 1** costs 0.09–0.14.
- **Truncating to ages 1–4** costs 0.10–0.12. Even so, it stays at or above the 0.70 gate at L12 and L16 (0.736, 0.713) and just below it at L15 (0.682).
- **Ages 1–3** fall to 0.58–0.65.
- **The level is a separate matter.** The level is mostly recovered from older ages (ages 4–6 alone give 0.19–0.30 of it). Since V55 and V56 rescale to the chain's level, and the pattern alone is worth 108 % of the oracle gain (§5.1), the pattern is the quantity a carrier must deliver.
- **Variance ratios** run from 0.64 (age 1 only) to 0.96 (all ages).

5.5 HTQB star → V56 (graded)

The Round-18 proposal "Hermite Tree Quartic Bank" (HTQB) uses the identities of §2.5 to read κ_4 from existing legs without a carried κ_4 state. It has two parts:

- a cubic **star** (Cu1: 24 Σ_a c_a P X^3, with c_a = −α_aφ_a/6, over the last 6 ages);
- a quadratic **path** (Q2: 48 V^T C V, over 3 ages).

Its proposer priced it at 27.2 u.

Ceilings with teacher μ and C (`round18_htqb.json`), at L12 / L15 / L16:

Form	Correlation
Star + path, h = 6	0.637 / 0.559 / 0.586
Shippable form (star h6 + path h3)	0.616 / 0.538 / 0.541
Star alone	0.521 / 0.489 / 0.471
Path alone	0.074 / 0.024 / 0.150

Alone, HTQB is borderline. It is, however, **complementary** to the V55 features. With leave-one-layer-out fits on MLP #0, still using teacher μ and C (`r18_htqb_combo.json`):

Features	Correlation, L7–L15
V55 feature set alone	0.48–0.56
+ star	0.65–0.74 (0.70–0.74 at L12–15)
+ star + path	0.71–0.76 (L12 0.761, L15 0.711)

V56 ships the zero-FLOP half. The cubic star is computed from the chain's **own** young dense source legs (ages ≤ 6) and the per-source coefficient c = −αφ/(6·var) recorded at birth. It enters as an 8th feature of the V55 map. The path is not shipped.

The 8-feature map reaches 0.53–0.71 per layer on MLP #0 and 0.54–0.66 on the synthetic net, depending on the fitting network (`k4s_fit.json`), against 0.41–0.55 for the 7-feature map. Utilisation is 0.1857, against 0.1855 for the V55 reference.

V56 grades.

Submission	Class	Score	Raw	Note
#334805 v56s_loop	B	3.6829e-9	1.9706e-8	−4.4 % vs the best B
#334811 v56As_loop	A	3.6597e-9	1.9539e-8	
#334869 v56s_fs (synthetic-fit map)	B	3.6723e-9	—	−0.29 %
#334870 v56As_fs	A	**3.6521e-9**	1.950e-8	−0.21 %, the chain this paper analyses
#334871 fs + star ×1.25	B	3.6724e-9	—	no extra gain

Tuning is closed (`kaggle_v56_tuning.txt`). Ratios are against v56_ref (raw 2.0409e-8):

Change	Ratio
Star ×0.75 / ×1.5	×1.0110 / ×1.0111 (score)
Star ages 4	×1.0045
s3sq ×1.1 / ×1.25	×1.0028 / ×1.0093
AGE_OLD 6	raw ×0.9782 but score ×1.0138

The AGE_OLD 6 result is consistent with §5.4: a sixth dense age helps accuracy, but not at its present price.

5.6 Can Hermite trees replace the κ_3 chain? (measured; no)

Since the tree κ_3 diagonal has pattern correlation 0.975–0.986 with the truth, we tested whether a Hermite-tree estimator could drop the source chain altogether. The test is teacher-forced, one Edgeworth closure per layer (`r19_tree_ceiling.json`):

Layer	Teacher floor	Chain	Tree κ_3 + κ_4 (h = 6)	Chain κ_3 + teacher κ_4	Chain κ_3 + tree κ_4
8	4.3e-11	1.445e-9	2.99e-8	6.6e-10	1.08e-8
12	5.2e-11	2.004e-9	3.40e-8	9.5e-10	1.91e-8
16	6.2e-11	2.072e-9	4.18e-8	1.03e-9	1.93e-8

The tree is 10–20× worse. An oracle level rescale does not rescue it: L12 still gives 2.80e-8. The closure needs κ_3 to about 1 %, and a 0.98 correlation is far from that. The tree κ_4 is also harmful as a direct replacement of the chain's κ_4; it is useful only as a *feature* inside the rescaled pattern map.

5.7 Summary of the graded κ_4 gains

Stage	Score (id)
Before V55 (A)	4.0784e-9 (#332621)
V55	3.8582e-9 (#334591)
V55, s3sq ×1.25	3.8331e-9 (#334763)
V56	3.6597e-9 (#334811)
V56, synthetic-fit map	**3.6521e-9 (#334870)**

In total this is −10.5 % on the score at unchanged cost, all of it from making the κ_4-diagonal *pattern* better at zero FLOPs.

5.8 Beyond the star: the κ_3-dressed C·T channel (measured 9–10 Oct; oracle and component level, not graded)

The V56 star is the *Gaussian* cubic birth. The Round-17 hub-pair C·T births were killed alone at 0.43–0.44 (§5.4). A design from a 9 Oct consultation, the "lagged κ_3-dressed vertex" (LKV), reads the chain's κ_3 through the quadratic and cubic Hermite vertices of §2.5. Every number below is our own measurement on public MLP #0.

Readouts. Take a vertex layer m and a target pre-activation layer t, at age j = t − m. Write u = z_m − μ_m and the K = 3 response of the ReLU at m as

$h - E\,h = \Phi\,u + g\,(u^2 - v) + c\,(u^3 - 3\,v\,u)$, $g = \varphi/(2\sigma)$, $c = -\alpha\varphi/(6\sigma^2)$.

With the one-step linear response Ystep_l = diag($w_1^{(l)}$) $W^{(l)}$ (legs move as L ← Ystep_l^T L, §1.2), define:

- the transports $\dot{Y}$ = Ystep_m ⋯ Ystep_{t−1} and $\bar{Y}$ = $W^{(m)}$ Ystep_{m+1} ⋯ Ystep_{t−1} (the hub leg, with no w_1 on the hub);
- the C-leg X = C_m $\dot{Y}$;
- the κ_3 object Q_ai = Σ_bc κ_3(z_m)_abc $\dot{Y}$_bi $\dot{Y}$_ci.

The single-vertex births then read out on the diagonal at t (readout factors from §2.5):

star_i = 24 Σ_a c_a $\bar{Y}$_ai X_ai^3 (the Gaussian cubic star, shipped in V56) ct_i = 24 Σ_a g_a $\bar{Y}$_ai X_ai Q_ai (κ_3 converted at a quadratic vertex: the C·T channel) tt_i = 36 Σ_a c_a $\bar{Y}$_ai YL_ai Q_ai, YL = (D21 + diag d3) $\dot{Y}$ (κ_3 squared at a cubic vertex: T·T)

The prediction sums the readouts over ages j = 1–6.

Ceiling (`fable_ceiling_2026-10-09.py`, $2{\cdot}10^6$ samples through the true network). Q, X and the exact births are computed from the samples. Project metric at L12 / L15 / L16:

Readout, ages 1–6	Pattern correlation	Level ratio
Sampling floor: sample κ_4 against the teacher	0.938 / 0.949 / 0.963	1.00 / 0.99 / 1.00
Exact one-vertex births + linear carry (telescoping identity, rel. err. 4.8e-6–1.0e-5)	0.938 / 0.949 / 0.963	1.00 / 0.99 / 1.00
Exact one-vertex births	0.937 / 0.919 / 0.932	0.77 / 0.63 / 0.62
K = 3 births with exact inputs	0.926 / 0.893 / 0.918	1.35 / 1.10 / 1.08
Star alone	0.521 / 0.489 / 0.471	−0.84 / −0.88 / −0.82
C·T alone	0.442 / 0.431 / 0.424	0.88 / 0.93 / 0.85
T·T alone	−0.159 / −0.133 / −0.057	−0.04 / −0.05 / −0.05
Star + T·T	0.490 / 0.445 / 0.436	
Star + C·T	**0.846 / 0.835 / 0.824**	0.04 / 0.05 / 0.03
Star + C·T + T·T	0.849 / 0.840 / 0.830	0.00 / 0.00 / −0.02

Readout, ages 1–6	Pattern correlation	Level ratio
Same, hub index of Q projected on the top-r eigenvectors of C_m, r = 32 / 64 / 128	0.67–0.71 / 0.74–0.75 / 0.78–0.81	

Reading the table:

- **Star + C·T matches the Gaussian-restart cubic-birth ceiling** of set A (§5.4: 0.848 / 0.799 / 0.817). It stays below the K = 3 births with exact inputs in the same run (0.89–0.93). Over ages 1, 2, 4 and 6 it builds up as 0.57 → 0.71 → 0.82 → 0.85 at L12 (0.43 → 0.65 → 0.80 → 0.84 at L15).
- **The Round-17 C·T kill was too narrow.** C·T alone sits exactly where the hub-pair carrier was killed (0.42–0.44), but it is complementary to the star.
- **T·T is dead.** It is negative alone and adds ≤ 0.006 on top of star + C·T.
- **The readouts carry the pattern, not the level.** The star and C·T levels cancel (sum level −0.02 to 0.05), so the prediction must enter as a *feature* of a rescaled map, as in V55/V56, not as a raw κ_4 diagonal.
- **The single-piece identities hold on the samples.** The exact one-vertex births split into joint cumulants of the linear, quadratic and cubic images of the response. Per age, the quadratic piece Q1 = 4 κ(quad, lin, lin, lin) correlates 0.934–0.974 with C·T. The cubic piece Cu1 = 4 κ(cub, lin, lin, lin) correlates 0.993–0.998 with star + T·T (plus a diagnostic C·K term).

As a feature: oracle end-to-end test (`fable_affine_fits.json`, `fable_affine_e2e.json`). At each layer 7–16 an affine map

$\kappa_4/v^2 \approx a \cdot \kappa_4\text{^chain}/v^2 + b \cdot \text{pred}/v^2 + c$

(three numbers per layer) is fitted by least squares to MLP #0's teacher, with pred taken from the sample-exact ceiling readouts. The fitted values replace the κ_4 diagonal in the plain chain's mean closure on the same network. **This is an oracle in two ways:** the map is fitted to the teacher on the network it is evaluated on, and the feature's FLOPs are not metered (util stays 0.2476).

κ_4 diagonal in the mean closure, layers 7–16	Fit corr, L7–L16	Final MSE	Ratio
Plain chain (own κ_4)	0.26–0.40	2.273e-8	1.000
Teacher κ_4 diagonal	—	1.591e-8	0.700
Affine(chain) only: the fair baseline	—	2.467e-8	1.085
Affine(chain, star)	0.37–0.51	2.296e-8	1.010
Affine(chain, star + C·T)	**0.82–0.86**	**1.758e-8**	**0.773**
Affine(chain, star + C·T + T·T)	0.81–0.86	1.801e-8	0.792

C·T recovers (1 − 0.773)/(1 − 0.700) ≈ 75 % of the perfect-κ_4 gain. Against the same-maturity affine baseline (×1.085), star + C·T is ×0.713 and star alone is ×0.931.

Component test: the chain already holds the inputs (`fable_component_2026-10-09.py`). The test uses captures of the plain chain with every source dense. Each of a source's three blocks has three legs, and each of a block's three Sym placements (a "role"; nine per source) pairs one leg at m (the anchor L) with the Hadamard product H of the other two legs at t. Then $Q_m(t) = (1/3)\,\Sigma_s\,\Sigma_{roles}\,L_s(m)\,H_s(t)^T$. The legs at t are the legs at m transported by $\dot{Y}$ (checked to 1.3e-7 and 8.2e-7).

Against the sample-exact readouts on decisive neurons, the chain's own readouts correlate:

- C·T: 0.993 (pair 14 → 15, age 1) and 0.996 (11 → 15, age 4);

- star: 1.000;
- T·T: 0.991–0.994.

The C·T content is therefore available from the chain's existing κ_3 legs. The obstacle is cost: formed per (source, role, age), Q costs one dense n^3 product each.

Age-collapse identity (derived; measured exact). Write G_m = diag(g_m)($\bar{Y}$_{m→t} ∘ X_{m→t}), an n × n hub-by-target matrix. Then

ct_i = 8 Σ_m Σ_s Σ_roles Σ_a (G_m)_ai (L_s(m) H_s(t)^T)_ai (8 = 24/3).

Let a_s be the first window layer that holds source s. Its anchor legs are transported, L_s(m) = (Ystep_{a_s} ⋯ Ystep_{m−1})^T L_s(a_s), so the sum over ages collapses onto one matrix per anchor layer:

ct_i = 8 Σ_s Σ_roles Σ_h (L_s(a_s)^T Γ(a_s))_hi H_s(t)_ih, Γ(a) = G(a) + Ystep(a) Γ(a+1).

Γ is built once per target by a backward recursion over the window. On the captures, the collapsed form matches the per-age route to a relative error of 2.9–3.0e-7 at L12 / L15 / L16 (`acds_ct_probe.json`) and 2.9–3.9e-7 at L7–L16 (`acds2_ct_probe.json`), which is float32 rounding.

Anchor grouping (derived; measured exact). The nine anchor legs of a source are linear in its two dense legs, A_s and P_s, plus thin legs (the low-rank feedback and residual columns). Grouping the roles by dense anchor gives

ct_i = 8 Σ_s [Σ_h (A_s^T Γ(a_s))_hi (H_A,s)_ih + Σ_h (P_s^T Γ(a_s))_hi (H_P,s)_ih + thin_s,i],

where H_A,s and H_P,s collect the target Hadamards of all roles anchored on A_s or on P_s. The thin_s,i terms are evaluated exactly through the thin products Z_s^T Γ. The grouped form matches the per-age route to 2.6–3.2e-7 at every target L7–L16. What remains is **two dense products per source and target**, plus building G_m and Γ.

Sketches. The dense products are replaced by A^T Γ_a ≈ (A^T $\tilde{S}$*a)(S_a^T Γ_a), with $\tilde{S}$ = S (S^{T}S)$^{-1}$. S_a^T Γ_a =* Σ{m≥a} S_m^T G_m when the rank-r basis S is transported like a leg. Bases tested:

- "eig": the top-r eigenvectors of C at the window start;
- "rad": C Ω with a Rademacher Ω;
- "direct-Γ": an orthonormal basis of Γ_a Ω per anchor, optionally with target-aware column weights ω_i = dec_i $\sqrt{\lambda_i}$ / v_i^2 and one power step.

The captures hold dense source stacks only at a few layers, so these tests use the consistent source subset: sources born between stack layers are absent until the next stack. On that subset, the exact chain C·T correlates 0.82–0.96 with the sample-exact one, against 0.993–0.996 for the full source set in the component test.

C·T variant	Fidelity to the exact chain C·T (decisive neurons)	Teacher-metric corr of affine(chain, star + C·T), L12 / L15 / L16 or their range
Sample-exact (ceiling)	—	0.859 / 0.858 / 0.843
Exact chain C·T, source subset	1	0.725 / 0.760 / 0.778
Round 1 (all anchors sketched), eig / rad, r = 64	0.93–0.94 / 0.89–0.93	0.60–0.63 / 0.60–0.61
Round 1, eig / rad, r = 128	0.95–0.97 / 0.94–0.96	0.67 / 0.65–0.66

C·T variant	Fidelity to the exact chain C·T (decisive neurons)	Teacher-metric corr of affine(chain, star + C·T), L12 / L15 / L16 or their range
Round 1, eig / rad, r = 256	0.97–0.98 / 0.97–0.98	0.71 / 0.70–0.72
Round 2 (grouped, low-rank anchors exact), eig r = 64	0.91–0.97 (L7–L16)	0.659 / 0.665 / 0.674
Round 2, eig r = 128	0.955–0.983	0.691 / 0.700 / 0.715
Round 2, direct-Γ weighted + power step, r = 64	0.89–0.94	0.660 / 0.672 / 0.674
Round 2, direct-Γ, r = 64	0.87–0.92	0.613 / 0.539 / 0.590

Two lessons:

- **Once the low-rank anchors are exact, the plain eigen-sketch of C is the best basis we tried.** At r = 64 the best direct-Γ basis ties it on the teacher metric at L12–L16, has lower fidelity and a worse end-to-end ratio (×0.885 against ×0.865, below), and costs more. The probe's thin-product count over targets 7–16 is 37.6–51.3 u against 27.6 u.
- **High fidelity is not enough.** In round 2 at L12–L16, sketches that correlate 0.96–0.98 with the exact chain C·T still lose 0.03–0.10 of teacher-metric correlation. The information C·T adds lies in the part the sketch drops.

End to end with the chain's own C·T (`acds_e2e.json`). This uses the same oracle affine map and protocol as above, with the chain's own star and the C·T variant from the captures:

C·T input to the affine feature	Final MSE	Ratio
Teacher κ_4 diagonal (reference)	1.591e-8	0.700
Sample-exact C·T (ceiling)	1.758e-8	0.773
Exact chain C·T, source subset	1.863e-8	0.819
Eigen-sketch, r = 128	1.917e-8	0.843
Eigen-sketch, r = 64	1.966e-8	0.865
Direct-Γ weighted + power step, r = 64	2.012e-8	0.885

The gap from ×0.773 to ×0.819 is at least partly the capture subset, since the full-source component fidelity is 0.993–0.996. That share has not been measured separately.

Layer windows (10 Oct, `results/summaries/oct10/ctwin_{1,2}.json`). The same oracle affine feature is injected only at the target layers of a window. Elsewhere the chain keeps its own κ_4. The base is 2.2732e-8.

C·T input to the affine feature	Window	Final MSE	Ratio
Sample-exact (ceiling, above)	L7–16	1.758e-8	0.773
Sample-exact	L11–16	1.811e-8	0.797
Sample-exact	L13–16	1.951e-8	0.858
Sample-exact	L15–16	2.135e-8	0.939
Sample-exact	L7–10	2.235e-8	0.983
Eigen-sketch, r = 64 (above)	L7–16	1.966e-8	0.865
Eigen-sketch, r = 64	L11–16	2.074e-8	0.912

C·T input to the affine feature	Window	Final MSE	Ratio
Eigen-sketch, r = 64	L13–16	2.164e-8	0.952

L11–16 carries (1 − 0.797)/(1 − 0.773) ≈ 89 % of the ceiling's gain, and L7–10 alone is nearly worthless (×0.983). With the r = 64 sketch, however, L11–16 keeps only (1 − 0.912)/(1 − 0.865) ≈ 65 % of the gain, while it holds 72 % of the probe's thin-product count (derived from `acds3_ct_probe.json`; the later targets hold more sources). Cutting target layers therefore saves less than it loses. Cutting rank is the better cut (round 3, below).

Round 3: cheaper bases and source subsets (10 Oct, `acds3_ct_probe.json`, `acds_e2e3.json`). The captures, protocol and oracle map are those of round 2. The new variants are:

- "rsi": a randomized subspace iteration on C at the window start (q = 2, three thin products with C), transported like the eigen-basis;
- "top-6": only the six sources with the largest ||A_s||||H_A,s|| + ||P_s||||H_P,s|| are kept, and the others are dropped;
- "young only" / "old only": only the sources anchored after / at the window start are kept.

The cost column is the probe's thin-product count summed over targets 7–16. It excludes forming G_m and Γ.

C·T variant	Thin-product count	Teacher-metric corr of affine(chain, star + C·T), L12 / L15 / L16	Final MSE	Ratio
Eig, r = 64 (round 2)	27.6 u	0.659 / 0.665 / 0.674	1.966e-8	0.865
rsi, r = 64	27.6 + 1.9 u	0.659 / 0.654 / 0.668	1.965e-8	0.864
Eig, r = 48	20.7 u	0.643 / 0.661 / 0.659	1.993e-8	0.877
Eig, r = 32	**13.8 u**	0.620 / 0.629 / 0.637	**1.987e-8**	**0.874**
Eig, r = 64, top-6 sources	20.9 u	0.665 / 0.625 / 0.642	2.063e-8	0.907
Eig, r = 32, top-6 sources	10.4 u	0.631 / 0.613 / 0.610	2.076e-8	0.913
Eig, r = 64, young sources only	—	0.675 / 0.660 / 0.694	2.051e-8	0.902
Eig, r = 64, old sources only	—	0.503 / 0.474 / 0.391	—	—

On every target the age-collapse and grouping identities again hold to 2.6–3.9e-7.

Reading the table:

- **Rank is cheap to cut.** r = 32 keeps ×0.874 against ×0.865 at r = 64, for half the thin-product count. r = 48 is no better than r = 32 (×0.877).
- **rsi ties the eigen-basis** (×0.864) and adds 1.9 u.
- **Dropping sources costs more than dropping rank.** Top-6 gives ×0.907–0.913 and young-only ×0.902. The old-anchored sources carry signal that no cheap subset keeps.
- **The three-layer teacher metric does not rank sketches end to end.** Young-only has the highest correlation at L12 / L15 / L16 (0.66–0.69) but ends at ×0.902, behind eig r = 32 (0.62–0.64, ×0.874). Sketch choices must be made on the end-to-end mean.

Cost and verdict. The score is MSE × max(0.1, FLOPs/2^{41}). V56 uses about 190 u (util 0.186), and our estimate for a deployable eigen-sketch C·T at r = 64 is about 40 u (estimate). Adding 40 u multiplies util by about 1.21, so a raw gain of ×0.865 gives a score of about ×1.05, i.e. roughly neutral. Break-even at ×0.865 needs less than about 30 u (estimate). The probe's count of thin sketch products alone already sums to 27.6 u over targets 7–16 at r = 64, before G_m and Γ are formed.

Round 3 changes this arithmetic. Scaling the 40 u estimate by the probe's thin-product counts (13.8 against 27.6 u) puts eig r = 32 at about 18–20 u (estimate). Util then rises by a factor of about 1.09–1.11, and the score ratio is about 0.874 × 1.10 ≈ 0.96–0.97, i.e. a gain of 3–4 % (estimate). The open risk is the metered op count, which no prototype has measured yet.

Moreover, every ratio in this subsection was measured:

- on one network;
- with an in-sample teacher-fitted map;
- on the plain chain rather than on V56, which already contains the star and the V55 features.

The increment on V56 with an offline-fitted map is **(untested)**.

Conclusion. The κ_4 C·T channel is real. Its ceiling is 0.82–0.85 together with the star, it is worth ×0.773 under an oracle map, and the chain's own legs reproduce it at 0.993–0.996. At r = 64 it is **not affordable**. At r = 32 (round 3) it is at the edge of affordability, at an estimated 3–4 % score gain, with every caveat above still applying. Per the joint oracle (§4.8), κ_3 is the larger lever: ×0.607 against ×0.686 on MLP #0, and ×0.716 against ×0.885 on the synthetic net.

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6. Negative results

Unless marked, ratios are final-MSE or score ratios against the chain, given as MLP #0 / synthetic net. "corr" is the decisive-neuron κ_4 pattern correlation. "Gate" is the pre-registered pass criterion.

The kill types are:

- **C:** a ceiling or oracle (an upper bound);
- **P:** a teacher-forced component test;
- **E:** an end-to-end prototype;
- **K:** a knob.

6.1 κ_3 representation and cost architectures

Idea	Measured (gate)	Kill
Low-rank Hadamard products of source legs	A∘P needs 681–778 of 1024 directions for 99 % energy (flat spectrum)	C
Old content only in the last k layers	Keep last 1/2/3 layers: 4.85 / 4.17 / 3.16e-7 against a chain error of 2.1e-8; dropping one birth: 1.3–2.7e-7	C
Query surrogates: trace, diagonal Hermite, TensorSketch	Error at L14: 98.2 %, 95.4 %, 790 % (d = 2048)	C/P
Pooled or aggregate states: vector metric, Tucker/TT, CP	Rank 32/128/256 → 35 / 7.8 / 1.6 %; k = 128 → 20 %; CP rank 4608 → 1.07e-7	C

Idea	Measured (gate)	Kill
Slice-to-slice recursion	R^2 0.030 / 0.047; RSC D21 error 35.9 % vs the chain's 6.6 %	C
Cubic source-index coding	388 / 527 % error (q = 1); 83 / 130 % (32 channels)	P
Birth operator ≈ diagonal + rank k	73 / 57 % (k = 32); 22 / 16 % (k = 384)	C
Adjoint low-rank response	Low-rank path carries 14 % of the response	C
ISDF skeleton / landmarks	56–71 % (anchor), 79–94 % (transported)	P
Birth-column screening	−5.6 % cost for +25–30 % raw	K
Asymmetric contraction rank	Util +0.8 %, raw +2.9 %	E
Young tier at half cost	193 products in every order (§2.6)	count
Explicit κ_3 tensor / lazy circuit	≈ 1567 u per layer / ≈ 245 u against 249 u	count
Short age memory: age modes, shift carrier	h = 10 still 16 %; 6 modes 8.5 %; shift error 2.9 % (gate 2 %)	C
Source-free dressed hypertree (DHMP)	Next-layer D21 error 36–40 % (gate 2.5 %)	P
Diagonal-only κ_3/κ_4, diagonal + low-rank C	Next-d3 error 83–99 %; diagonal + rank-64 C 12–33 %	P
Deterministic moment-matched particles	D21 error 74–160 % against the chain's 6.6 %	E
Response-cashout propagator (use each source once)	Equals keep-last-1: +4.85e-7, about 23× the chain error	C
Richardson extrapolation in rank	Exponent 1.46 vs 2.07; every extrapolation worse	C
Strassen level 6 / 7	8.13 GB against the 8 GB cap; level 7 runs out of memory	cap
Compensating closure with no old tier	7.43e-7 (32×); every knob 6.8–9.9e-7	C
Hermite-tree κ_3 replacing the chain (§5.6)	Closure 10–20× worse	P
Single shared pool for the whole κ_3 (§6.6 (a))	D21 error at L11–15: 7–11 % (r = 384), 3.4–6.3 % (r = 512); only r = 768 reaches 0.2–1 % (gate 2 % at r ≤ 512)	C
Deferred exact join, DEJ (§6.6 (b))	Youngest source exact, rest at r = 352–512: 2.5–8 % (gate 2 %)	C
Age/rank re-split of V56, 100 MLPs (§6.6 (c))	Score ×1.026 to ×1.104 (gate < 1)	K
Checkpointed birth archive, CBA-13 (§6.6 (d))	κ_3 zeroed in the closures over a window: ×14.8 to ×125; no recovery	E
Trace projection of old κ_3 (Wu et al., §6.6 (f))	26–42 % of D21	C
Learned pair-memory student, PM364 (§6.7)	Held-out MSE to V56 −0.9 % after training; D21 58–70 % (gate MSE ≤ 1e-9, D21 and d3 ≤ 2 %)	E

6.2 κ_4 channel

Idea	Measured (gate)	Kill
Per-row λ fitted to the (3,1) slice	2.2911 → 2.2988e-8 (worse)	K

Idea	Measured (gate)	Kill
Depth multiplier, pairwise transport, 13-feature regression	×1.25 / ×1.16; corr 0.24–0.59	K/P
Young κ_4 carrier, h ≤ 3	G4-weighted error 0.188 / 0.173 (ideal)	C
Quartic response kernel	−6.4 / −2.6 % at ≈ 42 u; not re-tuned	E (parked)
K_3×C all-distinct term	−8.8 % (MLP #0) at ≈ 140 u; cheap form −4.2 % at ≈ 13 u, synthetic run unfinished	E (parked)
Nine (2,1,1) hypertree classes	R^2 ≤ 0.07	P
Carried pair-response modes	≤ 3 % error cut against a 15 % gate	P
SSC 4-mode closure + drift MLP	corr 0.25–0.33 at L15; held-out R^2 ≤ 0.004 (gate 0.10)	P
Role-split λ	×0.971 / ×0.998 (gate 0.97 on both); graded +0.77 % (#332382)	K
Per-source pair innovations γ, η (scalar form)	×0.92 in sample, ×1.05 out of sample	P
Exchange vertex $3YC^{-1}Y$	corr −0.07 to −0.23; error ×26–95	P
Source-resolved skew energy	+≤ 0.004 corr	C
Every second-order carrier in §5.4	−0.35 to 0.56 (gate 0.70)	C
Augmented dense κ_4 core (504aldo F105)	2.04e-8 vs 2.23e-8 at +3 dense units per source-layer	C (cost)
T·T: κ_3 squared at a cubic vertex (§5.8)	Alone −0.16 to −0.06; adds ≤ 0.006 to star + C·T; oracle E2E ×0.792 against ×0.773 without it	C/E
LKV readouts injected as the raw κ_4 diagonal	Level ratio of the ages 1–6 sum −0.02 to 0.00 (§5.8); usable only as a feature	C
Low-rank C·T sketches, r = 32–256 (eig, Rademacher, direct-Γ)	Teacher-metric corr 0.60–0.63 at r = 64 against 0.73–0.78 exact (round 1); oracle E2E ×0.843–0.885 against ×0.773; ≈ 40 u (estimate) against a ∼30 u break-even. Round 3 reopens it: eig r = 32 gives ×0.874 at an estimated 18–20 u (§5.8)	E (parked, cost; r = 32 reopened)

6.3 Correctors and non-cumulant families

Idea	Measured
Covariance patches	2–3e-6 RMS against an error of 6.5–9e-5
True D21 in the feedback legs only	×0.9999
Low-rank covariance-error correction	Needs rank ≈ 256
Exact old tier	−3.9 % MSE for +19 % cost
Online mean-correction refit	R^2 0.02 (chance level 0.013)
Learned residual: ridge / GBM	×0.997 / ×0.995
Monte-Carlo blends and control variates	Weight ≈ 0.002; surrogate corr 0.19 at L16

Idea	Measured
Copula, TT-cross (60–500 % at width 32), TAP/AMP, SOS	Dead (§4.7)
ARC reference K3 at width 1024	1523 u, raw 3.7–4.5e-8: 5.9× dearer and 1.7× less accurate than V29

6.4 Outside-AI proposals

We consulted outside models in 20 numbered rounds (the last ones on 10–11 Oct, including Round 20, §6.7); ten internal Claude-subagent reviews followed on 19 Oct (§3.9). Below, "In PDF?" states whether the proposal itself contained a measurement on our data. Every verdict is our own measurement.

Round, author	Proposal	In PDF?	Our verdict
R1–2 (8 models)	Compensating closure without the old tier	No	Dead: 32× worse
R3, GPT Sol / Astra	Source-index coding; low-rank birth operator	No	Dead (§6.1)
R4, GPT Sol	**Spherical-input reparameterisation (from a public repo)**	No	**Shipped (V31), −1.1 % raw**
R5, Sol / Astra	20 families (TAP/AMP, copula, TT-cross, SOS, …)	No	Dead
R7, Astra	Quartic κ_4 kernel; teacher-forced closure diagnostic	No	Kernel parked; the diagnostic shaped §4
R7, Grok	Monte-Carlo surrogate, "40×"	No	Measured ×1.0–2.2
R7, unknown pasted AI	"Closed-form" (1,1,1,1), "30×" MC	No	Wrong formula; claim baseless
R7–8, Fable (internal agent)	Error decomposition, FB scale, $K_3 \times C$	**Yes**	FB 1.25 shipped (#332110)
R9, GPT Sol	DHMP, FHQR-2, QCG ("promotion targets, not predictions")	No	Dead (36–40 %, 16 %, by arithmetic)
R10, several validators	Young-tier contraction search	Yes (counting scripts)	Confirmed 193 products; one answer's 49–55 u claim was an arithmetic error
R11, Copilot	Response-state closure	No	D21 35.9 % even from the true state
R11, SciMaster/Bohrium	Diagonal κ_3/κ_4 + low-rank C, "115 u at 1.18× chain MSE"	**Claimed, no run exists**	d3 error 90–99 %
R11, Astra	Channel-history state	Declined ("no update rule")	—
R12, Opus (internal agent)	Deep κ_4 is carried memory	**Yes**	Basis of §5.2
R13, unsigned	Skew-lift carrier	No	corr 0.03
R13, Astra	Dense pair core	**Yes**, self-rejected	corr −0.35 to −0.22
R14, unsigned	Response-cashout; exchange vertex	No	Dead (§6.1, §6.2)

Round, author	Proposal	In PDF?	Our verdict
R15, Astra	Skew energy	**Yes**, reproduced identically	+0.004
R16, unsigned	Signed bilinear quartic memory	Attempted; data host unreachable; declared none	0.33–0.39
R16b, Astra	Pair-innovation carrier	**Yes**, reproduced exactly	0.36–0.44
R17, Claude (ours)	Hub-pair C·T births	Yes	0.43–0.44
R17, unsigned	GHQ-6 hypertree bank	Script, own gate KILL	0.47–0.56
R17, SciMaster	Factorised curvature-birth carrier, "confirms ≥ 0.70, passes gate"	**Claimed; the script crashes** (`KeyError 'D21'`); patched, its own gate prints KILL	0.39 / 0.32 / 0.40
R17, SciMaster	Curvature-source factorised carrier, "exceeds 0.7 (PASS)"	**Claimed, no script or output**	The identical design measured 0.43
R18, unsigned	**HTQB star + path**	Algebra self-test only (n = 4)	**Borderline alone; star shipped as V56**
10 Oct consultation	κ_3 first-order audit: RLL term and the γ-patch	No (derivations only)	Both killed (§6.5 (a), (b))
10 Oct consultation	Cost-first κ_3 representation at 60 % of the cost	No; declared that no representation qualifies	Negative; slice-state obstruction (derived, §6.6 (g))
10 Oct consultation	Checkpointed birth archive (CBA-13)	No	×14.8 to ×125 (§6.6 (d))
11 Oct consultation	Network-conditioned solver routing	Yes (audit of our archived runs, with script)	Oracle ceiling 0.80 %; not built (§6.6 (g))
11 Oct consultation	Learned pair-memory student (PM364)	No	Implemented by us; killed (§6.7)

Two patterns stand out:

- **Renamed dead families.** Several proposals re-derived a dead family under a new name. The response-cashout propagator is keep-last-1; QCG is an n-term CP; FHQR is a short horizon.
- **Claims without runs.** Three documents reported passing measurements that no run supported.

Including the kill list and the project metric in every prompt is what filtered these out.

6.5 Hypotheses killed on 10 Oct

§4.10 places the remaining κ_3 error in the carried late-layer slice and in the d3 level. Three mechanisms for it, and for the FB 1.25 paradox of §2.4, were tested on 10 Oct. Each is killed as measured.

(a) A missing cubic-response C×κ_3 term ("RLL"; P). The hypothesis, from a 10 Oct outside-AI audit, was that the chain misses half of the term with one cubic and two linear Hermite vertices, at first order in the pre-activation κ_3 T.

Derivation (derived). With the K = 3 response h − E h = l u + q (u^2 − v) + r (u^3 − 3vu) (§5.8), the connected partitions give

$\kappa(R_i, L_j, L_k) = r_i\, l_j\, l_k\, [3\, C_{ij}\, T_{iik} + 3\, C_{ik}\, T_{iij}]$ (+ κ_5, dropped).

The proposal also contained a C_jk T_iii term. That partition is disconnected, because the block {i, i, i} does not meet {j, k}, so it does not belong in the joint cumulant. We tested both the connected form and the form as proposed.

Test (`astra_rll_test.py`). With the teacher's inputs at layer l, we map the term to its image dD on the (2,1) slice at l + 1. We compare dD with the chain's residual E = teacher − chain at l + 1 (the captures of §4.10), off the diagonal and on d3 separately. We report:

- $\beta = \langle E, dD\rangle / |dD|^2$;
- $\rho = \mathrm{corr}(E, dD)$;
- r_half = $|E - dD/2| \,/\, |E|$, the residual left after adding half of the term.

The pre-registered kill was $\beta \approx 0$ or negative, or $\rho < 0.1$ at $L \geq 9$. Values are given for the transitions $l \to l + 1$, l = 2–15, with $l \geq 9$ in brackets:

Form	Part	Size relative to the teacher's slice	ρ	r_half
Connected	Off-diagonal	4.1–9.5 %	−0.04 to +0.07 (−0.04 to +0.02)	1.08–1.42 (1.15–1.31)
Connected	d3	4.5–10.0 %	−0.24 to +0.07 (−0.24 to −0.12)	1.23–2.03
As proposed	Off-diagonal	8.5–13.7 %	−0.06 to +0.02	1.23–1.89
As proposed	d3	8.9–15.1 %	−0.23 to +0.01	1.54–2.95

Verdict: KILL.

- The connected term is comparable in size to the residual itself (4–10 % of the slice, against a residual of 4–7 %), yet it is uncorrelated with the off-diagonal residual ($\rho \approx 0$, β −0.04 to +0.02 at $l \geq 9$).
- On d3 it is anti-correlated at depth.
- Adding half of it makes the off-diagonal slice residual worse, by 15–42 % from L3 on (8 % at L2 → 3), and makes the d3 residual worse by 23–103 %.

This is consistent with the chain already carrying the term in its feedback legs.

(b) A feedback-scale confound ("γ-patch"; E). The same audit proposed the following explanation for the paradox of §2.4. The FB scale multiplies not only the linear K×C coefficient but also a quadratic Sym(Xt, P, Yt) cross term of the thin feedback legs, which grows as FB^2. FB 2.0 would then restore the linear term correctly but overshoot the quadratic one.

The patch (`experiments/variants/v56_fbg.py`) sets the linear coefficient to FB and multiplies only the Xt·Yt contributions by $\gamma = (FB_{base}/FB)^2$, which holds them at FB_{base}^2. We ran it on the 100-MLP harness (§8.2) on top of the V56 A-class estimator (v56As_fs, FB 1.25). No MLP failed in any run. Paired ratios are against the base:

Variant	Linear FB	Quadratic Xt·Yt term held at	Raw MSE	Score	Paired geo-mean ratio (90 % CI)	Wins
base (V56 A, shipped)	1.25	1.25^2	2.0194e-8	3.7586e-9	1	—

Variant	Linear FB	Quadratic Xt·Yt term held at	Raw MSE	Score	Paired geo-mean ratio (90 % CI)	Wins
Unpatched FB 2.0	2.0	2.0^2	2.0689e-8	3.8508e-9	×1.0244 (1.0211–1.0277)	11/100
γ-patch, FB 2.0	2.0	1.25^2	2.0686e-8	3.8522e-9	×1.0243 (1.0209–1.0273)	10/100
γ-patch, FB 2.0	2.0	1.0^2	2.0686e-8	3.8522e-9	×1.0243 (1.0212–1.0274)	11/100
γ-patch, FB 1.75	1.75	1.25^2	2.0403e-8	3.7995e-9	×1.0103 (1.0080–1.0124)	25/100

Verdict: KILL.

- Holding the quadratic term at 1.25^2, at 1.0^2, or letting it grow to 2.0^2 gives the same result (×1.0243–1.0244), so the quadratic cross term is inert.
- The loss follows the linear coefficient: ×1.0103 at 1.75 and ×1.0243 at 2.0. The linear K×C restoration itself hurts the final mean.

The paradox of §2.4 remains open, with this explanation removed.

(c) The $\kappa_4 \to \kappa_3$ feed channel (C/E). A 10 Oct consultation (`consultations/fable_2026-10-10_k3_design.md`) proposed the following. Among the first-order terms of the all-distinct post-ReLU κ_3, the only one whose input the chain lacks is the feed g_c Φ_i Φ_j κ_4(z_m)_ccij from the pre-activation κ_4 at a quadratic vertex c (derived in the design note). According to the design note's reading of the code, the chain computes this feed from its memoryless κ_4 model (§1.2), whose only pattern is C and which does not depend on the vertex neuron c. The design would replace it with a carried cubic κ_4 (star + C·T, §5.8) in thin legs.

Ceiling (`fable_k3feed_ceiling_2026-10-10.py`; Kaggle CPU in five parts; N = 10^6 exact samples of MLP #0; targets L9–16; `k3feed_mlp0_part{1..5}.json`). The script forms several versions of the feed read on the (2,1) slice at t:

- **exact:** the exact feed of every vertex;
- **bornJ4 / bornJ6:** only the κ_4 born within J layers, which is the ceiling of any carrier with J-layer births;
- **star_J4 / star_J6:** the Gaussian-star-fed feed in closed form.

The sampled C·T part of the design's representable feed was not run. As a check, the feed from the Gaussian layer z_1 sits at its own noise level (0.13–0.18 against noise 0.14–0.20).

Quantity, L9–16	Value
Exact feed size, off-diagonal, relative to |Y21|	22.5–24.2 %
Exact feed size, diagonal, relative to |d3|	23.3–23.6 %
Split-half reliability of the exact feed, off-diagonal / diagonal	0.871–0.966 / 0.939–0.981
Sampling noise of the exact feed, relative to |Y21| (derived)	3.3–6.0 %
bornJ6 against exact, cross-half corr, off-diagonal / diagonal	0.871–0.960 / 0.938–0.976
bornJ4 against exact, same	0.861–0.949 / 0.932–0.969
star_J6 against bornJ6, same	−0.871 to −0.921 / −0.927 to −0.956
star_J6 size relative to |Y21|	12.9–15.9 %

The pre-registered validity sub-gate, a split-half reliability of at least 0.90 at every target, is not met at L9 (0.871) or L10 (0.894).

End to end (`experiments/variants/r20_feed_oracle.py`). The variant removes the chain's own feed at the slice readout in a window and adds a feed read from file. It runs on the plain chain of §4.8, MLP #0. Hook validity: the plain configuration reproduces 2.2732e-8 exactly (`teacher_eval_feedor1.json`). The util of these runs is not comparable with the chain's, because the hook adds a second readout.

Feed in the window	Window	Final MSE	Ratio
None (chain feed removed)	L9–16	1.715e-7	7.54
None	L1–16	2.724e-7	11.98
Exact (sampled)	L9–16	2.535e-8	1.115
bornJ6	L9–16	2.955e-8	1.300
bornJ4	L9–16	4.409e-8	1.940
star_J6	L9–16	4.783e-7	21.0
star_J6, sign flipped	L9–16	4.255e-8	1.872
Chain's own feed × 0.80 / 0.90 / 0.95	L9–16	2.777 / 2.371 / 2.284e-8	1.222 / 1.043 / 1.0045
Chain's own feed × 1.00 (dumped and re-injected)	L9–16	2.2725e-8	0.9997
Chain's own feed × 1.10 / 1.20	L9–16	2.487 / 3.011e-8	1.094 / 1.324

The removal runs are in `teacher_eval_feedor2.json` and the injections in `results/summaries/oct10/feedorE2E_{3,4,5,6}.json`. The pre-registered gates were: exact ≤ 0.80 PASS and > 0.92 KILL; bornJ6 ≤ 0.85 PASS and > 0.95 KILL. For reference, the teacher's κ_3 in L9–16 is worth about ×0.69 (§4.10).

Slice-level attribution (measured; `results/summaries/k3feed/feed_attribution.json`, script `experiments/probes/k3_error_profile.py`).

- At L9–L15, the chain's own feed correlates +0.95 to +0.97 with the exact feed. The exact feed is 12–26 % larger.
- The correction (exact − chain feed) correlates +0.41 to +0.59 with the chain's slice residual.
- Yet replacing the chain's feed with the exact one still increases the slice residual: from 5.2–6.2 % to 6.0–6.8 % of |Y21|.
- The best scalar rescaling of the chain's feed (β = +0.03 to +0.08) lowers the slice residual by only 0.6–3 %.

Reading:

- **The feed is large, and the chain needs it.** It is 22–24 % of |Y21|, and removing it costs ×7.5 (L9–16) or ×12 (all layers).
- **The chain's feed has the right shape and the wrong size**, and the correction points along the residual.
- **Every injection is nevertheless worse than the base, the exact feed included** (×1.115, KILL by its own gate). The chain's feed sits at a sharp optimum at scale 1.0: −5 % costs 0.45 % and ±10 % costs 4.3 % / 9.4 %. Scaling it up by 10 % hurts, although the exact feed is 12–26 % larger.
- **The sampled feed is noisy at the level that matters.** Its sampling noise, 3.3–6.0 % of |Y21|, is comparable to the chain's whole slice residual at L9–16 (about 5–7 %, §4.10). This would explain why the exact feed raises the slice residual although its correction is correlated with it.

- **The Gaussian star has the opposite sign to the sampled born feed** (corr −0.87 to −0.92). Neither sign helps: ×21.0 as computed and ×1.872 flipped. The sign is physical, not a convention error (measured 11 Oct on the κ_4 diagonal of the §5.8 ceiling run, $2{\cdot}10^6$ samples of MLP #0, ages 1–6 summed at L9–16; `experiments/probes/star_sign_check.py` → `star_sign_check.json`):

- **No convention error.** The closed-form star has the sign of the sampled cubic-vertex birth Cu1, of which it is the Gaussian part: corr +0.97 to +0.98. It is negative at every neuron. Its coefficient $c = -(\mu/\sigma)\varphi/(6\sigma^2)$ is negative wherever $\mu > 0$.
- **The quadratic-squared birth dominates.** Q2 = 6κ(B, B, A, A) has the opposite sign (corr with the star −0.91 to −0.93) and 1.8–2.5× its norm. Its Gaussian part on the diagonal, $48\ \Sigma_a\ g_a^2\ v_a\ \bar{Y}_ai^2\ X_ai^2$ (derived: $\kappa(He_2, He_2, u, u) = 8\ v\ C\ C$ by Wick), is never negative.
- **The total follows Q2.** The exact births correlate −0.48 to −0.60 with the star alone, +0.64 to +0.74 with Q2 and +0.59 to +0.80 with star + Q2.

A star-only feed therefore carries the wrong-signed part of the births. The feed contraction is linear in the same κ_4, so we read the feed-level sign the same way (derived reading; a Q2-fed feed was not computed).

Verdict: KILL as measured. The experiment is decisive about the $N = 10^6$ estimate of the exact feed, not about the true feed. The ceiling is noise-limited: reliability is 0.87–0.97, and the validity sub-gate failed at L9–10. Bringing the feed noise well below the residual would need about 10× more samples. With noise $\propto 1/\sqrt{N}$, that would put it at about 1.0–1.9 % of |Y21| (estimate). A definitive ceiling, and the sampled C·T part, remain **[OPEN]**.

6.6 Cost-side architectures killed on 10–11 Oct

The ledger of §3.6 puts 133 u of V56's 194 u into per-source families whose size grows with depth. On the 10 Oct leaderboard our raw MSE was already in the range of entries ranked 11–40, while our util was 0.187 against 0.10–0.14 at the top (§7.3). The tests below ask whether the per-source accumulation can be removed, shortened or deferred.

(a) Single-pool κ_3 ceiling (C). A carrier of fixed size would keep the whole κ_3 in one shared rank-r basis U, for young and old sources alike. Its best case is to project every leg of every source, $X \to UU^TX$, and read D21 and d3 from the projected legs. `experiments/probes/pool_ceiling.py` does this on the plain-chain captures of §5.8 (every source dense) at L11, L14 and L15. U is either the top-r eigenvectors of the chain's own covariance at that layer ("C basis"), or the top singular vectors of the mode-1 unfolding of κ_3 from a randomized range finder ("unfolding basis"). Errors are relative Frobenius errors of the readout against the unprojected sources (`pool_ceiling.json`):

Rank r	All sources pooled: D21 off-diagonal	All sources pooled: d3	Only sources older than 5 layers pooled: D21 off-diagonal
128	19.6–28.1 %	13.2–20.5 %	6.2–6.6 %
256	11.4–17.3 %	7.2–11.2 %	2.4–2.5 %
384	6.6–10.8 %	4.1–6.9 %	1.2–1.3 %
512	3.4–6.3 %	2.2–4.0 %	0.6–0.7 %
768	0.2–1.0 %	0.1–0.6 %	0.03–0.09 %

All values use the C basis. The unfolding basis is worse at every rank: at r = 384 it gives 8.2–14.0 % for the whole κ_3.

Verdict: KILL. The pre-registered gate was a D21 error ≤ 2 % at L11–15 for some r ≤ 512. Only r = 768 passes, and a rank-768 pooled leg keeps three quarters of a dense leg's columns, so it saves little (derived). The table also shows where the obstruction sits. Pooling only the old sources at r = 384 costs 1.2–1.3 %, which is today's split. It is the **young** sources that no shared basis holds.

(b) DEJ-4: a deferred exact join (C). The deferred exact join (DEJ) would keep only the youngest source exact and fold every older source into a shared basis at once. The DEJ-4 falsifier runs the same script with the youngest k = 1, …, 4 sources exact and all older sources projected at r = 352, 448 or 512 (C basis; `oct10/dej_young{1..4}.json`). The older legs are projected from their exact values, so this is the best case of an exact join at age k + 1. D21 off-diagonal error at L11–15:

Youngest sources kept exact	r = 352	r = 448	r = 512
k = 1 (the DEJ design)	5.7–8.1 %	3.6–5.3 %	2.5–3.9 %
k = 2	3.7–5.2 %	2.4–3.3 %	1.7–2.4 %
k = 3	2.6–3.2 %	1.6–1.9 %	1.1–1.5 %
k = 4	1.9–2.2 %	1.2–1.4 %	0.8–1.0 %

Verdict: KILL for k = 1 (2.5–8 % against the 2 % gate). The settings that pass, such as k = 4 at r = 448 (1.2–1.4 %, d3 0.9–1.2 %) and k = 3 at r = 512 (1.1–1.5 %, d3 0.9–1.3 %), are not a new architecture. They are other values of V56's own age gate and old-basis rank, so they were graded end to end in (c).

(c) The age/rank grid on V56, 100 MLPs (K). The 100-MLP harness (§8.2) ran five settings of the young-age gate AGE_OLD and the old-basis rank R_OLD on the A-class V56 (`kaggle_agegrid_100mlp.json`). The base is the shipped AGE_OLD 5, R_OLD 352, run in the same harness for the γ-patch test (`base_fbg` in `kaggle_fbg_100mlp.json`). No MLP failed. The ratios are ratios of the harness's aggregate scores.

Variant	AGE_OLD	R_OLD	Raw MSE	Util	Score	Score ratio
base (V56 A, shipped)	5	352	2.0194e-8	0.1861	3.7586e-9	1
age4_r448	4	448	2.0090e-8	0.1920	3.8569e-9	×1.026
age4_r352	4	352	2.2271e-8	0.1781	3.9666e-9	×1.055
age4_r512	4	512	1.9849e-8	0.2021	4.0111e-9	×1.067
age3_r512	3	512	2.0569e-8	0.1988	4.0889e-9	×1.088
age3_r448	3	448	2.2268e-8	0.1863	4.1479e-9	×1.104

Verdict: KILL; the one-for-one law holds (§3.5).

- The best ceiling point, age4_r448, gains 0.5 % raw for 3.2 % more util.
- age4_r512 gains 1.7 % raw for 8.6 % more util.
- Saving cost at r = 352 (age4_r352, util −4.3 %) costs 10.3 % raw.

The ceiling ranking of (b) predicts the raw ranking, but every move away from the shipped setting costs more in one factor of the score than it gains in the other.

(d) CBA-13: a checkpointed birth archive (E). A 10 Oct consultation proposed deferring κ_3 work: run the early layers with a cheap κ_3, keep the births in an archive, and rebuild the exact sources at a checkpoint. The falsifier (`experiments/variants/r20_cba.py`) gives the proposal its best case. Inside a window of layers the closures see $\kappa_3 = 0$ (both D21 and d3), while the source bank keeps evolving **exactly**, so the exact κ_3 is available again for free after the window. In mode "zero", the births' feedback and repeated-index inputs see zero too. In mode "zero_closure", the births keep the true D21. The run uses the plain chain on MLP #0, with base 2.2732e-8 (`oct10/cba_{1,2,3}.json`):

Window with $\kappa_3 = 0$ in the closures	Final MSE	Ratio	Per-layer MSE at the window's last layer → at L16
L1–4	3.358e-7	×14.8	7.55e-7 → 3.36e-7
L1–8	1.448e-6	×63.7	1.98e-6 → 1.45e-6
L9–12	1.060e-6	×46.6	1.11e-6 → 1.06e-6
L5–12	2.333e-6	×102.6	2.43e-6 → 2.33e-6
L1–12	2.855e-6	×125.6	2.96e-6 → 2.85e-6
L1–12, births keep the true D21 ("zero_closure")	2.851e-6	×125.4	—

Verdict: KILL.

- **The damage does not recover after the window.** With the exact sources restored, the per-layer error falls by at most a factor of 2.2 by L16 (window L1–4), and by only 3.5–4.5 % for windows that end at L12. The mean and covariance error made inside the window is carried to the output (§4.1).
- **The damage enters through the closures.** Keeping the true D21 in the births changes nothing (×125.4 against ×125.6).
- **Early κ_3 is needed although it is not the lever.** Replacing the chain's κ_3 by the teacher's at L1–4 is worth nothing (×1.004, §4.10), yet removing it at L1–4 costs ×14.8.

(e) κ_3 level scaling, at zero FLOPs (K). §4.10 found that the chain's d3 level shrinks by 1.4 % at L16. The same harness (`r20_cba.py`, mode "scale") multiplies d3 and/or the D21 slice by a fixed factor in a window. The run uses the plain chain on MLP #0 and on the synthetic net, with bases 2.2732e-8 / 1.9201e-8. Results: `results/summaries/oct10/k3scale_{1,2,3}.json`.

Change	Window	MLP #0	Synthetic net
d3 ×1.02	L9–16	×1.032	×1.015
d3 ×0.98	L9–16	×1.038	×1.047
d3 ×1.01	L1–16	×1.005	×1.013
D21 ×1.02	L9–16	×1.057	×1.016
D21 ×0.98	L9–16	×0.984	×1.021
d3 and D21 ×1.02	L9–16	×1.074	×1.017

Verdict: KILL. The κ_3 level sits at its optimum. Raising d3 toward the teacher's level, the direction that §4.10 suggests, is worse on both networks. The one gain, D21 ×0.98 on MLP #0 (×0.984), does not transfer to the synthetic net (×1.021). This is one more instance of §7.5: a local correction toward the teacher loses end to end.

(f) The reference paper, read for unused ideas (Wu et al., arXiv:2605.05179). We checked the paper's K = 3 ideas against `submissions/v56As_fs/estimator.py` (`consultations/wu_paper_deep_read_2026-10-10.md`):

- **Already in V56:** merging factor groups that share two legs (V56's M block fuses $P{\cdot}s + 3A{\cdot}e + Z\,L^T$); the final-layer trim to the D3 diagonal (V19); the symmetric-output product that computes three of four blocks (`_sym_product`); a κ_4 diagonal carried with memory ($dG = (W \circ W)\, g_{prev} + \lambda$-term, with only the off-diagonal core regenerated); and the mixed covariance-closure terms (V56's K3-simple term tables are a superset of the paper's $k \le 3$ frontier).

- **Cost:** the paper's factorised K = 3 costs about 4,150 u at our size (derived from its Table 1), about 21× V56 (§3.6).
- **Killed: the trace projection of old κ_3** (the paper's rotation-invariant part). On the captures at L11, L14 and L15, the old sources carry 35–57 % of |D21|. Their trace projection $\mathrm{Sym}(h_1 \otimes I)$ leaves 74–75 % error in the old part, i.e. 26–42 % of the total D21, against a 2 % tolerance.
- **Untested:** the augmented K = 2 chain at about 88 u (the paper's estimate), which would sit under the 0.1 floor. The paper's own error theory makes it unlikely to reach our accuracy (estimate).

(g) Cost-first consultations (10–11 Oct; negative). We asked outside models for a κ_3 representation at 60 % of V56's cost (116.4 u) that keeps D21 and d3 within the 2 % channel tolerance. Three results came back.

1. **No qualifying representation.** The consultation declared that it had none, and it declined to present a measured-dead compression or slice recursion as one. The budget arithmetic is that of §3.6.

2. **A slice-only state cannot be exact to first order (derived in the consultation; we restate it because it is short).** Let S(T)_ik = T_iik be the (2,1) slice, and let $U = W^T \mathrm{diag}(w_1)$ be the transport. Take three distinct indices a, b, c and the perturbation $H = \varepsilon\, \mathrm{Sym}(e_a, e_b, e_c)$. Then S(H) = 0 and $\mathrm{diag}_3(H) = 0$, so the current mean, covariance and slice cannot see H. After one transport, with u_a the a-th column of U,

   $$S(U^{\otimes 3} H) = (\varepsilon/3)\,[\,(u_a \circ u_b)\,u_c^T + (u_a \circ u_c)\,u_b^T + (u_b \circ u_c)\,u_a^T\,],\quad \mathrm{diag}_3(U^{\otimes 3} H) = \varepsilon\, u_a \circ u_b \circ u_c,$$

   which is generically nonzero at first order in ε. A recursion on the slice alone therefore cannot be exact through first order for general third cumulants. A linear state that is closed under every invertible transport and returns every diagonal must be injective on Sym^3, which takes $n(n+1)(n+2)/6 \approx 1.8 \cdot 10^8$ numbers (compare §2.6). The argument does not exclude nonlinear encodings such as the source chain itself, restricted families, or approximate ones. Its measured counterparts are: 91–96 % of the chain's slice error sits in the all-distinct entries (§4.5); slice-to-slice recursion has R^2 0.03–0.05 (§6.1); and PM364 fails (§6.7).

3. **Routing between budgets has a ceiling of 0.8 %.** The proposal was to inspect each network's weights cheaply and select one of several complete chains of different cost. An audit of our archived 100-MLP runs (`Architecture Analysis/routing_oracle_audit.py` → `archived_routing_audit.json`, supplied with the consultation) covers 86 variants matched on the same 100 MLPs. A selector that knew every variant's true per-MLP score would improve on the best fixed variant (3.7586e-9) by only 0.80 %, before any selector overhead (0.86 % over the 10 V56 tuning variants). The cheapest archived variant costs 188.9 u, so the bank holds no cheap expert. A bank with a genuinely cheaper expert is untested. The router was not built.

6.7 Round 20: a learned pair-memory student, PM364 (11 Oct; killed)

§6.6 shows that the per-source state cannot be pooled, deferred or shortened. The remaining route is a **source-free** state of fixed size whose missing content is *learned*. The design of a pair-memory student came from an 11 Oct consultation. We adapted it, implemented it (`experiments/student/`) and distilled it from V56.

Design. The student carries the mean, the covariance C and two n×n **pair memories** H_1 and H_2, with no per-ReLU sources. Per layer, with w the layer's weight in the row convention z = h w:

1. **Transport.** The mean and C are transported as usual. Each memory is transported by the restricted cubic operator $P_k = (w \circ w)^T H_k w$, which is the exact image of a κ_3 supported on its slice entries.
2. **Readout.** D21 = offdiag(P_1) + O, where $O_{ij} = o_i (P_2)_{ij} - o_j (P_2)_{ji}$ is an explicitly odd correction. d3 uses the exact restricted diagonal rule $(1 + g_i) \cdot [3((P_1)_{ii} - c_i) + c_i]$, with $c = (w \circ w \circ w)^T \mathrm{diag}(H_1)$. V56's dead-column rule is applied.

3. **κ_4 inputs.** Exactly V56's: the memory-carried diagonal, the adaptive λ, and the V55 pattern map.
4. **Closure.** V56's K3-simple closure, ported exactly. It returns the post-ReLU mean, variance, d3, κ_4, covariance K11 and slice K21.
5. **Memory update.** The post-ReLU slice is the closure's own output, P_full = K21 + diag(K3), which already contains the births, the $\kappa_4 \to \kappa_3$ feed and the linear dressing. The memories are updated as $H_1 \leftarrow$ P_full + $(\rho - 1) \odot L(H_1) + \eta \odot L(H_2)$ and $H_2 \leftarrow (\text{P_full} - \text{P_full}^T)/2 + \eta \odot L(H_1) + \rho \odot L(H_2)$, with L(H)_ij = $w1_i^2$ w1_j H_ij.

The four heads ρ, η, o and g are quadratic in 12 bounded per-neuron features, giving 4 × 91 = **364 parameters**. They are initialised at zero, so the zero model is the pure slice recursion.

Engineering checks (measured).

- **Closure port.** With teacher-forced inputs, the PyTorch port of the V56 closure matches the chain to about 5e-8 relative error in the post-ReLU mean per layer, and to 4e-6 relative in the final prediction (MSE 1.5e-11).
- **flopscope version.** `pm_flops.py` matches the PyTorch version to 2.8e-7.
- **Cost (estimate).** It issues about 9.7k operations. Extrapolated from n = 128, it costs about 93 u, below the 102.4 u floor. A successful student would therefore score about raw MSE × 0.1 (derived), against V56's × 0.186.

Data and training.

- **Labels.** 768 self-generated He networks (W ~ N(0, 2/n), float32, seeded with numpy `default_rng`) were labelled by one run each of the frozen V56 with a dump hook (`v56_dump.py`, `teacher_dump.py`). The seed ranges are 100000–100511 (train), 200000+ (validation) and 300000+ (test). The labels are V56's own per-layer quantities, so the student is distilled to V56, not to the Monte-Carlo truth.
- **Training.** `train_pm.py` uses AdamW with warm-up, cosine decay and gradient clipping. Every step is a free-running 16-layer rollout from layer 1, with no teacher forcing. The loss weights the final-layer MSE, the means, sampled covariance and D21 rows, d3 and κ_4 terms, with the D21 and d3 terms weighted 4× more at L9–16 than at L1–8.
- **Hardware.** One T4 GPU: 0.95 s forward and backward per network, 7.5 GB.

Result (seed 11; 448 training networks; 64 held-out networks from the training seed range, never trained on; 16 epochs; `results/summaries/student/pm364_s11_history.json`):

Held-out quantity	Zero model (epoch 0)	After 16 epochs
Final-layer MSE to V56	9.64e-7	9.56e-7 (−0.9 %)
D21 relative error, L2	57.6 %	57.6 %
D21 relative error, L3–16	63–70 %	63–69 %
d3 relative error, L2–16	25–39 %	25–39 %
Training loss (epoch 1 → 16)	747.5	743.6 (−0.5 %)

A second optimisation seed (23; same data and protocol; `results/summaries/student/pm364_s23_history.json`) reproduces this: held-out MSE to V56 goes from 9.64e-7 to 9.55e-7, and the D21 error stays at 57.6 % at L2 and 67–70 % at L15–16.

The pre-registered gate was a held-out MSE to V56 ≤ 1e-9 and D21 and d3 errors ≤ 2 %. *Verdict: KILL* (both seeds). The student's MSE to V56 is about 50× V56's own error against the truth (derived). The test split was

not evaluated after the kill.

Interpretation. The D21 error at L2 is 57.6 % at the first transport and is not moved by training. At L1 the memory holds only the closure's slice. The share of the next slice that the transport would draw from the all-distinct post-ReLU κ_3 entries is therefore missing, and no per-row head (a function of per-neuron features) can rebuild it. This agrees with:

- the first-order obstruction for slice-only states (§6.6 (g));
- the location of the chain's error in the all-distinct entries (91–96 %, §4.5);
- the age-memory table of §4.5, where keeping only ages ≤ 2 leaves 84 % D21 error.

A source-free state must carry all-distinct content in some form. No fixed-size form we know does so at the 2 % tolerance (§6.1, §6.6 (a)).

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7. What a better method must satisfy

7.1 Structural constraints (from §2 and §4–§6)

1. **Keep the κ_3 chain at full fidelity.**
   - The mean closure needs κ_3 to about 1 % (§5.6) and D21 to about 2 % (tolerance law).
   - Cheap κ_3 cannot be bought back with κ_4 (§5.1).
   - The κ_3 memory is ≥ 10 ages long, coherent, and not low-rank (§4.5).
   - No compressed linear κ_3 state is closed under transport (§2.6), and none of the more than 20 κ_3 representations of §6.1 came within 5× of its gate; the later pooled and deferred designs of §6.6 missed by about 1.25–3×.
2. **Carry odd information.** Any closure built from even statistics of the weights loses the deep κ_3 entirely (§2.2).
3. **Deliver the κ_4 pattern, not the level, on decisive neurons.**
   - The pattern alone is worth 108 % of the κ_4-oracle gain.
   - In the birth families we measured, reaching correlation ≥ 0.70 at L12 / L15 / L16 needed births that contain **both the quadratic and the cubic Hermite response** of each ReLU.
   - These births must cover at least ages 1–4, and ideally 1–6 (§5.4, set C).
   - Second-order births of any form are capped near 0.45–0.55 in our ceiling model.
   - The V56 star plus a κ_3-dressed C·T readout meets this (ceiling 0.82–0.85), and the chain's own legs already carry its inputs (§5.8). It must enter as a feature of a rescaled map, because its level is near zero.
4. **Improve the κ_3 input first.**
   - The teacher's κ_3 slice and d3 are worth more than the κ_4 diagonal on both networks (×0.607 against ×0.686, and ×0.716 against ×0.885), and the two together give ×0.272 (§4.8).
   - A new κ_3 representation needs at least the cubic response of each ReLU: quadratic births make 10–21 % D21 and d3 error in one step, against a 2 % tolerance (§4.9).
   - The target is the carried late-layer error: net of teacher noise, 5–7 % in the slice at L8–16 and a 1.4 % shrink of the d3 level at L16. Teacher κ_3 helps only at L9–16 (§4.10). A missing $C\times\kappa_3$ term, the feedback scale and the $\kappa_4\to\kappa_3$ feed (at $N = 10^6$) are ruled out as the cause (§6.5). A plain rescaling of the level is ruled out too (§6.6 (e)).
5. **Carry the all-distinct κ_3 content, at every layer, from the first layer on (§6.6–§6.7).**

- A state built from the slice alone cannot be exact to first order (derived, §6.6 (g)). A learned 364-parameter student on such a state stays at 58–70 % D21 error (§6.7).
- One shared basis cannot hold the young sources: the whole κ_3 needs rank 768 for ≤ 2 % (§6.6 (a)), and keeping only the newest source exact leaves 2.5–8 % (§6.6 (b)).
- κ_3 cannot be skipped early and restored later: a zero window costs ×15 to ×125, and the damage persists after the window (§6.6 (d)).

7.2 Budget constraints

- **Ops and memory.** At most about 1,000 extra warm-call ops above the B-class base, which keeps the total ≤ ~18k for nomination. Total memory under 8 GB, measured under a subprocess runner. No host synchronisations.

- **FLOPs: break-even arithmetic.** A carrier worth 74 % of the oracle gain would cut raw to about ×0.78 (estimate, from the 50 %-blend oracle). On our ≈ 192 u:

  - at +30 u, the score improves by about 10 % (estimate);
  - at +60 u, it is about 2 % worse (estimate).

  So the carrier must cost ≤ ~30 u. - **Starting point.** The V56 star already supplies part of the cubic signal at 0 u. A new carrier is worth only what it adds on top of V55 + star: 0.65–0.74 leave-one-layer-out, measured with teacher μ and C on one network.

7.3 Score targets (raw × util arithmetic)

Target	Requirement
3.0e-9	κ_4-oracle raw (1.61e-8) at today's util of ≈ 0.187, or today's raw (1.95e-8) at util 0.154 (≈ 158 u)
Rank 12 on the 8–9 Oct board (2.2e-9)	raw 1.18e-8 at util 0.187, or 1.47e-8 at 0.15
Leader class (10 Oct board)	raw 1.14–1.65e-8 at util 0.106–0.144

Cost alone cannot get there. Today's raw at the 0.1 floor would give 1.95e-9. That needs a 47 % cost cut, which we did not find inside the sum-of-sources family (§2.6, §6.1); e30_a, also a sum-of-sources chain, is 14 % cheaper than ours (158.8 u against 184.1 u, §3.9).

The 1024-wide leaders therefore pair a raw MSE 1.2–1.7× lower with a cost 1.3–1.8× lower (derived from the 10 Oct board). Their mechanisms are not public.

The 10 Oct board (`consultations/competition_scan_2026-10-10.md`).

- The six highest entries have raw MSE 1.14–1.65e-8 at util 0.106–0.144.
- Our leaderboard entry shows raw 1.97e-8 at util 0.187 (rank 42 on 10 Oct). The row links #334805, which displays the same 3.7e-9 as our then-best #334870 (§8.4). Entries ranked 11–40 have raw MSE comparable to ours (1.7–2.5e-8 among those listed).
- The entry at rank 11 has V29-level raw MSE (2.10e-8) at util 0.102, the floor.
- At our raw MSE, util 0.10 would score about 2.0e-9 and util 0.14 about 2.8e-9 (derived).

On that board the leaders are ahead of us on both factors, and the entries ranked 11–40, whose raw MSE is comparable to ours, differ from us mainly in cost. This motivated the cost-side tests of §3.6 and §6.6–§6.7. Within the sum-of-sources family, they confirm that cost and accuracy trade one for one (§6.6 (c)). Every

source-free or pooled alternative we could build fails its 2 % D21 gate, by factors from about 1.25 (DEJ at r = 512) to about 35 (PM364).

7.4 Open design target [OPEN]

The design target is **a factorised κ_4 carrier whose births come from the cubic (and quadratic) Hermite response of each ReLU over ages 1–6, read through the existing κ_3 source legs, for ≤ 30 u and ≤ ~1,000 ops.**

Its ceiling is 0.80–0.85 (set C, ages 1–6) and 0.68–0.74 for ages 1–4. The value of reaching the oracle is about 3.0e-9 at the current cost (estimate).

Status on 10 Oct (§5.8). The readout side of this target now exists:

- star + C·T has a ceiling of 0.82–0.85, and the chain's own κ_3 legs reproduce C·T with fidelity 0.993–0.996;
- through an oracle affine map it is worth ×0.773 end to end;
- the age-collapse identity and A/P anchor grouping make it exact at two dense products per source and target.

The missing piece is cost:

- the eigen-sketch at r = 64 keeps ×0.865 at about 40 u (estimate), which is roughly score-neutral on V56's ≈ 190 u; at r = 128 it keeps ×0.843, with twice the probe's thin-product count (55.3 u against 27.6 u over targets 7–16);
- break-even needs ≤ ∼30 u at that accuracy, or more accuracy at the same cost;
- round 3 (§5.8) found such a point: eig r = 32 keeps ×0.874 at an estimated 18–20 u, about a 3–4 % score gain (estimate), with the metered op count still unmeasured.

No estimator implementation exists yet, and the increment on top of V56, with a map fitted offline and transferred, is untested. In parallel, the joint oracle (§4.8) makes a better κ_3 input the larger open lever. Its first candidate mechanism, the $\kappa_4 \to \kappa_3$ feed, is killed at N = 10^6 but is not ruled out at larger N (§6.5).

Status on 11 Oct (§3.6, §6.6–§6.7). The cost side has been mapped, and it is negative:

- the 133 u of per-source readout and transport cannot be pooled into one basis, deferred to a checkpoint or shortened in age without a one-for-one accuracy loss;
- routing between existing budgets is worth at most 0.8 %;
- a source-free learned student fails at the first layer, because its state has no all-distinct κ_3.

What remains open on the cost side is a representation that carries all-distinct κ_3 content at a cost that does not grow with the number of sources. We do not have a candidate.

Smaller open items:

- the cheap K_3×C term on the synthetic network (−4.2 % on MLP #0 at ≈ 13 u, unfinished);
- the HTQB path term end to end (+0.04–0.06 leave-one-layer-out, untested in the chain);
- the quartic kernel re-tuned jointly with λ and FB.

7.5 A lesson: local corrections against coupled errors

Again and again in this project, a change that makes one channel locally more correct made the final mean worse. All of the following were measured:

- **Full FB restoration** (§2.4) brings the slice closer to the derivation at L4–10, but the final mean gets 3.4 % / 3.1 % worse on the teacher pair. FB 2.0 is ×1.024 on 100 MLPs (§6.5).
- **The γ-patch** (§6.5), which keeps the linear restoration and removes its quadratic side effect, is no better (×1.0243).
- **Scaling the chain's $\kappa_4 \to \kappa_3$ feed** (§6.5) by ±5–10 % in either direction costs 0.45–9.4 %. This holds even upward, toward the exact feed, which is 12–26 % larger and whose correction correlates with the residual. Injecting the sampled exact feed gives ×1.115.
- **The analytic RLL term** (§6.5), of the right order and comparable size, makes the slice residual 15–42 % worse when half of it is added.
- **Single κ_4 inputs:** injected alone, the off-diagonal κ_4 slices give ×1.03–1.22 (§4.2), and the chain's κ_4 pattern at the teacher's level gives +21.5 % (§5.1).
- **The κ_3 level** (§6.6 (e)): the chain's d3 is 1.4 % too small at L16, yet raising it by 2 % at L9–16 costs ×1.032 / ×1.015 on the two networks. The one rescaling that helps on MLP #0, D21 ×0.98 (×0.984), hurts on the synthetic net (×1.021).

Only **joint** input oracles give large gains. The teacher's κ_3 and κ_4 together give ×0.272, against ×0.607 and ×0.686 alone (§4.8). In the channel oracle of §4.2 (base 2.290e-8), the true variance, κ_3 diagonal and κ_4 diagonal together reach the noise floor (×0.057).

The pattern is consistent with the chain's channel errors partly cancelling in the mean closure: correcting one channel uncovers the errors it was offsetting in the others. In the feed case, Monte-Carlo noise in the injected quantity is a competing explanation.

Direct one-step measurement (11 Oct; `experiments/probes/channel_cancellation.py` → `channel_cancellation.json`). Setup:

- **Network and layers:** MLP #0, every layer L2–16, against the $6 \cdot 10^7$-sample teacher.
- **Closure:** the Edgeworth mean closure to order 4. On the teacher's own inputs it reproduces the teacher's ReLU mean to 0.8–1.5e-5 (relative), against the chain's one-step error of 2.6–2.9e-4.
- **Chain inputs:** μ, var and d3 come from the frozen V56 chain. Its effective κ_4 diagonal is back-solved from its own post-ReLU mean, reproducing that mean to 1–4e-6; on the 5 % of neurons where κ_4 has no effect on the mean, it is set to the teacher's.
- **Swaps:** each fresh input (var, d3, κ_4 diagonal) is replaced by the teacher's at the chain's own μ, and the change in the post-ReLU mean is that channel's error.

Findings (measured):

- **The fresh channel errors cancel.** Their norms add up to 1.9–2.6× the norm of their joint error. The cancellation ratio $\|\Sigma F\|^2 \,/\, \Sigma\|F\|^2$ is 0.41–0.54 at L3–L10 and 0.62–0.83 at L11–16 (1 would mean no cancellation). The main pair is var with d3: cosine −0.40 to −0.83 at L2–L11 and −0.17 to −0.31 at L12–16. d3 with κ_4 is −0.19 to −0.28 from L5 on.
- **Correcting one channel alone makes the step worse.** The fresh error grows when var alone is corrected at L2–L9 (×1.01–1.31), and when d3 alone is corrected at L4–L10 (×1.03–1.13). The latter matches the end-to-end cost of raising d3 (§6.6 (e)). Correcting var and d3 together always helps (×0.56–0.79). κ_4 alone helps (×0.50–0.91).
- **Most of the one-step error is carried, not fresh.** With the true var, d3 and κ_4 at the chain's μ, 23–55 % of the error remains, and that part comes from the mean error carried in from earlier layers.

Limits:

- **Noise.** The teacher's own closure error, which bounds its sampling noise, is at most 15 % of the fresh error in norm. That moves a fix-one ratio by at most about 1 %, so all the excesses above except var at L9 (×1.01) are larger than the noise.
- **Scope.** This is a one-step measurement at the closure. How the cancellation propagates through depth to the final mean was not measured; the end-to-end reading remains a conjecture.

The practical lesson for moment-chain estimators is narrower, and it rests only on the measurements above:

- a local test (a slice residual, a one-step error, a pattern correlation) is a necessary gate, but it is not a sufficient one;
- candidates must be promoted on the end-to-end mean;
- a single-channel correction should be expected to need joint re-tuning of the channels it is coupled to (λ, the FB scale, the feed scale).

FB 1.25 is the only correction in this list that gained (§2.4), and it did so at a quarter of the derived restoration.

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8. Reproducibility

8.1 Versions and data

- Software: whestbench 0.16.1, flopscope 0.12.1 (grader on 0.12.3 from 17 Oct, including #337457; identical billing, §3.9).
- Dataset revision: v2-phase2.
- Every graded number in this paper carries its AIcrowd submission id. The shipped estimators are in `submissions/<variant>/estimator.py`, each with a lineage header naming every change; for example, `submissions/v56As_fs/estimator.py` (#334870) and `submissions/v56s_fs/estimator.py` (#334869).

8.2 Harness

The 100-MLP paired harness runs on the public mini split with its 10^9-sample truth, using Kaggle CPU runners A–E (`kaggle/runners/`). It is compared with `experiments/probes/kaggle_pair_compare.py`, which reports:

- the geometric-mean ratio;
- wins;
- a bootstrap 90 % interval;
- the score MSE × max(0.1, util);
- the residual profile.

The base is bit-identical across runners, and paired standard errors are 0.01–0.15 %.

Calibration against the grader:

Change	Harness	Graded
V55 B / A	Projected 3.8764e-9 / 3.8562e-9	3.8762e-9 / 3.8582e-9 (within 0.06 %)
Transport join	×0.959	−3.4 %
R_OLD 352	×0.9930	−1.1 %
v52 safe B / A vs v51	+2.1 % / +0.9 %	+1.9 % / +0.8 %

Change	Harness	Graded
V56 synthetic-fit map	×0.9964	−0.29 % (B) / −0.21 % (A)
Star ×1.25	×0.9972 additional	No additional gain

Up to 16 Oct every predicted sign matched the grade (in §3.9, c1's non-significant joint ×1.0035 graded −1.1 %, and N2's ×0.935 graded ×0.951). For single changes worth more than about 1 %, the predicted size was accurate to within 0.05–0.7 percentage points. For gains under 0.5 % the size was unreliable: the graded gain came in at roughly 0 to 2× the predicted one.

The harness measures residual time but does not enforce the cap. Its in-process mode did not reproduce the V56 memory failure; only `whest run --runner subprocess` did (§3.4).

8.3 Scripts and result files

Result files are in `results/summaries/` unless a path is given.

Section	Scripts → results
§2.5 identities	`experiments/probes/r17_curvature_identities.py` → `r17_curvature_identities.json`
§3.1 ledger	`experiments/probes/m1_matmul_ledger.py` → `m1_matmul_ledger.json`, `m1_matmul_ledger_callers.json`
§4 error map	Teacher scripts in `kaggle/k4/` (pre- and post-ReLU teachers), the error-map probes listed in `docs/build_notes.py`, and `opus12_errmap.py`
§4.8 joint oracle	`experiments/variants/r20_joint_oracle.py` → `r20_joint_oracle.json`
§4.9 one-step channels	`experiments/probes/r20_onestep_channels.py` → `r20_onestep_channels.json`
§4.10 κ_3 error profile and windows	`experiments/probes/k3_error_profile.py` → `chain_k3_error_profile.json`, `teacher_k3_noise_estimate.json` (and the §6.5 (c) slice attribution `k3feed/feed_attribution.json`); κ_3 windows through `r20_joint_oracle.py` (Kaggle wrappers `kaggle/k3loc/esfer_{1,2,3}/run.py`) → `oct10/k3loc_{1,2,3}.json`
§5.1 oracles	`experiments/chain/k4_oracle_run2.py` → `k4oracle2.json`, `k4oracle2_joint_*.json`, `k4oracle2_target_*.json`
§5.2 memory	`opus12_k4_carry.py`, `q18_k4_horizon.py`, `opus12_q18_noise.py`
§5.3 V55	`experiments/variants/v55_k4c.py`, `experiments/probes/k4_pattern_features.py` → `k4c_fit.json`, `k4_pattern_features*.json`, `teacher_eval_v55_k4c.json`, `kaggle_v55_deep.txt`, `kaggle_v55_sweep1.txt`, `kaggle_v55_sweep2.txt`
§5.4 ceilings	`r16_sbqm_ceiling.py`, `r16b_astra/`, `experiments/probes/r17_hermite_family_ceiling.py` → `data/kaggle_out/r17_ceiling/r17_hermite_family_ceiling.json`, `r17_hermite_family_ceiling_B.json`, `data/kaggle_out/r17_ceiling_c/r17_hermite_family_ceiling_C.json`
§5.5 HTQB and V56	`r18_htqb/`, `r18_htqb_combo.py` → `round18_htqb.json`, `r18_htqb_combo.json`; `experiments/variants/v56_star.py` → `k4s_fit.json`, `kaggle_v56_tuning.txt`
§5.6 tree test	`r19_tree_ceiling.py`, `r19_tree_rescale.py` → `r19_tree_ceiling.json`, `r19_tree_rescale.json`

Section	Scripts → results
§5.8 C·T channel	`fable_ceiling_2026-10-09.py` → `fable_ceiling_2026-10-09.json`; oracle affine fits and E2E through `r20_joint_oracle.py` (Kaggle wrapper `kaggle/fable_e2e2/kernel/run.py`) → `fable_affine_fits.json`, `fable_affine_e2e.json`; `fable_component_2026-10-09.py` → `fable_component_2026-10-09.json`; `acds_ct_probe.py`, `acds2_ct_probe.py` → `acds_ct_probe.json`, `acds2_ct_probe.json`, `acds_e2e.json`; layer windows (Kaggle wrappers `kaggle/acds/ctwin_{1,2}/run.py`) → `oct10/ctwin_{1,2}.json`; round 3: `acds2_ct_probe.py --lite 1 --ranks 32,48,64` (wrapper `kaggle/acds/kernel/run.py`) → `acds3_ct_probe.json`, E2E (wrapper `kaggle/acds/e2e3_kernel/run.py`) → `acds_e2e3.json`
§6.5 (a) RLL	`experiments/probes/astra_rll_test.py` → `astra_rll_test.json`
§6.5 (b) γ-patch	`experiments/variants/v56_fbg.py` (runner sets `kaggle/fbg/set_{1..5}`) → `kaggle_fbg_100mlp.json`; per-MLP outputs `data/kaggle_out/fbg_{1..5}/`, compared with `kaggle_pair_compare.py --ref base_fbg`
§2.2 parity theorem and floor	`experiments/probes/parity_check.py` → `parity_check.json`
§3.6 V56 call-site ledger	`experiments/probes/site_ledger.py submissions/v56As_fs/estimator.py <weights.npz>` → `site_ledger_v56As_fs.json`
§3.6, §6.6 (f) Wu et al. read	`consultations/wu_paper_deep_read_2026-10-10.md` (cost table, idea-by-idea check against V56, trace-projection kill; no separate result file)
§6.5 (c) star sign	`experiments/probes/star_sign_check.py data/kaggle_out/fable_ceiling_all/fable_ceiling_all.npz` → `star_sign_check.json`
§7.5 channel cancellation	`experiments/probes/channel_cancellation.py` → `channel_cancellation.json`
§3.8 knob sweep	`kaggle/tune/`, `kaggle/tune2/` (variants, runners, `local_check.py`) → `kaggle_tune_100mlp.json`, `kaggle_tune_compare.txt`, `tune_local_check.json`; per-MLP outputs `data/kaggle_out/tune_{1..10}/`, `tune2_2/`
§3.7 E-series	github.com/hamidhosen42/ARC-White-Box: `experiments/results.csv`, `experiments/submissions.csv`
§3.9 final days	`kaggle/x8*`, `kaggle/x10/`–`x14/`, `experiments/variants/e30k_x8.py`–`e30k_x14.py`, `submissions/e30*`, `submissions/e299safe/` → `data/kaggle_out/{fs123_*,ledger,exp5_*,e30k31_*,e30tune_*,x8_*,x8c_*,x10_eval,x11_*,x12_*,x13_*,x14_*}`
§3.8 E299-B	`kaggle/e299b/warm_check.py`, `mem_check.py` → `e299b_warm_check.json`, `e299b_mem_check.json`; `submissions/e299b/`, `submissions/e299b1/`
§6.6 (a) single pool	`experiments/probes/pool_ceiling.py --layers 11,14,15` → `pool_ceiling.json`
§6.6 (b) DEJ-4	`pool_ceiling.py --ranks 352,448,512 --young k` (Kaggle wrapper `kaggle/dej/kernel/run.py`) → `oct10/dej_young{1..4}.json`
§6.6 (c) age/rank grid	`submissions/v56As_fs/estimator.py` with `V21_AGE_OLD` / `V21_R_OLD` (runner sets `kaggle/agegrid/set_{1..5}`, `variants.json`) → `kaggle_agegrid_100mlp.json`; base `base_fbg` in `kaggle_fbg_100mlp.json`
§6.6 (d) CBA-13	`experiments/variants/r20_cba.py` (`CBA_MODE=zero` / `zero_closure`, `CBA_LMIN`, `CBA_LMAX`; wrappers `kaggle/cba/k1`–`k3`) → `oct10/cba_{1,2,3}.json`
§6.6 (e) κ_3 level scaling	`r20_cba.py` (`CBA_MODE=scale`, `CBA_D3S`, `CBA_D21S`; wrappers `kaggle/k3scale/k1`–`k3`) → `oct10/k3scale_{1,2,3}.json`

Section	Scripts → results
§6.6 (g) consultations	`Architecture Analysis/Astra_Cost_First_Architecture_Assessment_2026-10-10.pdf` (slice-state argument), `Architecture Analysis/Astra_Conditional_Budget_Architecture_2026-10-11.md`, `Architecture Analysis/routing_oracle_audit.py` → `Architecture Analysis/archived_routing_audit.json`
§6.7 PM364 student	`experiments/student/`: `pm_student.py` (PyTorch student), `pm_flops.py` (flopscope version), `v56_dump.py` + `teacher_dump.py` (V56 labels; Kaggle label kernels `kaggle/student/lab_*`), `train_pm.py` → `student/pm364_s11_history.json`, `student/pm364_s11_theta.npy`
§6.5 (c) $\kappa_4 \to \kappa_3$ feed	Design `consultations/fable_2026-10-10_k3_design.md`; ceiling `experiments/probes/fable_k3feed_ceiling_2026-10-10.py` (CPU wrappers `kaggle/k3feed/cpu_{1..5}/`) → `k3feed/k3feed_mlp0_part{1..5}.json`; E2E `experiments/variants/r20_feed_oracle.py` (wrappers `kaggle/feedor/k1–k6`) → `k3feed/teacher_eval_feedor1.json` (hook validity), `k3feed/teacher_eval_feedor2.json` (removal), `oct10/feedorE2E_{3,4,5,6}.json` (injections)

The kill register is `docs/architecture_research/KILL_AUDIT.md` (27 Sep, 74 entries). The round verdicts are `consultations/round14_verdicts.md` through `round18_verdicts.md` and `consultations/new_architecture_step1.md`. The experiment queue is `docs/experiment_queue/queue.json`.

Unless a path is given, probe scripts are in `experiments/probes/`. All scripts will be released under MIT with the write-up after the competition closes.

8.4 Known inconsistencies in our records

- The κ_4-diagonal oracle appears as 1.600e-8 and 1.609e-8 in two runs.
- The κ_4-diagonal oracle ratio is ×0.702 in §4.2 (base 2.290e-8), ×0.686 in §4.8 (base 2.273e-8, every layer) and ×0.700 in §5.8 (same base, layers 7–16 only). The bases and injection windows differ; none of the three is a typo.
- The chain's κ_4 correlation is 0.78–0.83 over all neurons but 0.26–0.34 on decisive neurons. Both figures are correct for their populations.
- Submission #332395 (last-block λ ×1.25, 26 Sep) graded 4.6768e-9 (raw 2.1682e-8), +1.0 % against v48 (#332391, 4.6294e-9); re-read from the grader on 21 Oct.
- The id of the failed v54A submission (7 Oct) is not recorded in our files.
- The chain's slice error is "6.7 % at L15" in §4.5 (base 2.290e-8) and 6.6 % at L15 in §4.10 (the §5.8 captures). Its d3 error is 3.2 / 3.8 / 4.6 / 4.7 % at L7 / L11 / L15 / L16 in §4.3 and 3.0 / 3.6 / 4.4 / 4.6 % in `chain_k3_error_profile.json`. The two come from different chain runs; both include teacher noise.
- Our working notes priced eig r = 32 at "≈ 18 u". The probe's thin-product count is 13.8 u (`acds3_ct_probe.json`), which excludes G_m and Γ. §5.8 therefore quotes 18–20 u as an estimate, scaled from the 40 u estimate for r = 64.
- The slice-level feed attribution of §6.5 (c) was first quoted for L10–L14 only (+0.96, 14–24 %, +0.45 to +0.58). The saved file `k3feed/feed_attribution.json` covers L9–L15: +0.95 to +0.97, 12–26 %, +0.41 to +0.59.
- The feed-oracle runs report util 0.3108–0.3425, because the hook performs a second slice readout. These values are not comparable with the chain's 0.2476.
- The κ_3 level-scaling ratios of §6.6 (e) were first read from the run logs; the result files were saved afterwards as `results/summaries/oct10/k3scale_{1,2,3}.json`.
- The PM364 engineering checks of §6.7 (closure-port parity 5e-8 / 4e-6 / MSE 1.5e-11, flopscope parity 2.8e-7, 9.7k operations, 93 u extrapolated) were measured in the 11 Oct session and recorded afterwards in `results/summaries/student/pm364_engineering_checks.json`.

- The V56 ledger counts 20,586 operations in the warm call (§3.6), against ~19.9k–20k quoted for the A class in §3.3. The ledger counts each outermost flopscope call as one operation on one network; the two counts are not from the same instrument.
- The ledger's 193.8 u is one network; the harness mean for the same estimator is util 0.1861 (≈ 190.6 u).
- On 16 Oct the board showed the team at rank 46 (tied) with 3.4e-9, linking #335834 (E296, 13 Oct), the earliest submission with that displayed score; #336377 (3.3946e-9) displays the same value.
- The leaderboard lists our entry at raw 1.97e-8, while the grader reported 1.950e-8 for #334870. Resolved 11 Oct: the board's row links #334805 (8 Oct; score 3.6829e-9, raw 1.9706e-8, util 0.1869). #334870 (3.6521e-9, raw 1.9497e-8) is graded in the same Phase-2 round (1429), as are #334869 and #334871. All four display as 3.7e-9. Entries with equal displayed scores share a rank on the board (rank 43 on 11 Oct, tied with another 3.7e-9 entry), so the board appears to keep the earlier submission (inferred, not documented by the organisers). On 20–21 Oct the board showed rank 34 (tied) with 2.8e-9, linking #337301; #337457 (2.7858e-9) displays the same value.
- The CBA ratios are quoted to one decimal from the files (×63.7, ×125.6, ×102.6); our working notes rounded them to ×64, ×125 and ×103.

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9. Disclosure of LLM use

Primary assistant. The main engineering and research assistant was Claude Code (Claude Opus 5 and 5.5), directed by the team. Apart from the public code we build on (504aldo, LordProcciuto, kai_leal_gonzalez) and the E-series layer (below), it wrote essentially all code in our repository: every estimator variant, oracle, probe, harness runner and Kaggle pipeline (including the 17–21 Oct table dump-and-refit kernels) and the PM364 student. The team chose what to build, ordered every run and reviewed the results. The new table features of §3.9 (N1–N4a) were proposed by Claude Code and selected by held-out measurement. Internal Claude subagents did specific derivations and measurement tasks; every number they reported was measured. These subagents were:

- Fable 5, an Anthropic Claude model run as a Claude Code subagent (Rounds 7–8: the all-distinct κ_3 decomposition, the FB scale and the $K_3 \times C$ term; the parity argument; the 9 Oct C·T/LKV design of §5.8 and the 10 Oct $\kappa_4 \to \kappa_3$ feed design of §6.5 (c));
- Opus (Round 12: κ_4 as carried memory);
- Sonnet (Round 10: the young-tier count).

On 19 Oct ten further Claude subagents reviewed outside documents, the forum and our records (§3.9); their joint judgement that a top-10 jump was infeasible in the time left was an AI judgement, not a measurement. On 21 Oct five Claude subagents reviewed this paper and the main write-up for numbers, consistency, rules and attribution; the team checked each finding before accepting an edit.

Outside models consulted. We sent prompts to:

- GPT-5.6 (Sol, Terra);
- GPT Astra 6;
- Gemini 3.1 Pro, Grok, DeepSeek and Copilot;
- SciMaster/Bohrium;
- several unsigned answers.

Origin of shipped ideas.

Shipped idea	Origin
Spherical-input reparameterisation	GPT Sol R4, pointing to a public repository
HTQB star	Round-18 proposal, built on our Round-17 ceilings
Parity argument	Fable 5 consultation

Verification. Every outside claim was re-derived or re-measured by us before entering this paper. Several claimed measurements were found to be unsupported, and §6.4 says which. The 10–11 Oct consultations (§6.6 (g), §6.7) ran no network inference. Their one numerical result, the routing ceiling, is an audit of our own archived run files. The PM364 student was implemented, trained and evaluated by us. No leaderboard feedback entered any fitted coefficient. The V55 map was fitted against our own Monte-Carlo teacher for MLP #0 (the synthetic network held out); the shipped V56 maps (#334869, #334870) were fitted against our own teacher for the synthetic network. The refitted e30_a tables N2–N4a (§3.9) were fitted on public mini nets 0–69 and validated on 70–99. Discrete choices (corrector scale, A or B class, which variant to keep) were made on the harness and, in a few cases, between graded submissions.

Verified versus derived. The split of our mathematical claims into numerically verified, derived-and-consistent, derived-only and conjectural is given in the main write-up, §9.

E-series layer. The calibration and cost layer of §3.7 (E282–E299, including #336377) and its fits are mdhamidhosen's work in his own repository (github.com/hamidhosen42/ARC-White-Box; private until release, available to the organisers on request). This disclosure covers our repository; the E-series is covered by its author's own statement. The numbers in §3.7 are taken from his ledger and re-checked against the grader.

Text. Almost all of the text of this paper and of the main write-up was drafted by Claude (Claude Code) from our result files and notes, and revised by the team, which takes responsibility for every claim.

Licensing and attribution. Our own lineage (#334870 and earlier) starts from 504aldo's V29 (MIT), with its copyright header and LICENSE retained. The submission this write-up refers to, #337457, starts from LordProcciuto's e30_a (MIT; Copyright 504aldo and LordProcciuto) and uses kai_leal_gonzalez's two-term (3,1) closure (MIT). #337049 also uses e30_a's exact cost savings, fixed old basis and C_pre compaction, and kai_leal_gonzalez's closure at mix 0.25. The spherical-input idea is from barnobarno666's public repository (idea only). The E-series layer (§3.7) is mdhamidhosen's. §1.4 lists which parts of #337457 are whose; only the parts marked as ours there, and the analysis of §2–§7, are claimed.

References

1. Wilson Wu, Victor Lecomte, Michael Winer, George Robinson, Jacob Hilton, Paul Christiano. *Estimating the expected output of wide random MLPs more efficiently than sampling.* arXiv:2605.05179 (v2, 14 May 2026) (the K3 "simple" chain, §4.3, and the reference implementations used in §6.3).
2. 504aldo, *whest-p2-cumulant-k3* (GitHub, MIT licence): the V29 factorised third-cumulant chain with memoryless κ_4 regeneration, D21 feedback, an age-gated shared basis and pooled Strassen–Winograd. Forum post #18218. Finding F105 on the r = 1 κ_4 core.
3. WhestBench Phase 2 forum threads:
   - topic 18218 (504aldo chain);
   - topic 18216 (yujigon, zero-constant closure with CP κ_3);
   - topic 18214 (venture_agent, exact bivariate ReLU covariance);

    - topic 18200 (baghyeonsig, blended MC);
    - topic 18219 (nkosi_ndwandwe, MC economics);
    - topic 18199 (keenanpepper, no heavy tail across MLPs);
    - topic 18175 (Puffi, Phase 1 SSC stack).
4. barnobarno666, public Phase-2 repository, Phase 8 (spherical-input reparameterisation).
5. LordProcciuto, *e30_a*, github.com/siming767-creator/whestbench-lordprocciuto (MIT), and Phase-2 write-up (graded #335981).
6. kai_leal_gonzalez, two-term closure of the (3,1) slice of κ_4, github.com/al07-alt/whest-p2-k31-two-term (MIT), forum topic 18246.
7. Organiser ruling on building on other participants' public MIT code, forum topic 18240 (5 Oct 2026).
8. Organiser ruling of 11 Sep 2026 that Strassen–Winograd is legal at any recursion depth (competition forum).